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DOCTORAL THESIS

**Remote Sensing Image Classification and
Annotation based on Graphs**

Defended by

Khitem AMIRI

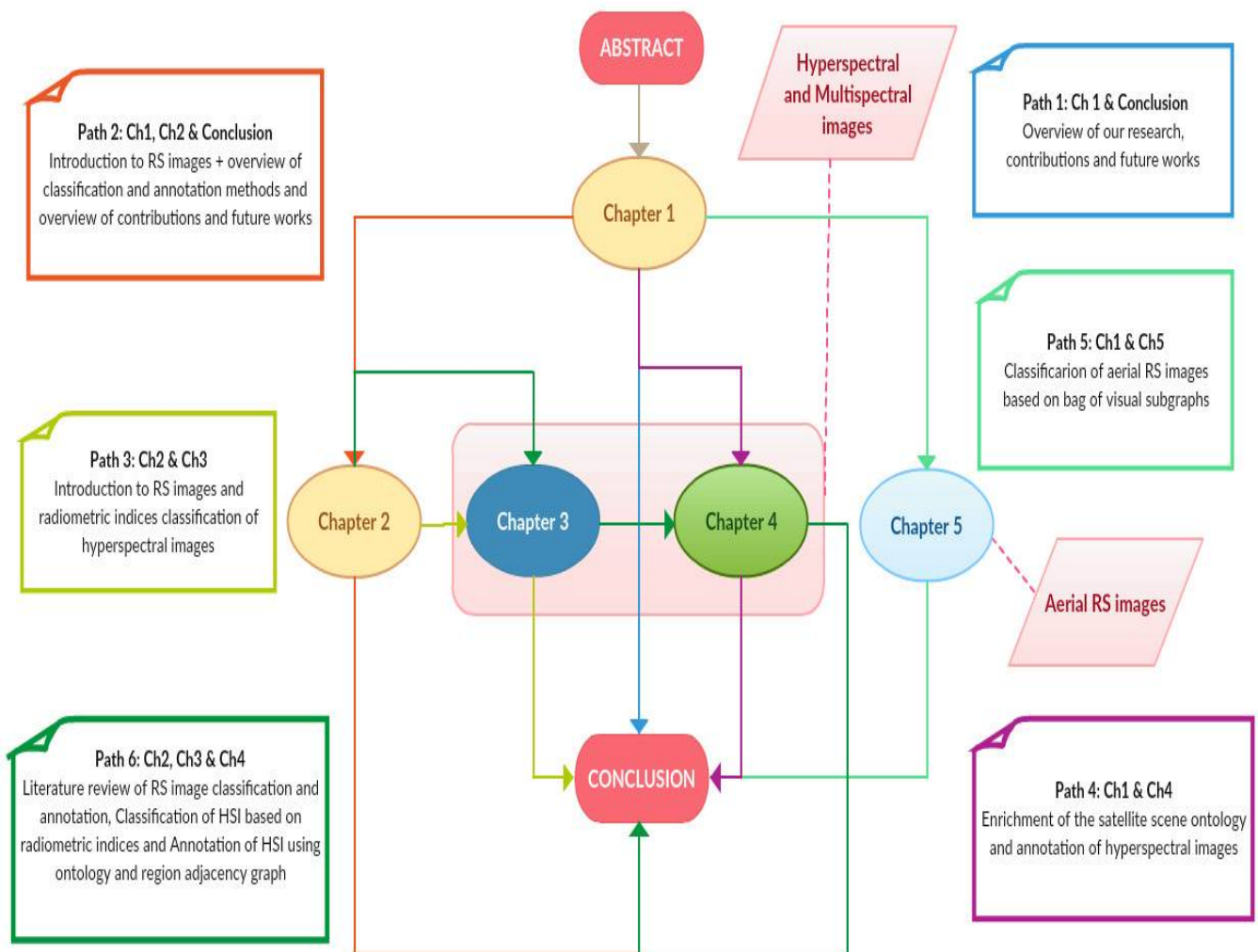
*A thesis submitted in fulfillment of the requirements
for the degree of Doctor of Computer Science*

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MANOUBA UNIVERSITY

Abstract

Faculty Name

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Now more than ever, the information has become available in different format ranging from text to images, audios and videos. In addition to the storage challenge, there is a matter of efficient understanding and treatment of these information resources. Among the needs, the organization and indexing of information is of primarily importance. With the growing development of digital imagery, efficiently searching for images has become one of the biggest challenges. In fact, with the recent development of sensors and the availability of remotely sensed images, the annotation of these images has gained a growing interest. Therefore, RS images are used in several fields and having a collection of accurately labeled data can affect positively the performance of learning based methods.

The first attempt to annotate images was done manually based on human experts. This solution gave interesting results; however it is not scalable to the actual large volumes of data, time consuming and subjective. This had lead to the use of Text Based Image retrieval (TBIR) methods. TBIR methods suffered from the semantic gap problem since the low level features of the image are completely ignored. As a result, Content-based image retrieval (CBIR) were proposed and used for decades to index and search for images. CBIR methods face also the semantic gap problem because of the difficulty of extracting semantic information from the images by computers as humans do. In order to reduce the semantic gap problem, the automatic image annotation was considered as a prominent solution in the recent years.

The purpose of this thesis is to explore the existing approaches for automatic annotation of images and specifically remotely sensed images. It starts with the definition of satellite images. Then, it reports the existing methods for image annotation which constitutes the starting point of this research. The thesis explores also the classification of satellite image methods since they are closely related to the annotation task. Two recent taxonomies of image annotation and image classification methods were presented and a set of representative approaches were explained. Generally, these methods rely on the use of spatial and/or spectral information of RS images. Due to the complexity of RS images, the combination of spatial-spectral information generally outperforms methods that only used either spatial or spectral information. The complexity of the annotation process and the desire to get closer to the human annotation have led to the incorporation of a priori knowledge. The later were frequently encoded as domain ontologies and were largely applied in several domains such as the semantic web.

On the other hand, one common and important step between image annotation and image classification is feature extraction. Convinced by the importance of this step, we propose to use radiometric indices as spectral features of hyperspectral and multispectral images for the classification task. We have got promising results confirming the discriminative power of radiometric indices on the majority of datasets.

Since, we aim to propose a faithful automatic annotation of images; we consider almost all the relevant information about RS images via the use of the largest domain ontology (*SatelliteSceneOntology-V3*) combined with the available spectral libraries and the low level features of the images. Before annotating the images, we proceed to enrich the ontology conceptually and intentionally (by adding the radiometric indices values among other features). Thus, the comparison of image regions features and ontology concepts became possible. An image is first oversegmented using SLIC algorithm, then features are extracted for each region and the image is represented by a region adjacency graph. The later store the spectral and spatial attributes of image regions. The spectral features of graph nodes (image regions) are compared to those stored in the ontology and the closest concepts are retained. At this step, even if we are satisfied with the resulting annotation, we still think that an

image is not just about the content or the objects it contains, but what's the image is about is more interesting. Consequently, we propose to represent the resulting image annotation by a graph of concepts that is not only easier to understand, but allows to store and to infer several information about image regions.

Finally, we have tried to use region adjacency graphs (RAG) for the classification of remotely sensed scenes. Hence, we oversegment images always using the SLIC algorithm and we represent the image by a RAG. Then, we search for a set of cliques inside the images that can be considered as specific patterns that have a different frequency depending on the scene category or class. Based on the frequency of subgraphs, we computed the histogram of subgraphs and we used it as a feature vector for the classification of three popularly used RS scene datasets. The obtained results were very satisfactory.

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Khitem AMIRI

Dedicated to:

Mom, Dad

My sisters Sarra and Amoula

My brother Meher and Imed

*All the family **AMIRI***

And to all my friends.

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Chapter 1

Introduction

In this research, we are interested in the automatic annotation of remotely sensed (RS) images. This involves many fields such as object recognition, object identification, image segmentation, image classification, machine learning and several other related fields.

The annotation of images used to be done manually and is obviously a complex task for humans as well as for machines. Manual annotation gives interesting results and contributes to the understanding of unlabelled images using classification and recognition algorithms. However the manual annotation cannot be used nowadays due to the overwhelming volumes of images and RS images available ([zhang2012review](#)). It is also incomplete and subjective. Moreover, most of the existing manual annotations are associated with the whole image and not with image regions. Thus we cannot identify precisely the objects therein. This is a real problem especially for RS images, because they contain many objects and many classes in the same image. These limitations encouraged researchers over years to propose new methods to automatically annotate images.

In the first proposals, the image is annotated based on the surrounding text, the name of the image file, the metadata, etc. This method is not efficient and can produce poor semantical descriptions since image features are completely ignored ([hanbury2008survey](#)). Consequently, Automatic Image Annotation (AIA) began using low level features of images. They use color, texture and shape features independently or by combining them. The most important issue AIA suffers from, is the semantic gap problem which is defined as the lack of concordance between the low level features extracted by the computer and the semantic understanding of the image by humans (Smeulders et al., [2000](#)). Image annotation has been considered as a solution for the semantic gap problem. This is one of the motivations which pushed us to study the annotation of remotely sensed images.

1.1 Motivations of RS image annotation

Being aware of the importance of RS images and their possible applications, we are working on their automatic annotation. This can be useful for classification, object recognition and several other tasks. Usually, we annotate data (add metadata) in order to allow accessing it. This is the key idea behind the semantic web. Thus, the metadata added to images or documents allows us to perform more effective searches. In the case of images, fully describing the image by textual annotation or keywords allows us to perform image searches effectively using text search techniques. The major drawback of the manual addition of metadata is that it is an extremely labor-intensive and time-consuming task. Thus, the big volumes of data, that keep increasing, preclude the use of manual annotation. We urgently need to automate the indexing and the organization of image collections.

In our research, we are interested by the proposal of a semantic annotation approach for RS images that does not involve humans and that guarantees a near human understanding of the image semantics. This will allow us to perform more precise RS image retrieval and a better understanding of the image content. Image annotation is also considered as a solution for the semantic gap problem and by the present research we aim to reduce this gap for RS images. On the other hand, spatial queries are emergent nowadays and constitute an interesting way of searching specific information inside the RS images. We can for example search for a vegetation region next to water region for a specific task or activity or even statistics. We also can specify the size of the regions and have more refined results. Hence, the image annotation can facilitate these tasks.

1.2 Issues

The first challenge that faces image annotation as well as other tasks (such as classification) is the "incomplete labeling". The datasets of annotated images which are available for training and testing are not annotated with all the relevant keywords. Moreover, keywords do not correspond to image regions but to the whole image. Therefore, we cannot have any idea about the spatial distribution of regions within the image.

Also, the "class-imbalance" constitutes another issue for large scale image annotation and retrieval. The "class-imbalance" is the existence of large variations in the frequency of different labels per image. This is the case of most of the existing hyperspectral images. The semantic gap is another challenge that is still unresolved in images as well as RS images.

The automatic semantic annotation of RS images is challenging since we do not have, yet relevant knowledge structures that contain all the concepts needed to annotate RS images. Ontologies, which are considered as the most meaningful knowledge structures, may play an important role for semantic image annotation. Nevertheless, the available RS ontologies are not detailed enough to represent all the concepts in RS images. Consequently, most of the existing approaches deal with image annotation and not semantic annotation. The semantic annotation requires the consideration of spatial relations between regions of the image, the semantic relations between concepts and a priori knowledge in order to make deductions and semantic interpretation of the image.

Our goal is to develop an automated semantic RS image annotation approach which takes advantage of a priori knowledge in order to exploit rich and high level concepts. We need to maintain the semantic relationships between concepts and to allow inferring spatial relationships between objects/concepts.

1.3 Contributions

In this research, our main contribution is the proposal of a faithful representation of satellite images based on graphs. Once, we represented the image with a graph, we adapt several algorithms in order to annotate, to classify and to extract useful features related to graphs. We propose also another minimal representation for hyperspectral images based on a set of radiometric indices which are tested in the classification of hyperspectral and multispectral images and gave comparable results with the-state-of-the-art classification methods. We summarize our contributions as follows:

- Proposal of graph based representation for RS images. This representation considers the spatial and spectral information of image regions
- Conceptual and intentional enrichment of an existing satellite scene ontology
- Definition of satellite image annotation process and proposal of a semantic annotation method based on the proposed graph and the enriched RS ontology
- Proposal of a new RS scene classification method using the adjacency graphs to extract the frequent subgraphs

We are going to explain our contributions in the next chapters.

1.4 Dissertation Structure

This dissertation is divided into five chapters. We describe the structure in a chapter-by-chapter manner as follows:

- Chapter 1: Introduction of the context of the thesis
- Chapter 2: Automatic Satellite Image Annotation: This chapter constitutes a general introduction to the background of this research. It defines the automatic annotation of RS images and reports a representative set of image annotation methods organized under a recent taxonomy. In the second part, we present the classification problem for RS images and report the major methods dealing with this issue.
- Chapter 3: Classification of hyperspectral image based on Radiometric Indices: in this chapter we present our approach for the classification of hyperspectral images. We aim in this chapter to test the discriminative power of radiometric indices.
- Chapter 4: Graph Model for the Annotation of RS Images: We reserve this chapter to perform the annotation of RS images based on the use of region adjacency graphs. This chapter is divided into two main parts. The first part is reserved to the explanation of the enrichment of the satellite scene ontology. And the second part describes the annotation method and reports the annotation accuracy in terms of precision, recall, F1 and N+ scores.
- Chapter 5: Bag of SubGraphs for the Classification of large scale RS scenes: In this chapter, we present our approach for RS scene classification while exploiting the already defined Region Adjacency Graph (RAG). This involves the search of relevant features based on graphs. We found that the frequency of subgraphs in the different image categories constitutes a good feature for an efficient classification of RS scenes.
- Chapter 6: Conclusion and future works: This chapter concludes and discusses the overall findings and contributions. We also give insights about our future works especially regarding chapters 4 and 5.

Chapter 2

Automatic Satellite Image Classification and Annotation

2.1 Introduction

The human eye can be considered as a remote sensor, and can therefore capture images that can be stored in the brain and retrieved later. However, we can only reproduce them in a subjective way. Our eye is so efficient but it can only capture the visible radiation which covers a very small part of the electromagnetic spectrum. In order to overcome these limitations, several instruments have been invented and produced that allow to capture more details about our environment without being in a direct contact with it. Thus, remote sensing (RS) refers to the acquisition of information about an object or a phenomenon using a measuring instrument that has no direct contact with the object being studied. It refers to the remote use of any type of instrument (for example, an airplane, spacecraft, satellite or even a boat) allowing the acquisition of information about the environment. RS images constitute a powerful source of information that allows the monitoring of the environment. We can, for example, monitor the changes of population or the evolution of agriculture fields. We can create and update maps and we can even study areas that are unreachable for humans.

2.2 Remote Sensing Imagery

Remote Sensing images became available since 1972 with the launch of ERTS-1 satellite later called Landsat-1. Since then, there were many enhancements and we witnessed the proposal of several sensors that are monitoring the Earth. We can cite SPOT-1 HRV launched in 1986 and IKONOS in 1999. In order to acquire images in a remote way, we use a sensor or a satellite that can be seen, in the simplest ways, as an object moving around a bigger object. There are multiple classifications of sensors. We have equatorial and polar orbit sensors, active and passive ones, etc. The choice of the sensor depends on the mission and thus the produced images vary accordingly ranging from high to low spatial resolution and high to low spectral resolution images.

2.2.1 Acquisition process of RS images

The acquisition of RS images consists of measuring the reflected/emitted light from objects. Sensors record the electromagnetic energy reflected/emitted from the ground then transmit it to a receiving station on the earth and finally process it into an image. The acquisition process can be described by Figure 2.1.

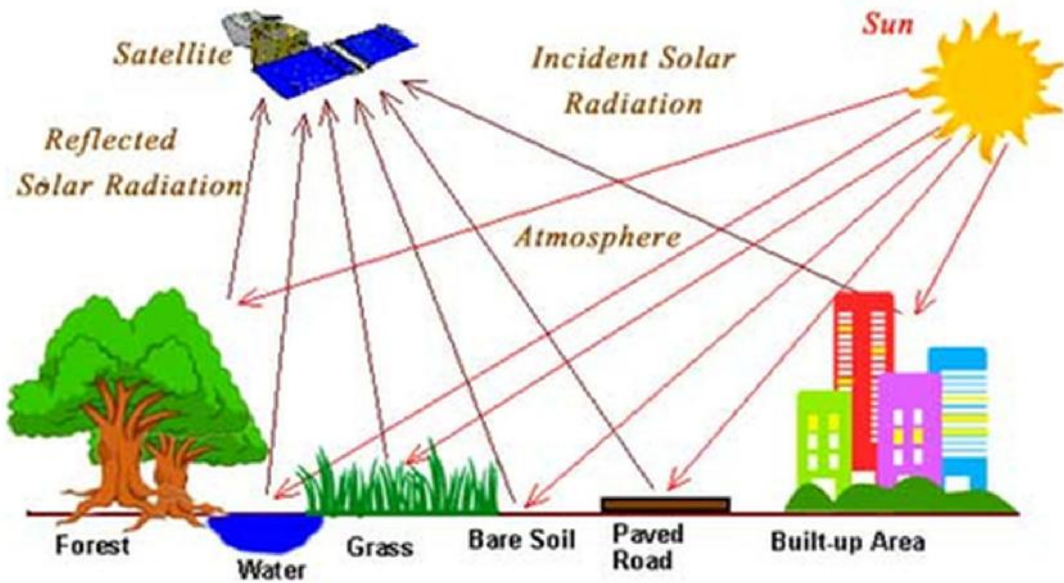


FIGURE 2.1: Process of remote sensing image acquisition

The light source can be either the sun or the sensor itself (in case of active sensors). Active sensors, unlike passive ones, are not dependent of the sun and are able to emit electromagnetic energy and can assess images at night time. There are many factors that influence the effectiveness of the measurement of reflected energy from objects. While the light is traveling from the energy source to the target or the inverse, it passes through the atmosphere and it interacts with it. Thus, the particles of the environment can absorb or reflect a part of the energy as well as the target objects and can consequently alter the target response. When the energy reaches the object, it interacts with it depending on its properties as well as the nature of the object. The latter can absorb a part of the Incident Energy (IE), diffuses a part and reflects another part (as illustrated in Figure 2.1). In fact, the sensor records the emitted plus the reflected energy from objects under his field of view. Then, the recorded energy is transmitted by the sensor to a receiving station at the Earth which processes the data into an image. Later, the produced image is interpreted in order to extract useful information about the studied zone to reveal new information or to monitor and solve a particular problem.

The amount of reflected light is specific to each object and allows us to discriminate between objects on the ground. The response of objects constructs their spectral signatures which can be recorded with different levels of details depending on the spectral range covered by the sensor. This can be highlighted by Figure 2.2.

2.2.2 Remote Sensing Image Resolutions

Over the recent years, sensors have been evolved and there are several amazing spectrometers that enable the monitoring of the environment with a greater precision and to prevent several risks.

In general, we need to find a compromise between the spectral and spatial resolutions of RS images. Sensors that are producing "high" to "very high" spatial resolutions are only able to produce a "low" to "medium" spectral resolutions.

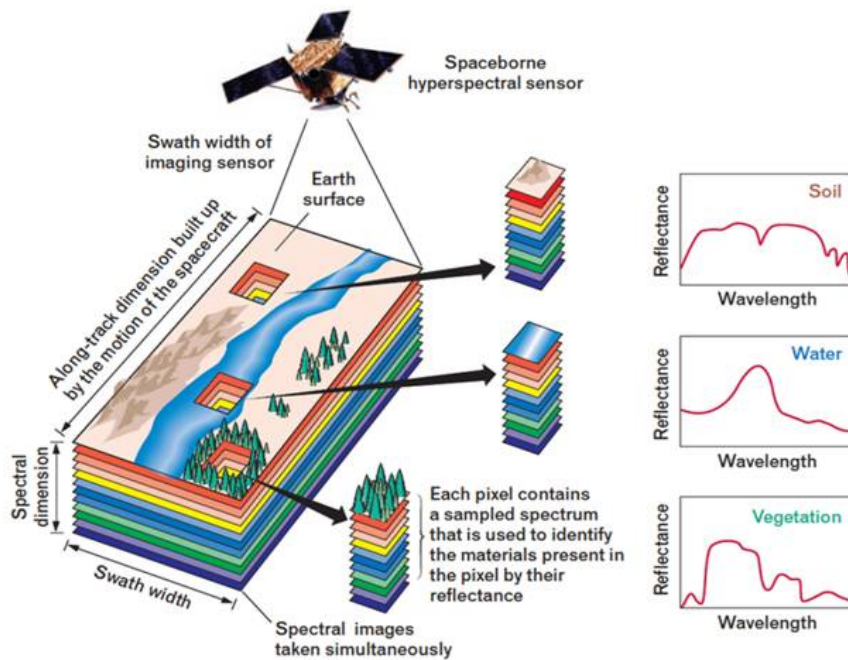


FIGURE 2.2: Imaging Spectroscopy (from Cossa Earth Observation Center)

An image is also characterized by its radiometric resolution. The latter is defined as the smallest change in intensity level that can be depicted by the sensing system. Generally, we use 8 bits which gives $2^8 = 256$ intensity levels that can be detected. The radiometric resolution depends on the signal to noise ratio of the detector. For a digital image, the radiometric resolution is limited by the number of discrete quantization levels that are utilized to digitize the continuous intensity value. A small number of quantization levels does not greatly impact the visual quality of the image, however the accuracy of numerical analysis will be affected by the reduced number of quantization levels.

Another important resolution that impacts several analysis tasks on RS images is the spatial resolution. It is frequently defined as the pixel size. But, it is not the case. Based on the spatial resolution, the image will be used for a particular task. A low spatial resolution image is characterized by 1 km and a medium resolution image by hundreds of meters. These images are only available in multispectral mode including visible and infrared portions of the spectrum. They are useful for the global and continental mappings, large disasters monitoring, atmosphere and ocean monitoring, etc.

On the other hand, satellite images with "high" resolution have a spatial resolution of approximately tens of meters. They can be used for the monitoring of land cover and land use, of the vegetation and geology among others. There are multiple sensors that provide high spatial resolution data such as SPOT with 5 m, EO-1 (Earth Observation) with 10 and 30 m and Landsat(1 to 5) with a minimum resolution of 30 m.

Satellite data with very high resolution have a spatial resolution at about 1 m. This resolution can only be found in the panchromatic image. High spatial resolution images are useful for multiple applications such as detailed mapping, urban studies and control of agriculture activities, etc. There are several sensors having a very high spatial resolution such as WorldView-2 with 0.5 m, QuickBird with 0.6 m

and Ikonos with 1 m for the panchromatic band.

A high to very high spatial resolution image is very useful but not sufficient for the precise identification of objects. Hence, the spectral resolution can help us greatly in performing object identification. The spectral resolution is specific to every sensor. As we stated earlier, objects respond to the light and reflect a specific amount of energy that is known as their spectral (or optical) resolution. The latter defines the maximal number of spectral peaks that can be determined by a given spectrometer. Thus, if we have a spectrometer with a wavelength range of 100 nm and a spectral resolution of 1 nm , then the system will be able to resolve at maximum 100 individual peaks across the spectrum.

In order to correctly deal with a given application, a researcher or an analyst have to choose correctly the spectral resolution. A compromise have to be found, since a "too low" spectral resolution implies a loss of information and a "too high" resolution can cause unnecessary computational load. There is no quantification of "too low", "low", "medium", etc, spectral resolutions. It depends on the application and the desired information. Thus, in order to identify rock types or mineral types, a "high" to "too high" spectral resolution is needed since there are very small differences between the spectral curves that may not be visible in low/medium resolutions. Consequently, such applications require sharp curves to discriminate fine classes or object types. However, identifying vegetation from water objects, for example, would need "low" to "medium" spectral resolution images.

2.2.3 Remote Sensing Image Types

There are several types of remotely sensed images that are useful for various applications. In fact, we are familiar to Google Earth and Google Maps satellite images, however there are other typers of images. We can cite Light Detection and Ranging (LiDAR), Radio Detection and Ranging (RADAR), Unmanned Aerial Systems (UAS), hyperspectral images, thermal images, aerial photography, etc.

RS includes aerial photography and satellite images. It refers to the collection of data from an important distance of the target. RS creates images of the Earth's surface that are much larger than those captured by humans on the ground. RS images, thus, show the position and the relations between objects and the geographic features within the target area.

The first kind of RS began in 1860 when geographers used balloons and kites to capture a wider region of the Earth from above. The introduction of airplanes allowed capturing images from much higher altitudes. Aerial photography is produced using sensors mounted on aerial crafts. Nowadays, the altitude of aerial photography ranges from a small distance above the ground to more than 60.000 feet. A low altitude allows to capture more details. However, a high altitude leads to loosing fine details while gaining in the size of the area captured and the relations between regions. In the beginning, aerial images were produced in black-and-white. Then they used natural-color which showed several problems with these color images such as haze, cloud, etc. Nowadays, several aerial images are still in black-and-white because their correction can be easily performed by adjusting the contrast (madry2013introduction).

Aerial images are also produced using Unmanned Aerial Systems (UAS), also called drones. UAS are aircrafts that don't have on-board pilots and that carry a payload and are connected to a remote operator. They are increasingly gaining popularity in the recent years due to their relatively low cost and to the quality of their

produced data. They are particularly suitable for agriculture monitoring since they record images at high spatial and temporal resolutions.

Infrared photography is another type of aerial photography. As the name suggests, infrared photography uses a sensor that is sensitive to the infrared light. It uses the wavelengths ranging from 700 nm to 900 nm corresponding to the near infrared region of the spectrum (which is the domain of thermal imagery). The film of the sensor (camera) used, in infrared photography, is also sensitive to the visible light. Thus, an infrared-passing filter is utilized in order to block the visible light and allow only the infrared light to pass through the camera. This kind of images is originally used by the military in order to identify suspicious targets.

Thermal imaging is one of the three night vision technologies in addition to the near infrared illumination and low light imaging. It uses the infrared radiation of objects in order to capture an image of them. This imaging method is particularly useful in dark environment having low visibility. It is able to work even without any ambient light and it can penetrate fog, haze and smoke.

Thermal imaregy is based on the fact that all objects emit infrared energy (heat) depending on their temperature. Thus, the heat signature defines the infrared energy emitted by an object. Generally, the radiation emitted by an object is proportional to its temperature value (the radiation increases if the temperature increases). A thermal camera can detect the small differences in temperature values. Since, the objects share rarely the same temperature as the objects around them, the thermal camera can detect them separately in the thermal image. The produced image is, generally, grayscale where black corresponds to cold objects, white to hot objects and the gray values indicate variations between them.

Therefore, the infrared camera is very useful for several applications such as security, surveillance and military uses, medical diagnosis, etc. However, the interpretation of thermal images is not evident because they constitute a representation of temperature values. The interpretation is dependant of the user's expertise.

On the other hand, LiDAR is based on aerial mapping. It uses the reflected laser returns from the ground to the aircraft. The latter is equipped with GPS (Global Positioning System) and IMU (Inertial Measurement Unit) sensors that allow the computation of the elevation and the geospatial location of objects precisely.

The key idea behind LiDAR technology, is to emit light at the target surface on the ground and to measure the time it takes to return to its source. Since we know the speed of the light (300.000 kilometers per second), we can easily compute the distance travelled by the reflected light based on the equation 2.1:

$$Distance = \frac{SpeedofLight * TimeofFlight}{2} \quad (2.1)$$

Then, the time taken by each pulse to return back to the scanner is calculated iteratively thus allowing the LiDAR instrument to build a map of the target surface. While the sensor is moving, we have to compute the height, the location and the orientation in order to compute the position of the laser pulse at sending and receiving times. This can be done using a GPS placed on the aircraft which measures the position each 0.5s. As shown in Figure 2.3 , the ground GPS allows the computation of the position of the aircraft every 0.5s based on a known ground position. The altitude is measured by the IMU sensor each 0.002s. The first return measures the highest elevations of the ground such as tree or buildings since the travelled distance is smaller than the rest of the objects. However, the last return represents the ground level of the bare Earth. The latter allows us to have an accurate idea about the elevation and objects on the target Earth surface.

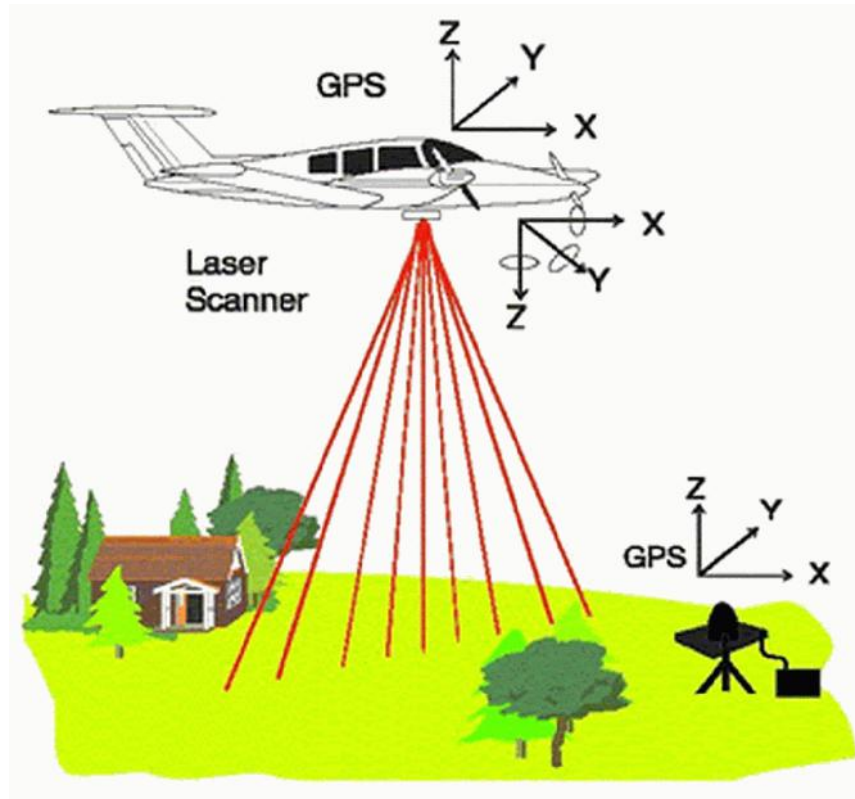


FIGURE 2.3: LiDAR acquisition

LiDAR utilizes lasers in order to send and receive the signals. However, RADAR imaging produces its light to illuminate a given area on the ground and captures an image of it at radio wavelengths. It uses the radio signals to determine the distance and the angle of inclination of objects on the ground. It is not affected by weather conditions such as clouds or fogs. RADAR uses longer wavelengths than LiDAR, thus it cannot detect small objects and their data can be distorted. It provides the distance from objects as well as their angular positions on the surface.

The RADAR pulses can consider multiple targets at the same time and returns data about many objects also at the same time. In contrast, many small and closely placed objects can be ignored or distorted. It also takes more time than LiDAR to reach objects on the ground since its signals travel on a lower speed. Therefore, the data produced by RADAR technology is less accurate than LiDAR data due to the incomplete data returned by RADAR concerning the objects on the ground. Thus, the data collected by RADAR has a lower resolution than those recorded by LiDAR. RADAR is mainly used to compute the range, the position and the velocity of the target objects while LiDAR is used to measure their position and their distance.

Unlike the aforementioned aerial images, hyperspectral images (HSI) are composed of hundreds of contiguous bands. They have spectral resolution that is very interesting for the identification of objects. In HSI, each pixel has three dimensions where two dimensions correspond to the spatial information and the third dimension corresponds to the spectral information. This 3-dimensional (3D) data cube is referred to as *hypercube data* or as an *image cube* (Graves et al., 2003). RGB color images constitute a simple and common illustration of 3D data cube in which each pixel has blue, green and red color values.

Multispectral images are composed of 4 to 20 color channels. However, it is

assumed that HSI are spectrally sampled at more than 20 equally distributed wavelengths (Scarcelli and Yun, 2012). The spectral range of HSI can include the infrared and ultraviolet regions in addition to the visible region. In both HSI and MSI, the color channels are stacked up in order to form the data cube where the size is dependent of the number of wavelengths used (Houlgrave et al., 2013). Therefore, HSI are characterized by a higher spectral resolution compared to MSI (Ostrum, 2006).

2.3 Image Classification

2.3.1 Formulation of image classification

Image classification has a growing interest among the research community. It has been used in several applications especially in medical and remote-sensing fields. In fact, the classification of hyperspectral images is formulated by (Guo et al., 2018b). Let $D = \{x_1, x_2, \dots, x_N\}$ be the image dataset, where $x_n = \{x_{n^1}, x_{n^2}, \dots, x_{n^S}\}$ is the n -th pixel with S bands or channels and N denotes the number of image pixels. To obtain a classifier, we have to build a training set $T = \{(x_1, y_1), (x_2, y_2), \dots, (x_M, y_M)\}$ where $y_m \in \{c_1, c_2, \dots, c_K\}$ which denotes the K labels and M denotes the number of samples ($M < N$). The objective of image classification is to find the class y_m that best matches a given $x_m \in D$.

2.3.2 Annotation vs Classification of Images

Image classification is the process of associating class labels with the image pixels or regions or images (in the case of scene classification). Both tasks, classification and annotation, can be performed manually, semi-automatically or automatically. However, nowadays we tend to use automatic approaches due to the big volumes of images. The main difference between image annotation and image classification is the number of class labels. Thus, in image annotation we mainly use a predefined vocabulary that contains a bigger number of labels than the number of classes used in image classification. Also, the classification can be viewed as a step in the annotation process. On the other hand, the annotation step is a prior step to the classification task since it allows to label the datasets of images. These labeled datasets will be used by classification approaches in order to model the co-occurrences between low level features and the classes or labels.

2.3.3 Overview of RS image classification approaches

Hyperspectral (HS) sensors such as *AVIRIS*, *Hyperion* and *HYDICE* record the reflectance from the surface of the Earth across the full range of solar wavelengths. They have high spectral resolutions that allow the collection of high dimensional data with rich information. In order to extract useful information from HS data, feature mining is widely addressed (Cheriyadat, 2014). Feature mining aims to identify the informative subspace of features. It is considered as a complex and ultimate task for Hyperspectral Image (HSI) classification. It includes Feature Generation, Feature Selection and Feature Extraction. The identification of the effective set of features that can efficiently discriminate between the classes in HS data is the key point to make a classifier powerful for HSI classification (Tan et al., 2014). In fact, the performance of a classifier is conditioned by the quality of the input features. Thus finding the effective features is a hot topic in HSI classification that caught the attention of several researchers in the last two decades.

Image classification can be divided into two categories based on the features considered. The first category is reserved to the original or also called knowledge based features. The second category concerns the transformed features which are computed based on dimensionality reduction techniques namely feature extraction techniques. The knowledge features are also called original or handcrafted features. They are computed based on the original image itself, i.e. using mathematical operations on the values of different bands. Knowledge-based methods rely on the understanding of the spectral characteristics of hyperspectral images. They use arithmetic operations on the relevant bands in order to distinguish the object classes. Generally, the radiometric indices use the blue, green, red, Near Infrared and Short Wave Infrared (SWIR) bands. As typical illustrations, we can find the Normalized Difference Vegetation Index (NDVI) and Normalized difference water index (NDWI) among several other radiometric indices (Jia et al., 2015).

Due to the huge number of features, some Feature Selection (FS) techniques are used. They consist of selecting an informative subspace from the original features based on some criterion such as entropy.

On the other hand, transformed features are produced using a specific kind of dimensionality reduction techniques which is Feature Extraction (FE). FE techniques can be divided into supervised, unsupervised and semi supervised techniques (Kavitha and Kayalvizhi, 2018) as shown in Figure 2.4.

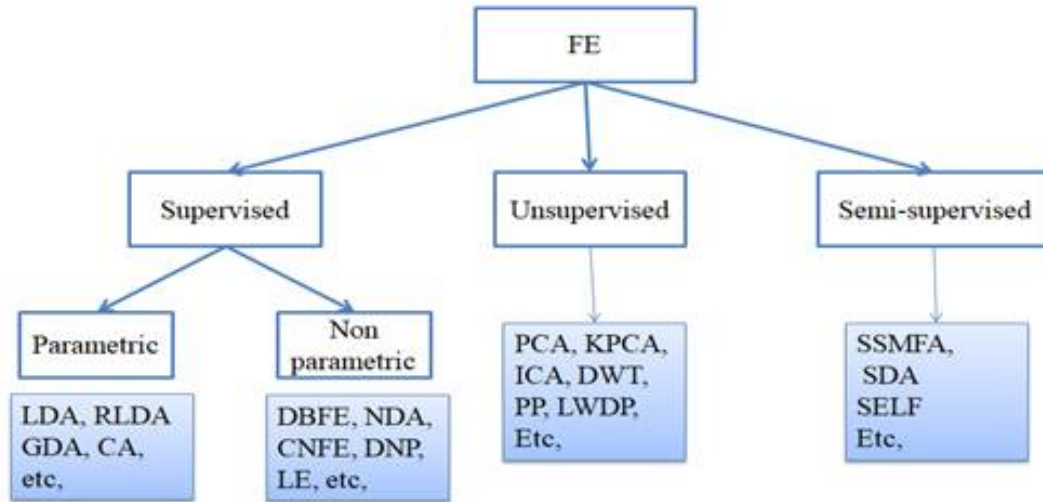


FIGURE 2.4: Feature extraction methods for HSI classification (Jia et al., 2015)

Supervised feature extraction methods are split into parametric and non parametric. In the parametric subclass, we can find the Linear Discriminant Analysis (LDA) (Paul and Kumar, 2018) and Canonical Analysis (Wang et al., 2014) which are based on the mean vector and covariance matrix of each class. The ratio of within-class to between-class scatter matrices is used as separability criterion for LDA.

One of the major limits of LDA concerns its dependence to the Gaussian distribution of classes. Also, LDA cannot handle cases where class data do not constitute a single cluster. Another limit concerns the number of produced features. Since the rank of the between-class scatter matrix is less than the number of classes (C), a maximum of $(C - 1)$ features can be extracted. Generally, $(C - 1)$ features are not sufficient for HSI classification (Wang et al., 2017).

In order to address these issues, Decision Boundary FE (DBFE) (Shang et al., 2018) and Nonparametric Weighted FE (NWFE) have been proposed, among others. In NWFE, different weights are assigned to each sample to measure the ‘weighted mean’. Then, it finds the distance between samples and their weighted means and computes the nonparametric between-class and within-class scatter matrices. The latter gives large weights to the samples that are close to the boundary and low weights to the samples far from the boundary (Guo et al., 2018b). There are nonlinear extensions of LDA that are based on kernels. Kernel Fisher Discriminant Analysis (KFD) (Kianisarkaleh and Ghassemian, 2016) and Generalized Discriminant Analysis (GDA) (Chen et al., 2015) have been proposed independently for kernel-based nonlinear extensions of LDA. KFD tries to handle the possible singularity problem of the total scatter matrix by considering singular value decomposition while GDA used regularization techniques. KFD and GDA involve Eigen decomposition which is very expensive to compute (Benediktsson et al., 2005) and (Fauvel et al., 2008).

Maximum Margin Projection (MMP) aims to find discriminant and geometrical structures of the data manifold. It computes a Nearest Neighbor (NN) graph that models the local geometric structure. Then, the graph is divided into two graphs (based on the class and neighborhood information): a within-class graph and between-class graph. Each tuple of data points in the within-class graph are connected if they have the same label or are too close to each other. The data points of different labels are connected in the between-class graph. Finally, a linear transformation matrix has to be found in order to map the image to a new subspace while preserving the local neighborhood and the discriminant information.

(Dalla Mura et al., 2011) proposed an HSI classification approach based on Independent Component Analysis (ICA) and Extended morphological Attribute Profiles (EAPs) as illustrated in 2.5. The role of the ICA is to map the HS images into a subspace where the components are as independent as possible. And a set of extended EAPs features are obtained via the features extracted using APs. Finally, the features produced based on the morphological processing are classified with an SVM classifier. Applying their approach on a *Pavia Center* and *Pavia university* datasets, authors got an overall accuracy of 94.47% and 98.8% on the *Pavia University* and *Pavia Center* respectively.

Under the **unsupervised class of methods**, we can find Principle Component Analysis (PCA) (Chen et al., 2014), ICA (Ma et al., 2016b), Discrete Wavelet Transform (DWT) (Aptoula et al., 2016) and several other methods (Cheriyadat, 2014). The key idea behind PCA and ICA is to map the original high dimensional data into a lower dimensional subspace using data transform. Unlike the supervised FE methods, the unsupervised FE methods do not require any a priori knowledge about the data.

On the other hand, semi-supervised FE methods for HSI classification consider both labeled and unlabeled samples trying to deal with the scarcity of labeled data challenge faced by supervised methods. We can cite the Semi-Supervised Marginal Fisher Analysis (SSMFA) (Wang et al., 2014), Semi-supervised Discriminant Analysis (SDA) (Cai et al., 2007) and SEmi-supervised Local Fisher discriminant analysis (SELF) (Sugiyama et al., 2010).

The unsupervised learning based methods create a set of basis functions in order to encode features to create learned features from handcrafted ones. Thus, the input of the function is the collection of handcrafted features and the output is the learned features. The Bag of Visual Words (BoVW) (Tsai, 2012) is an illustrative example of this class where SIFT features are the input and the learned codebook is the output. The BoVW can be illustrated by Figure 2.6.

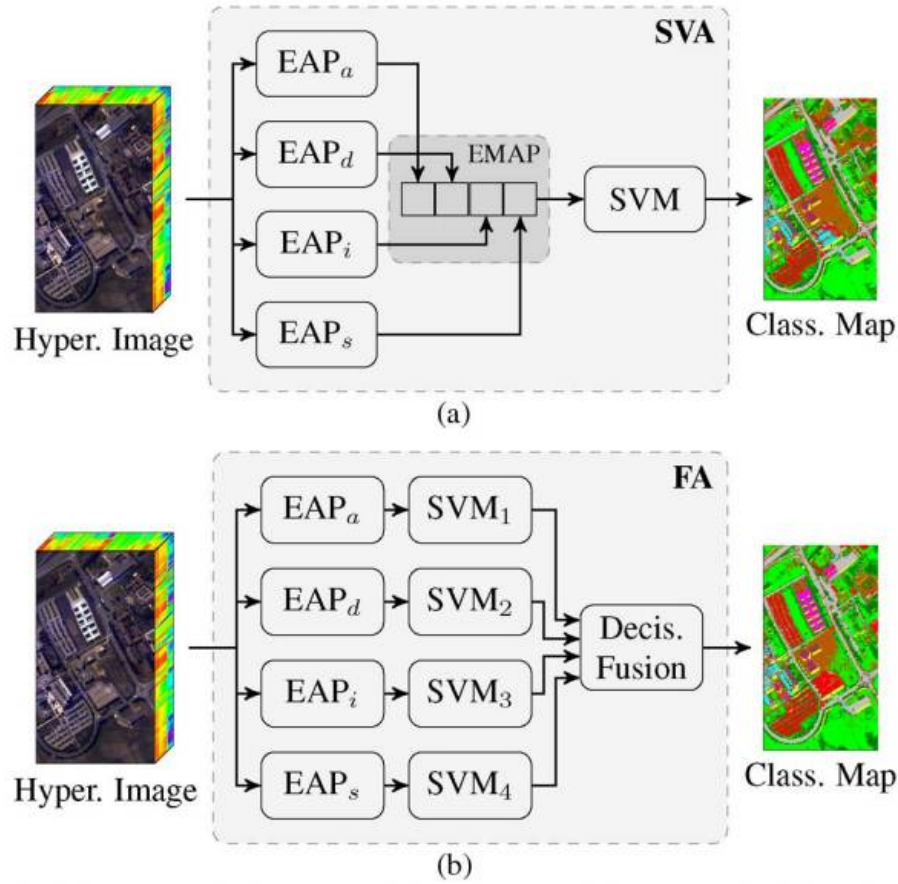


FIGURE 2.5: Approaches of multiple EAPs (a) Stacked Vector Approach (SVA) and (b) fusion approach (FA) (Dalla Mura et al., 2011)

There are several variations of the BoVW including those using handcrafted features and those using deeply-learned features. The later have interesting representation capability for RS images while the handcrafted features give less comparable results. Deep Learning (DL) based methods are recently gaining popularity through the proposed methods. Sharing the same idea of BoVW, the BoCF (Cheng et al., 2017b) is a recent method that exploits the discriminative power of convolutional features extracted from CNN layers in order to perform scene classification.

The BoCF method extracts the features from the layers of the pre-trained CNNs AlexNet, GoogleNet and VGGNet-16. The authors use the layers "conv5", "inception 4(e)" and "conv5-3" which are pretrained on the ImageNet dataset. The flowchart of the BoCF is illustrated by Figure 2.7. Then, once the features are extracted from the layers of deep networks, they cluster them using k-means algorithm in order to construct the codebook. Each cluster in the codebook corresponds to a visual word and each image is therefore represented by an histogram of visual words. Finally, a linear one-versus-all SVM classifier is trained for each class. Thereby, images belonging to the selected class are considered as positive instances whereas the rest of the images are considered as negative instances. For testing, the label with the highest score is assigned to each unlabeled image.

The features vectors computed have a big effect on the classification accuracy. Thus, many methods tried to address this issue.

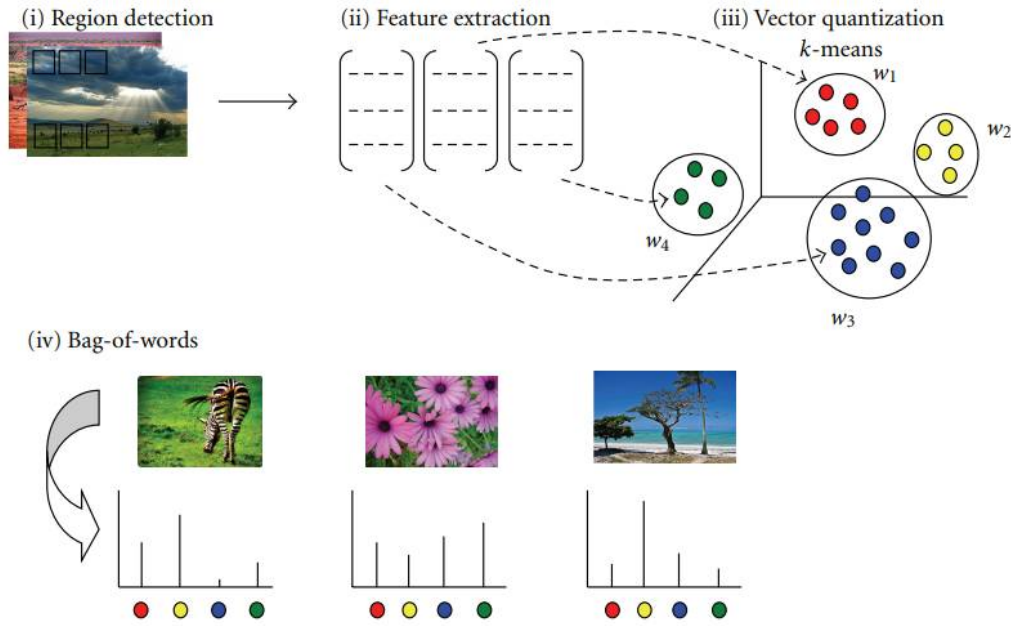


FIGURE 2.6: Steps of the bag-of-words for image representation

Recently, in the field of Remote Sensing images, the classification methods are divided into three main classes based on the features used (Cheng et al., 2017b). The first class represents handcrafted features based where the features are mined by experts in RS images which corresponds to the same class in the taxonomy of . The second class concerns the unsupervised learned features based classification methods. Since, the proposal of features is a hard task and time consuming, the third class was a necessity which concerns deep learning features based methods. The aim of deep learning is to skip the feature extraction step and to learn discriminative features through the layers of the network.

Due to the complexity of the feature generation, selection, or extraction, Deep Learning based techniques, especially Convolutional Neural Network (CNN) have been widely studied. Hence, deep-learning methodologies have been used for building ‘deep features’ for HSI and have been applied in spectral-spatial classification approaches based on Deep Auto Encoder [8, 9], Deep-Belief Network (DBN) (Jiang et al., 2018a), Attribute Profiles (Ma et al., 2016a), and CNN (Paoletti et al., 2018). Recently, CNNs based methods (depicted in Figure 2.8) are largely used for RS image

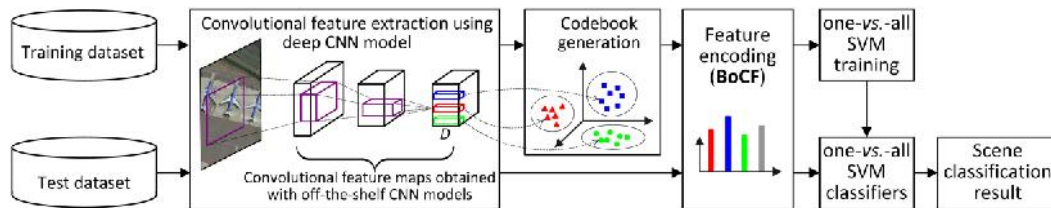


FIGURE 2.7: Illustration of the Bag of Convolutional Features method (Cheng et al., 2017b)

classification since the CNN architecture is suitable for images. CNN based methods are achieving high accuracy values despite the fact that they are "black box" and very hard to interpret.

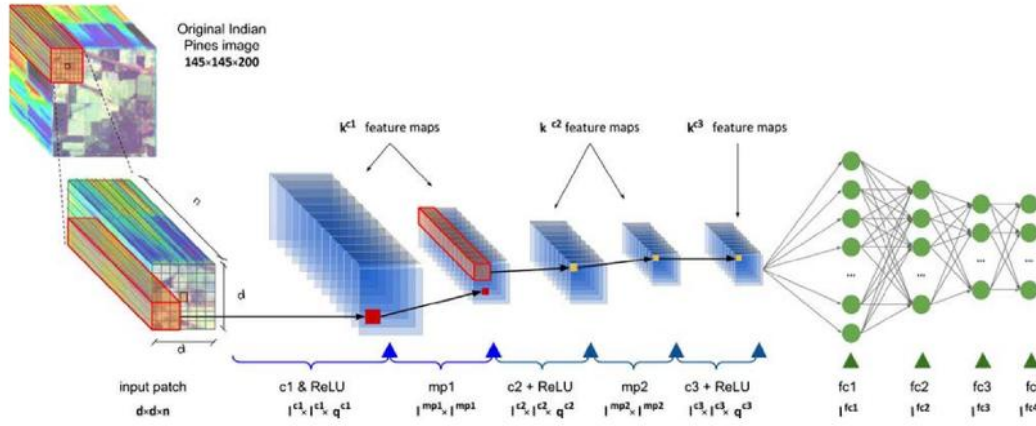


FIGURE 2.8: Illustration of Convolutional Neural Network for HSI classification (Paoletti et al., 2018)

Certainly, these methods achieve interesting results and allow us to skip the feature generation phase, however they have major limits. Many challenges limit the application of Machine and Deep Learning techniques for HSI classification. Among them, the scarcity of labeled data since there is few freely available HS datasets. On the other hand, using all the spectral bands in Machine Learning, not only causes unnecessary computational load, but also makes dealing with high dimensional data a challenging task. The results of class training have low generalization capacity when the ratio of the number of training samples available over the number of features is low which occurs for HSI if we do not consider feature reduction (Cheriyadat, 2014). Thus, we may obtain high classification accuracy on the training data but low accuracy on the testing data. This problem is known as Hughes effect (Zhao et al., 2016a) or the curse of dimensionality (Rouse Jr et al., 1973).

The majority of machine-learning and deep-learning algorithms typically require a large number of training samples. Nevertheless, building collections of labeled samples is a critical and time consuming task, especially for HSI. This encourages researchers to keep trying to improve the existing algorithms. In order to distinguish between image annotation and image classification, we define in the next section 2.4 the annotation problem and we report a subset of the existing methods under the taxonomy of the recent taxonomy of AIA.

2.4 Remote Sensing Image Annotation

Image annotation is the process of associating keywords or labels with the image (Carneiro and Vasconcelos, 2005), (Srikanth et al., 2005). There is sometimes confusion with image classification. However, there are some differences between them even if the goal is close. We are going to report their definitions in subsections 2.4.1 and 2.3.2. Then we present the recent taxonomy of image annotation approaches proposed by (Cheng et al., 2018) in subsection 2.4.3.

2.4.1 The Image Annotation Problem

"Automatic Image Annotation (AIA) is the task of automatically assigning keywords to an image that describe the content of the image" (Srikanth et al., 2005). This is a complex process that may involve several algorithms and classifiers. The annotation of images was addressed from several points of view thus giving rise to many definitions and formulations.

Generative and discriminative based methods are among the famous formulations of image annotation. AIA was considered as a retrieval problem inspired from the information retrieval field. Thus, the annotation is solved by finding the nearest neighbors of an unlabeled image in the feature space and to use their labels to annotate the test image (Makadia et al., 2010). Image annotation is also defined as a translation problem by translating image instances into keywords. To solve the translation model for image annotation two alternatives exist. The first solution is to solve it directly using language translation models. The second solution consists of solving the translation in an indirect manner by making inferences based on the co-occurrence of images and keywords. The annotation is also stated as a relevance model which starts by the computation of the joint probability of image features and keywords. Then, it computes the posterior probability of each label and the unlabelled image. Finally, the unlabelled image is annotated by the label with the highest probability value. From a discriminative view, image annotation is considered as a multi classification problem in which each image is labeled with multiple keywords or classes.

2.4.2 Semantic Image Annotation

With the increasing complexity of image content, especially RS images which contain many objects, the semantic annotation has emerged as a necessity. The semantic annotation of images (Carneiro and Vasconcelos, 2005) aims to assign a set of keywords (also called labels, or tags or textual descriptions) that best describe the content of an image. The keywords are extracted from a vocabulary L of semantic descriptors. The latter is learned based on a training set D containing pairs of (image, caption), $D = \{(I_1, w_1), \dots, (I_D, w_D)\}$.

The challenge of the semantic annotation of images is to map the extracted low level features to the high level concepts; thus to bridge the semantic gap problem. The semantic annotation depends on the a priori knowledge of the user and is very hard to efficiently reproduce by machines. The main difference between AIA and semantic annotation is the use of predefined knowledge structures such as ontologies or thesaurus or taxonomies. These knowledge structures allow the inference of deductions about the image semantics since the semantic is not only related to the components of the image but to its purpose (what is it about). Consequently, the resulting annotation is more efficient and closer to human's understanding or interpretation.

2.4.3 Taxonomy of Image Annotation Approaches

Image annotation was performed from multiple points of view which gives us such a large number of methods. Recently, an interesting review of Automatic Image Annotation (AIA) methods is proposed by (Cheng et al., 2018) that considers 138 papers published in well known journals and conferences. They divide the set of proposed techniques into 5 categories namely the Generative model, the Neighbrest Neighbors

model, the Discriminative model, the Tag-Completion model and the Deep Learning based models. This taxonomy is reported in Figure 2.9.

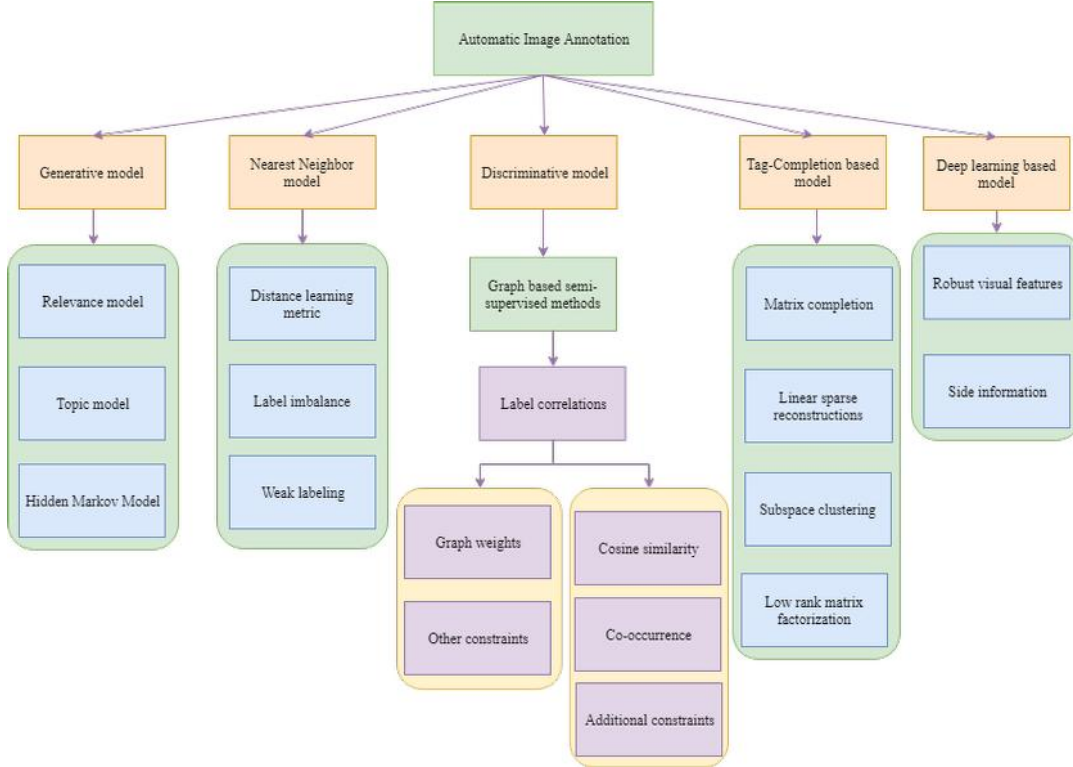


FIGURE 2.9: Taxonomy of Automatic Image Annotation techniques (Cheng et al., 2018)

2.5 Overview of Classical Remote Sensing Image Annotation Approaches

AIA was performed by several techniques and we present hereafter a comprehensive set of approaches. We present Generative (Subsection 2.5.1), Discriminative, NN based, Graph based and DL based models.

2.5.1 Generative Models

The main idea behind generative models is the maximization of the likelihood between image features and labels. For an unlabeled image, generative based models compute the probability of a label by estimating of the joint probability between image features and words from the training dataset. The generative models used for AIA are principally the relevance model, topic model, and Markov Random Field (MRF) model that can be summarized in Figure 2.10.

Relevance based models are performed in 3 main steps. They start by the definition of the joint distributions of image features and labels. Generally, Gaussian distributions are used for the visual features of images and multinomial distributions for labels. The second step consists in computing the posterior probability of each label for an unlabelled image. The last step is the annotation of a test image performed by choosing the label having the highest probability value.

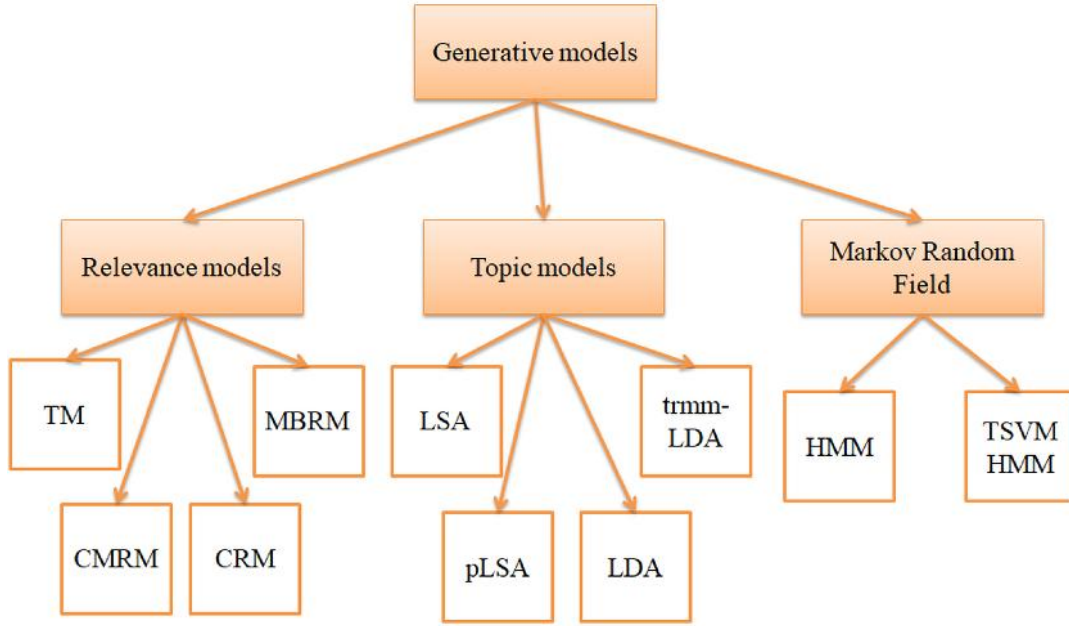


FIGURE 2.10: A summary of generative based models for AIA (Cheng et al., 2018)

Under the relevance based models we find Translation Model (TM) (Duygulu et al., 2002), Cross Mmedia Relevance Model (CMRM) (Jeon et al., 2003), Continuous space Relevance Model (CRM) (Lavrenko et al., 2004) and Multiple Bernoulli Relevance Models (MBRM) (Feng et al., 2004).

TM aims to create a one-to-one matching between a region and a word. The regions are clustered from the training set of images and represented by the index of the closest centroid of the cluster (blob or region). Then, each blob is associated with a word in the vocabulary by maximizing the joint probability using the expectation maximization algorithm. This process is similar to the process of learning a lexicon. However, this is computationally expensive and time consuming.

CMRM, however, uses blobs generated from the features of the image. It measures the joint distribution between the keywords and the whole image instead of the specific blobs as used in TM. This helps avoiding the errors that are caused by the vocabulary of blobs. In CMRM, the image I is represented by a collection of blobs $I = \{b_1, b_2, \dots, b_n\}$. Then, the conditional probability P of an image I belonging to a class w is computed using Equation 2.2.

$$P(w|I) = P(w|b_1, b_2, \dots, b_n) \quad (2.2)$$

The joint probability between a word w and the set of blobs $\{b_1, b_2, \dots, b_n\}$ is performed using a training set T of annotated images. For a given image j in the training set T , the joint probability is computed using Equation 2.3:

$$P(w, b_1, b_2, \dots, b_m) = \sum_{j \in T} P(j) P(w, b_1, b_2, \dots, b_m | j) \quad (2.3)$$

For a given image j , the prior probability $P(j)$ is constant for the training set T . If we assume that the words w and the blobs $\{b_1, b_2, \dots, b_m\}$ are independent, then a blob model and a word model can be built for every image j in the training set T . So the Equation 2.3 can be updated as Equation 2.4:

$$P(w, b_1, b_2, \dots, b_m) = \sum_{j \in T} P(j) P(w|j) \prod_{i=1}^m P(b_i|j) \quad (2.4)$$

where:

$$P(w|j) = (1 - \alpha_j) \frac{\#(w, j)}{\#j} + \alpha_j \frac{\#(w, T)}{\#T} \quad (2.5)$$

$$P(b_i|j) = (1 - \beta_j) \frac{\#(b_i, j)}{\#j} + \beta_j \frac{\#(b_i, T)}{\#T} \quad (2.6)$$

$\#(w, j)$ denotes the frequency of the word w in the image j and $\#(w, T)$ denotes the frequency of w in the entire training set T . Similarly, $\#(b_i, j)$ and $\#(b_i, T)$ represent the frequency of the blob b_j in the image j and the training set T respectively. $\#j$ is the number of all words and blobs present in the image j and $\#T$ is the size of the training set. Finally, α_j and β_j are parameters to be adjusted.

CRM, unlike TM and CMRM, uses continuous values of feature vectors of the image in order to avoid losing information due to the quantification phase. Also, in order to surpass the clustering errors of blobs used in CMRM, CRM model uses regions.

In CRM, we assume that an image is composed of n regions $\{r_1, r_2, \dots, r_n\}$. Each region r_i is composed of a set of pixels. In order to map image regions into real-word vectors, a function ϕ is used:

$$\phi(r) : R \leftarrow IR^k, r \leftarrow \phi(r)$$

The value $\phi(r)$ represents the feature vectors of the image region r . The joint distribution of the set of regions $\{r_1, r_2, \dots, r_n\}$ and the set of words $\{w_1, w_2, \dots, w_n\}$ is measured using Equation 2.7:

$$P(r_A, w_B) = \sum_{j \in T} P_T(j) \prod_{t=1}^q P_v(w_t|j) \prod_{i=1}^n \int_{g_i \in IR^k} P_R(r_i|g_i) P_\phi(g_i|j) dg_i \quad (2.7)$$

Where: $P_R(r_i|g_i)$ is the global probability distribution that maps the vectors g to image regions. The feature vectors of image regions are supposed to have a Gaussian distribution. And, the words are supposed to follow the multinomial distribution. Therefore, we have:

$$P_R(r|g) = \begin{cases} \frac{1}{N_g} & \text{if } \phi(r) = g \\ 0 & \text{Else} \end{cases}$$

$$P_\phi(g|j) = \frac{1}{N} \sum_{i=1}^n \frac{1}{\sqrt{2^k \pi^k} |\Sigma|} \exp \left\{ (g - \phi(r_i)) \Sigma^{-1} (g - \phi(r_i)) \right\} \quad (2.8)$$

$$P_w(w|j) = \frac{\mu P_w + N_{w,j}}{\mu + \sum_{w'} N_{w',j}} \quad (2.9)$$

Where: Σ is the covariance matrix.

MBRM (Lavrenko et al., 2004) uses multiple Bernoulli models instead of the multinomial distribution in order to model word distribution. The main idea is to emphasize the word, thus its presence or absence is meaningful for image annotation rather than its frequency. The conditional probability of a word w is given by Equation 2.10:

$$P_w(w|j) = \frac{\mu \delta_{w,j} + N_w}{\mu + N} \quad (2.10)$$

MBRM model is depicted by Figure 2.11 where $\{g_1, \dots, g_n\}$ represent the feature vectors, $\{r_1, \dots, r_n\}$ correspond to image regions $\{w_1, \dots, w_m\}$ to words.

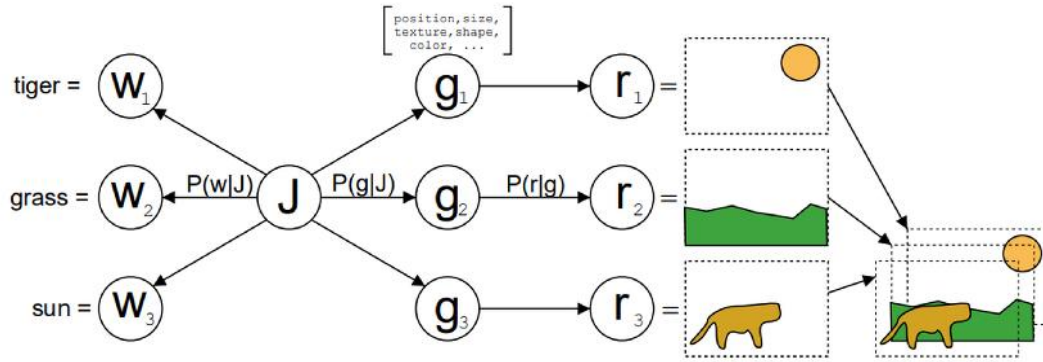


FIGURE 2.11: Illustration of MBRM model (Lavrenko et al., 2004)

Topic models are used for AIA from a different perspective. A Topic model considers the image as a mix of topics. Each topic is characterized by a probability distribution over the annotation words and over the image features. It outperforms relevance models because it considers visual feature co-occurrence and semantic content. Thus, unlike relevance models, words and visual features are not considered equally important. Among the topic based models, we can find LSA, LDA, pLSA and tr-mm-LDA.

The pLSA (probabilistic Latent Semantic Analysis) model (Monay and Gatica-Perez, 2004) supposes that a set of co-occurrence words is associated with a latent topic z which is a concept that is characterized by a set of related words. It was, first, proposed for text documents and it assumes that for each element w_i in the document d_j there exists a latent topic z that generates w_i . Thus the joint probability of the element w_i and the document d_j is calculated using Equation 2.11. It can be illustrated by Figure 2.12.

$$P(w_i, d_j) = P(d_j) \sum_k P(z_k | d_j) P(w_i | z_k) \quad (2.11)$$

Although pLSA is considered as a generative model, it does not allow to generate new documents.

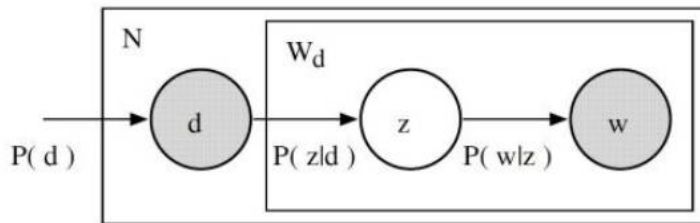


FIGURE 2.12: Illustration of PLSA model

Always under the generative based models category, the Markov Random Field (e.g. HMM, TSVM-HMM, SHMM, etc) is another interesting alternative that considered for AIA. In the Hidden Markov Model (HMM), the image is represented by

a set of feature vectors and is assumed to be generated by a HMM (Ghoshal et al., 2005; Yu and Ip, 2006).

Let T be the training set containing n images with their annotations $T = \{I_1, I_2, \dots, I_n\}$. An image is composed of a set of regions $I_i = \{r_1, r_2, \dots, r_m\}$ and is represented by a set of keywords $W = \{w_1, w_2, \dots, w_K\}$. The vocabulary V is constructed by collecting all the keywords of all images in the dataset T . The HMM is used in order to model the image-features and caption-text jointly as the realization of a generative process in Equation 2.12:

$$f(r_1^M, W_{w0}) = \sum w_1^M \in W \prod_{t=1}^M f(r_t w_t) p(w_t) \quad (2.12)$$

There are multiple generative based models proposed for AIA with interesting results. However, generative based AIA models cannot guarantee the semantic relevance of predicted tags since it is based on the estimation of generative likelihood of keywords and image features. Also, they cannot handle the relations between the features and the labels. In addition to the high computation cost implied by parameters configuration and the use of EM algorithm.

2.5.2 Discriminative Models

In a discriminative model, the AIA is considered as a multi label classification problem. The first step is to learn an independent binary classifier for each label and then to use the binary classifiers to predict labels for unlabeled images. Several methods were proposed that address the annotation as a multi-label classification task. We can cite the supervised multiclass labeling (SML) model (Carneiro et al., 2007) which deals with the semantic annotation and retrieval of images as classification problems. In this context, they define a class as "a group of images in the database that share a common semantic label". The flowchart of the SML model is shown in Figure 2.13. The images are represented by bags of features which are used by a GMM (Gaussian Mixture Model) in order to learn the mixtures of the classes. The GMMs that are learned from all the images belonging to the same class ('mountain' in the flowchart) are then used by a density estimate of the class. Thus, they learn a per class distributions for each label.

The SVM-DMBRM (Murthy et al., 2014) is an hybrid model that improved the annotation results by combining the advantages of generative and discriminative models for image annotation. The SVM model is considered as the discriminative model and is used in order to address the poor labeling problem (occurs when the images are not annotated with all the relevant keywords). While the Discrete Multiple Bernoulli Relevance Model (DMBRM) is considered as the generative model and is utilized in order to solve the class imbalance problem. They obtain good annotation results on the *Corel-5k*, *ESP-Game* and *IAPRTC-12* datasets as can be illustrated by Figure 2.14. However, the annotations are related to the whole image and not to regions and represents mainly the objects present in the image.

The main limit of multi-label classification approaches is that they build a binary classifier for each class. Consequently, they are not easily scalable to large number of classes which is generally the case of image annotation task. An interesting model exploits the hierarchy of the annotation by constructing a Predictive Clustering Tree (PCT) instead of using binary classifiers. the PCT is used to predict all the annotations of a given image "simultaneously". The main advantage of the HMC model is that it use only one tree to predict the classes of an image. This model is illustrated by Figure 2.15.

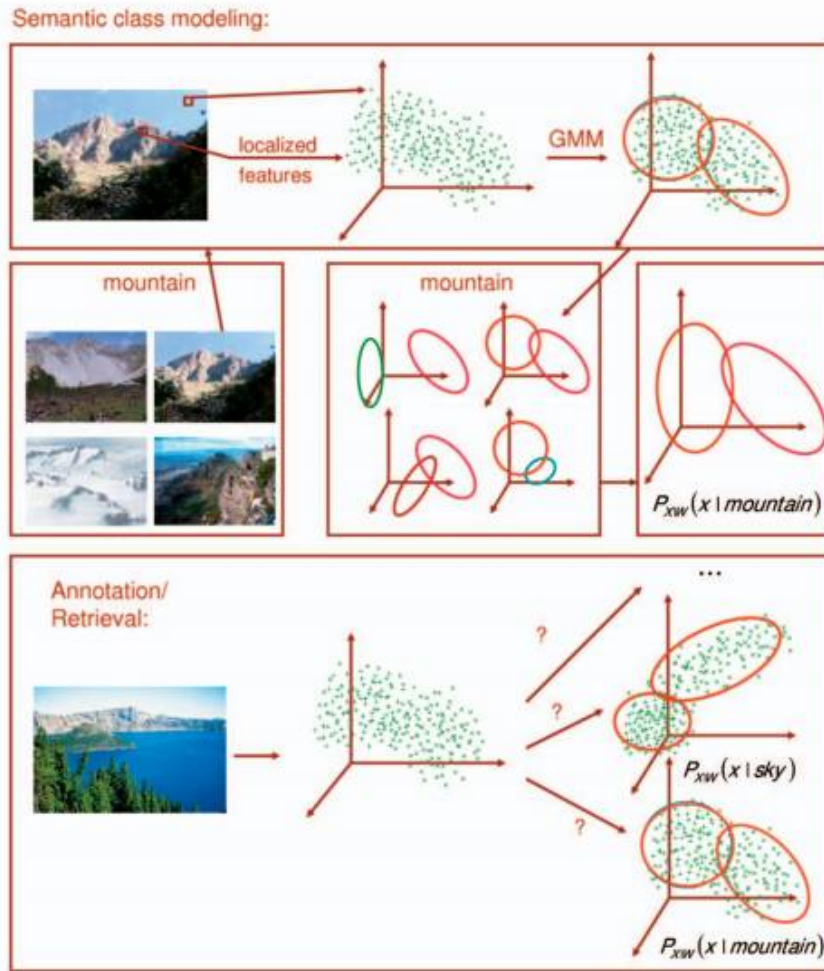


FIGURE 2.13: Illustration of Supervised Multiclass Labeling (SML) model (Carneiro et al., 2007)

In Figure 2.15, the table on the left summarizes the set of medical images with their visual features and their annotations or labels. In the right of the image, hierarchies of medical annotations are presented. The PCT can be seen as a generalization of decision tree. An example of the PCT is shown in Figure 2.16.

Generally, the size of the labeled images is lower than the number of images in the dataset. Consequently, image annotation was considered as a semi-supervised learning problem. The later is defined as a label propagation from labeled images to unlabeled ones. Therefore, the correlation between the set of keywords is considered which can enhance the annotation results. Recently, several methods used graphs in order to model the semi-supervised learning problem for AIA. The graph nodes represent all the images of the dataset and the edges represent the similarity or neighborhood relations. Then, label correlations can be naturally incorporated in the graph.

Discriminative based models are interesting and contributed to the advancement of the research on image annotation. They compute a separate classifier for each keyword and use these binary classifiers in order to decide whether an image belongs to a given class. However, these models are not scalable with respect to the number of keywords or labels since a classifier per label is not a practical solution.

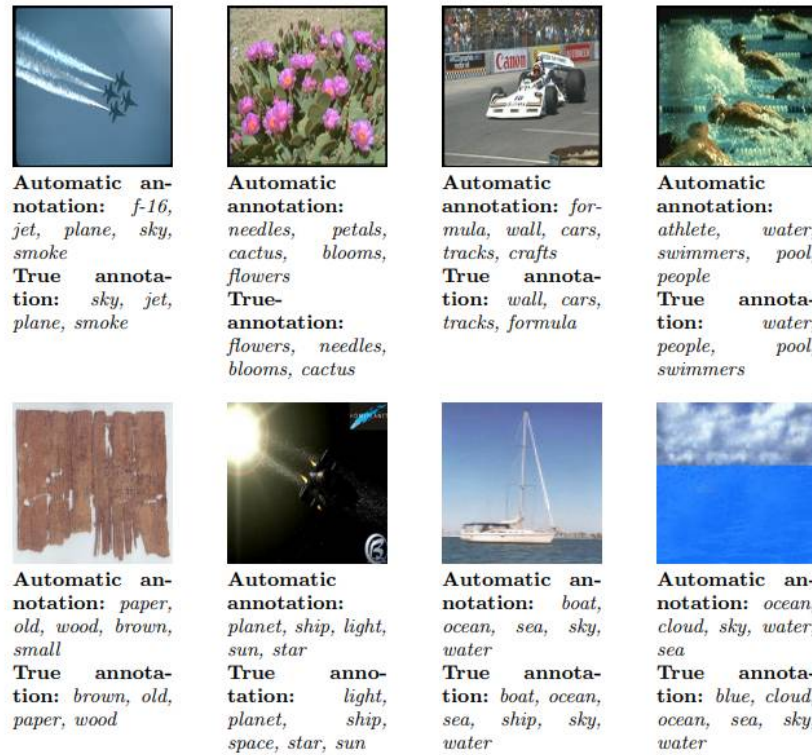


FIGURE 2.14: Annotation generated by the SVM-MBRM model (Murthy et al., 2014)

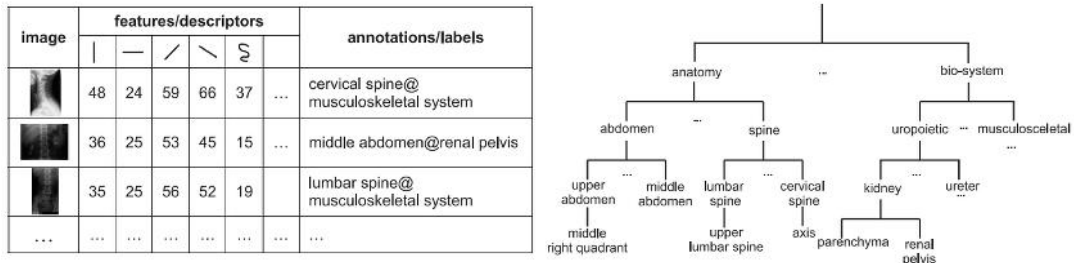


FIGURE 2.15: Hierarchical Multi-label Classification for medical image annotation (Dimitrovski et al., 2011)

2.5.3 Graph based Models

Graphs are widely used in several fields because of their power in modeling the relationships between objects. They are frequently used in the modeling of social networks, chemical compounds, medical image analysis, image segmentation, image classification and annotation, etc. They allow the representation of binary relationships between nodes (that can be persons in social networks, regions or images) such as adjacency and similarity or distance. The Region Adjacency Graph is one of the famous graphs and is used in many applications.

A RAG is formally defined as a graph $G = (V, E)$ where $V = \{v_1, v_2, \dots, v_n\}$ represents the set of vertices which correspond to image regions. $E = \{e_1, e_2, \dots, e_m\}$ is the set of edges. An edge $e = (v_k, v_l)$ links two regions v_k and v_l only if they are adjacent based on the 4 or 8 connectivity.

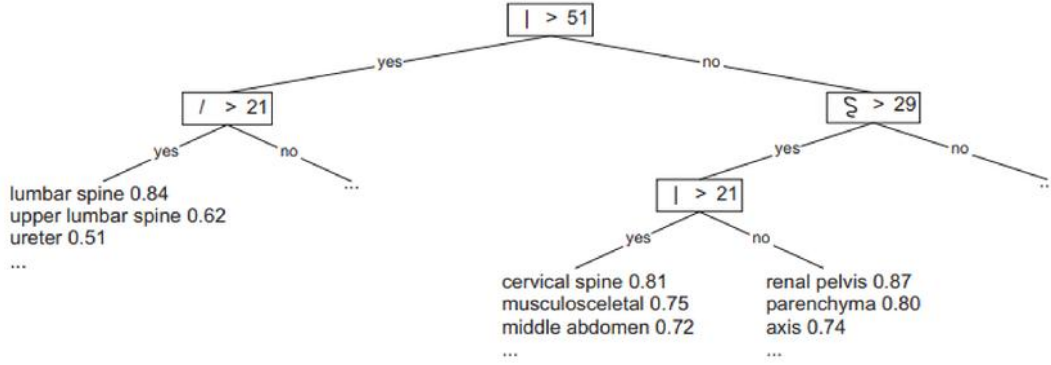


FIGURE 2.16: Predictive Clustering Tree of the Hierarchical Multi-label Classification for medical image annotation (Dimitrovski et al., 2011)

AIA methods used graphs in order to describe the objects in the image and to consider their spatial relations. The GCap (Graph-based automatic image captioning) model (Pan et al., 2004) is an illustration of graph based methods used to refine the image annotation. The proposed graph $G = (I \cup R \cup W, E)$ corresponds to all images in the training dataset T . The GCap model is illustrated in Figure 2.17. They use three types of nodes: image nodes I , region nodes R described by visual feature vectors and word nodes W representing a predefined vocabulary. The edges E between images are added considering the co-occurrence relation. The regions are linked based on similarity relation in the feature space using the Euclidean distance. These relations result from a learning process that exploits the dataset T . The main limitation of the GCap model is that it is extremely dependent on the training dataset T and the vocabulary W . Also the model does not give a direct association between regions and words, thus we cannot match a word with its corresponding region in the image.

The Multi-Criterion Graph (MCGraph) (Hueting et al., 2014) $G = (I \cup W, E)$ is proposed in order to represent the relationships between images I and words W of a training dataset T as shown in Figure 2.18. A feature vector f_i and a set of keywords $W_i = \{w_1, w_2, \dots, w_n\}$ correspond to each image $i \in T$. The proposed graph contains two types of nodes: image nodes I , and word nodes W taken from a predefined vocabulary. They define three types of edges. The first type relates pairs of images i and j and has weights that reflect the similarity measure $sim_f(i, j)$ between their corresponding feature vectors f_i and f_j . The second type of edges links images and words. Weights $sim(i, w_j)$ that are provided by human-experts, correspond to the perceived similarity of image i to word w_j . The last type of edges relates pairs of words w_i and w_j and the weights represent the semantic similarity measure $sim(w_i, w_j)$ between words w_i and w_j . We notice that every image node in the proposed graph is only linked to its k -nearest neighbors in the image feature space.

The graphs used for the annotation allow the representation of pairwise relationships only (adjacency, similarity). However, objects inside an image can share many types of relations. Thus, hypergraph based methods were proposed. For RS images, hypergraphs were proposed for the classification (Gao and Chua, 2013). Each pixel in the image corresponds to a vertex in the hypergraph and the hyperedge is defined based on the spatial neighbors of each pixel. In order to classify the hyperspectral image, authors used semi supervised learning on the constructed hypergraph as

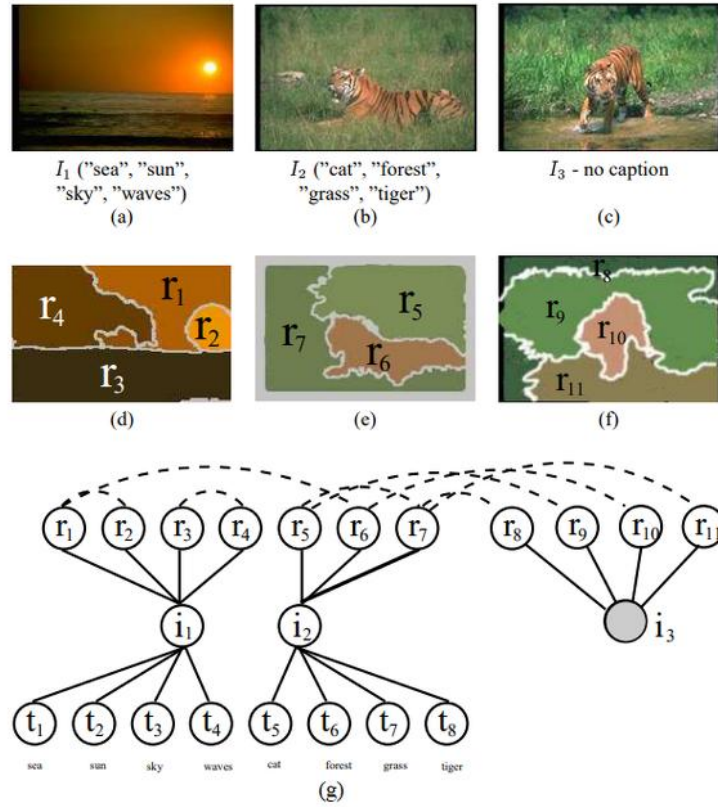


FIGURE 2.17: The GCap method presented by (Pan et al., 2004) for AIA

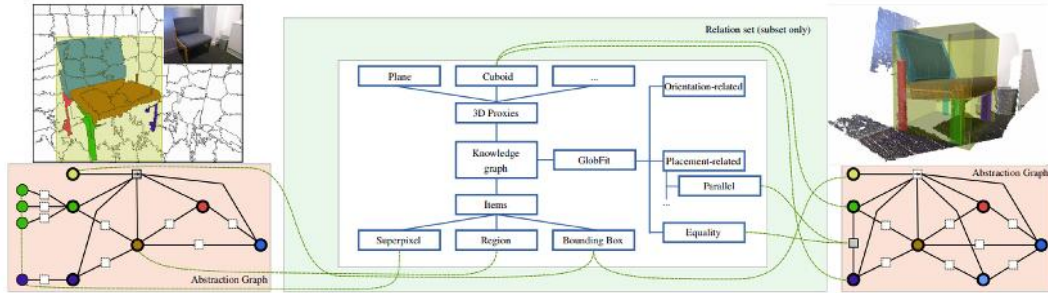


FIGURE 2.18: TMulti-Criterion Graph (MCGraph) (Hueting et al., 2014)

shown in Figure 2.19

Authors in (Wen et al., 2012) utilized hypergraphs for hyperspectral image classification in order to explore feature-based pixels relationship. They define the correlation between pixels based on the neighborhood clustering method, in which each pixel is connected to its neighbors. Then, they compute the relevance scores of pixels with a hypergraph learning procedure. Finally, hyperspectral image classification is performed as shown in Figure 2.20 using the computed relevance. scores.

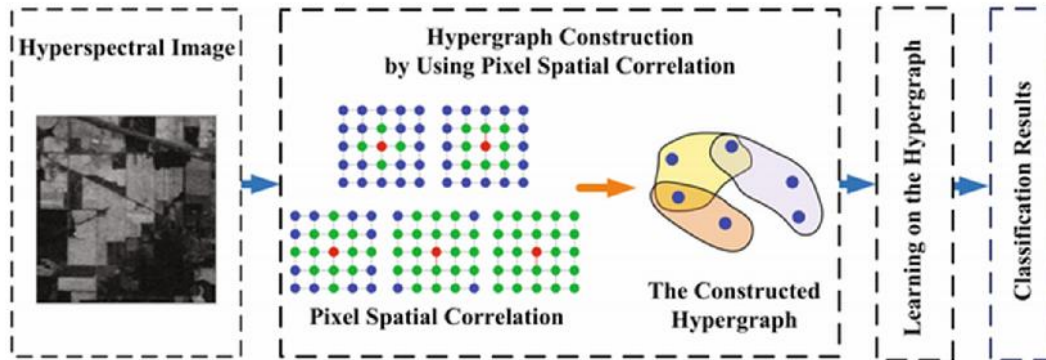


FIGURE 2.19: Flowchart of hyperspectral image classification method based on pixel spatial correlation (Gao and Chua, 2013)

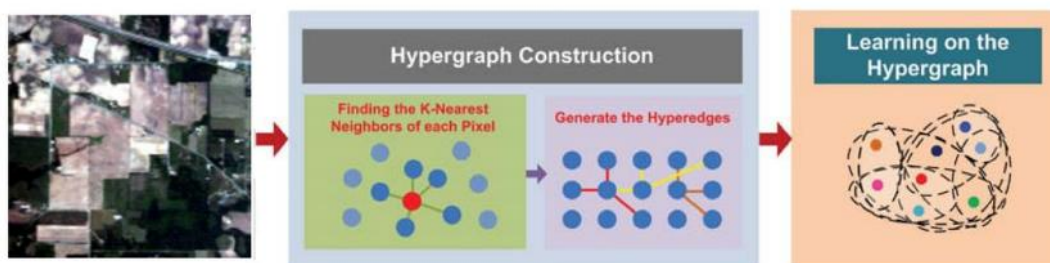


FIGURE 2.20: The flowchart of hyperspectral image classification (Wen et al., 2012)

2.5.4 Nearest Neighbor based Models

Nearest Neighbor (NN) based models are based on the assumption that similar images having similar visual features share similar labels. Thus, in order to annotate a new unlabeled image, NN based models start by searching for the set of similar annotated images in the feature space and then use their keywords in order to annotate the query image. The NN based methods are implemented in two steps. The first step consists of extracting the representative features of the image. The second step is the computation of a distance metric for the already extracted features. The Joint Equal Contribution (Makadia et al., 2008) is a classical NN model. It uses global low-level features and a basic distance to find the NN for a test image. The keywords are assigned based on a greedy label transfer approach that selects the keywords using the co-occurrence and the frequency.

NN based models face the distance metric learning, the label imbalance and the weak labeling challenges. The Distance Metric Learning (DML) aims to learn a distance measure in order to decide whether the two given images are similar or not (to be linked or not). Usually, DML methods require a learning task in the form of class labels for image classification (Bar-Hillel et al., 2005; Weinberger and Saul, 2009) or pairwise constraints for the clustering and retrieval tasks (Goldberger et al., 2005; Xing et al., 2003; Hoi et al., 2006). Using DML for automatic image annotation can enhance the search of NN images and consequently improve the annotation results.

The class imbalance is a real problem. It can be defined as the variance in the number of images corresponding to a given label in the training dataset. Generally, the class-imbalance causes poor labeling results for NN based methods because they

make use of adjacent images to annotate a new image. Thus, if a given label has a high frequency to describe its neighbors then its probability to describe the new image will also be high. Weak labeling is due to the manual annotation which can be incomplete. Thus, we can have images in the training set that are not annotated with all the relevant keywords. Moreover, obvious keywords might be omitted by the annotators. Also, we may have images tagged with irrelevant keywords.

Among the NN based models, we can cite the TagProp model (Guillaumin et al., 2009) which uses a weighted NN and a metric learning in a discriminative framework. It transfers labels based on a weighted combination of their presence and their absence in the tags of the neighboring images. In order to deal with the class-imbalance challenge, TagProp uses a word-specific discriminant models which allow to enhance the probability of rare keywords and decrease the probability of the frequent keywords.

2PKNN model (Verma and Jawahar, 2012) is considered as a classic solution to class-imbalance and weak labeling issues. Thus, for each label, 2PKNN finds the set of all the related semantic neighbors by considering the k nearest images in the vocabulary. Therefore, each label appears at least k times in the training set. Let w_i be a label, $T_i \in T$ be a subset of the training dataset where T_i contains the images that are tagged with w_i . For an untagged image J , the k similar images to J are extracted to constitute $T_{j,i} \in T$ with $T_{j,i} = \{T_{j,1}, T_{j,2}, \dots, T_{j,m}\}$ which represents all the semantic labels of the image j . The 2PKNN also proposes a metric learning method to learn the appropriate weights combination of features in the distance space. In order to annotate a new image J , the 2PKNN model computes the posterior probability of a label to be given to J using Equation 2.13:

$$P(w_k|J) = \frac{P(J|w_i)P(w_i)}{P(J)} \quad (2.13)$$

With $P(w_i)$ is the prior probability of the label w_i . The conditional probability of the untagged image J given a label $w_k \in W$ is expressed by Equation 2.14:

$$P(J|w_k) = \sum \theta_{J,I_i} P(w_k|I_i) = \sum \exp(-D(J, I_i)) \delta(w_k \in w_i) \quad (2.14)$$

Where $P(w_k|I_i) = \delta(w_k \in W_i)$ reflects the presence/absence of the label w_k in the set of labels of image I_i . The distance between an image I and J is expressed using Equation 2.15:

$$D(I, J) = \sum_{i=1}^n w_i \sum_{j=1}^{N_i} d_{IJ}^i(j) \quad (2.15)$$

The 2PKNN model (illustrated by Figure 2.22) achieved good annotation results. It is inspired the work of (Verma and Jawahar, 2012) which also considers that a label should appear at least k times in the training set to overcome the class-imbalance issue. (Verma and Jawahar, 2012) use also a weighted NN method in order to assign importance to labels using the similarity between images. Then, it computes the score for each label to each image. Unlike other methods (Verma and Jawahar, 2012; Guillaumin et al., 2009; Makadia et al., 2008), they use a variable number of keywords for unlabeled images.

2.5.5 Tag-Completion based Models

Unlike other image annotation methods, tag-completion based methods do not assume that images in the training sets are annotated with all the relevant keywords.

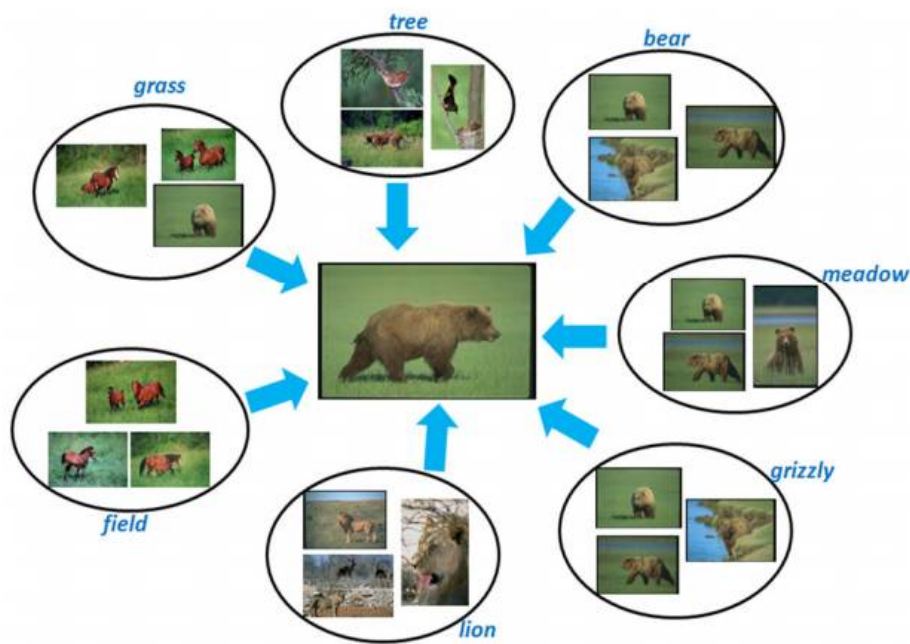


FIGURE 2.21: Illustration of the 2 Pass K Nearest Neighbors model (Verma and Jawahar, 2012)

Thus, the main purpose of tag-completion based methods is to automatically find the missing keywords and to correct the noisy ones without training processes.

Tag completion based methods have different frame structures and focus on the consistency of image content and the relations between tags. The images of the dataset are represented as a matrix where each row corresponds to an image and each column to a tag or a label. Tag-completion method updates the initial matrix by identifying the correct associations of images and tags. These methods can be divided into several classes such as subspace cluster-based methods and matrix completion-based methods, etc.

TMC model proposed by (Wu et al., 2013) is matrix completion-based in which the relations between images and tags are encoded by a matrix. Each entry corresponds to the relevance of a tag to an image. In addition to the initial tag matrix, they consider a correlation matrix that represents the correlation between tags and a visual features matrix that encodes the images with d dimensional features vectors. Based on these matrices, the TMC model searches for a output complete tag matrix by minimizing the discrepancy between the visual features correlation and the semantic tags correlation. The objective of the TMC model is to preserve the correlations in the feature space and between labels.

Lin et al. (20) used image-specific and tag-specific Linear Sparse Reconstructions (LSR) in a tag-completion method. The proposed LSR model uses a convex optimization problem in order to model the image-specific and tag-specific reconstructions while respecting the sparsity constraints. The image-specific reconstruction is computed based on the visual and semantic similarities between images. However, the co-occurrence between tags are used to compute the tag-specific reconstruction. Then, LSR model normalizes and merges the two linear reconstructions using a weighted linear combination in Equation 2.16.

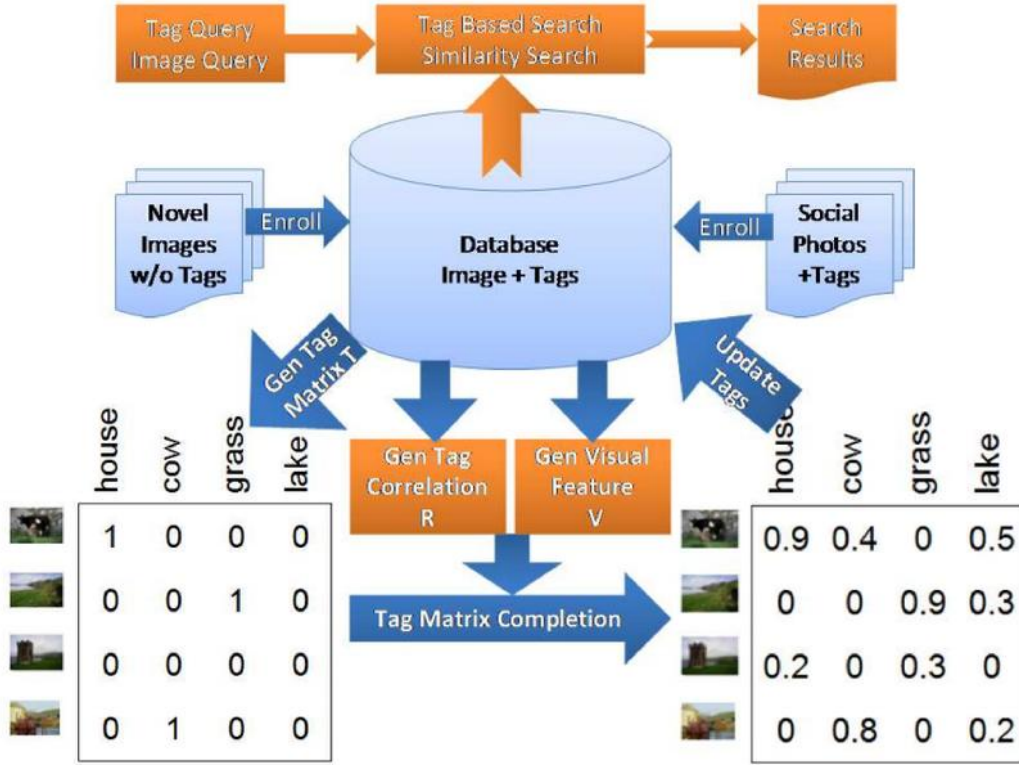


FIGURE 2.22: Matrix completion-based TMC model (Wu et al., 2013) for AIA

$$\omega = \theta T + (1 - \theta)R \quad (2.16)$$

Where ω is the final result, T and R are the normalized completion result of image-specific and tag-specific reconstructions respectively, and $\theta \in [0, 1]$ is a weighting parameter.

Tag completion based methods are largely studied and lots of methods were proposed for AIA. Tag completion methods do not require a training process and can complete missing and noisy labels. They are, also, efficient and independent to scale of the annotation. However, they transform the tag completion process into an optimization problem which is a complex process and time consuming. The optimization process does not guarantee a global optimal result.

2.5.6 Deep Learning based Models

Recently deep learning (DL) based methods are becoming largely used for several computer vision applications such as object identification, image classification, image annotation and retrieval. The main purpose of DL based methods is to skip the tedious step of feature engineering and extraction that should be efficiently performed by the rest of methods. Instead of extracting features, DL based methods propose to use deep neural architectures in order to perform the given task. In image classification, DL based approaches rely on big datasets composed of pairs of (image, class) to train the model to recognize the classes and to extract by itself relevant features to do that.

(Niu et al., 2018) proposed to use DL for the AIA task. They study two main challenges. First, they treat the problem of learning a set of features that should be suitable for the prediction of the visual concepts. The concepts cover object, scene and abstract levels. Thus, they propose to new multi-scale deep model in order to extract the visual features. As shown in Figure 2.23, the model includes four steps "visual feature learning", "textual feature learning", "multi-class classification", and "label quantity prediction".

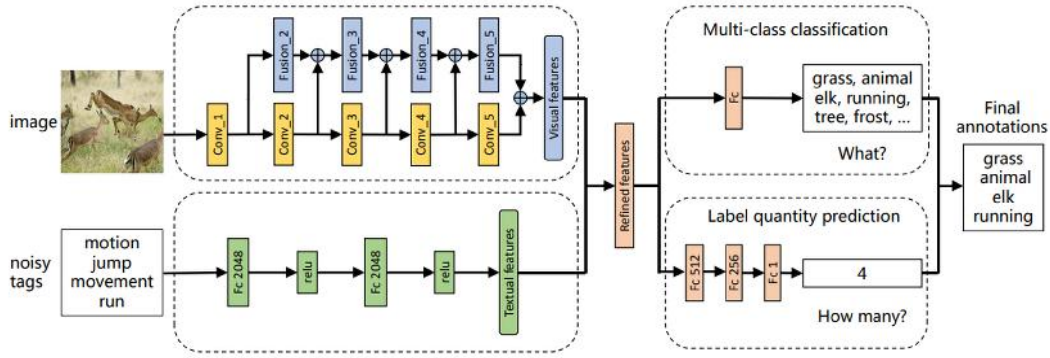


FIGURE 2.23: Illustration Multi-Modal Multi-Scale Deep Learning model for AIA (Niu et al., 2018)

The second challenge concerns the optimal number of class labels used for the annotation of images. Unlike traditional methods that pick the best k labels, (Niu et al., 2018) try to optimize the number of labels since it differs according to the image. This problem is addressed based on the component "label quantity prediction" which estimates the optimal number of labels for each image. They tested their model on NUS-WIDE (Chua et al., 2009) and MOSCOCO (Vinyals et al., 2017) datasets and they got interesting results. In figure 2.24, they illustrate examples of the produced annotations that exist also in the ground truth annotations in green color and reserve the blue color to correct label but that do not exist in the ground truth.

(Wang et al., 2016) proposed a new CNN-RNN framework to address the multi label classification problem as illustrated by Figure 2.25. The proposed framework is divided into two modules. The CNN module allows the extraction of the semantic representations from images while the RNN module allows to model the relations between images and labels and the dependencies of labels. The recurrent layers depicts the information about the previously predicted labels. They obtained interesting results on the NUS-WIDE and MS-COCO datasets.

2.6 Discussion

Despite the differences in application and formulation, all the proposed approaches for AIA basically attempt to learn the correlations between the features of the image and words based on the examples of annotated images in the dataset. Several methods were proposed to address the AIA problem.

The generative, discriminative and deep learning based methods are learning-based. The discriminative based approaches define the AIA problem as a multi label classification problem. While the generative based approaches resolve the annotation by computing the conditional probability between images and labels. On the

Test images	Ground-truth	Annotations	Test images	Ground-truth	Annotations	Test images	Ground-truth	Annotations
	clouds lake reflection Sky sun water	water lake reflection sky clouds sun		airport clouds military plane sky	clouds plane airport sky military		clouds person sky water	water sky clouds person
	buildings cityscape nighttime sky	buildings sky cityscape nighttime		animal tiger	animal tiger		person police	person police
	animal	grass horses animal sky		water	sky boats clouds sunset water		animal water	water animal coral fish
	grass	grass animal elk		sky	window sky vehicle		clouds sky	airport plane clouds sky vehicle

FIGURE 2.24: Illustration of results on NUS-WIDE obtained by Multi-Modal Multi-Scale Deep Learning model (Niu et al., 2018)

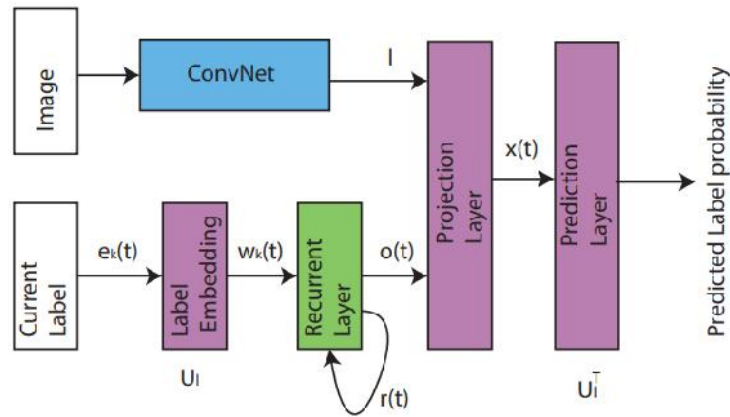


FIGURE 2.25: The architecture of the RNN-CNN model for multilabel classification (Niu et al., 2018)

other hand, deep learning based methods basically use the robust CNN features in order to resolve the annotation task. Nearest neighbor based methods for AIA are divided into two steps. They find the similar images to a query unlabeled image then use their keywords to annotate the image. In order to represent and exploit the spatial relations between images or objects, graphs are used for the image annotation task. The Region Adjacency Graph is largely used. All the above mentioned categories supposed that the annotations of the learning datasets are correct and complete. However, tag-completion based methods assume the inverse and aim to complete the missing tags and correct the noisy ones.

The majority of proposed methods cannot efficiently deal with the semantic gap problem leading to annotate images with irrelevant keywords. The major advantages and limits of the five categories are presented in Table 2.1. Some of them require large training datasets, a serious problem in RS. We have few available datasets

of images with their ground truths, so applying the learning based methods is not an easy task. Moreover, they do not consider semantic relations between keywords nor spatial distribution of regions in the image. So, the resulting annotation does not faithfully reflect the semantics of the image. One major drawback of generative models is that they maximize the generative data likelihood, which is not necessarily optimal for predictive purposes. Because the same low level feature can have completely different semantics, generative models may generate semantically irrelevant annotations. Furthermore, the multi-label classification approaches cannot be extended to large number of categories since a binary classifier should be built for each category (Cheng et al., 2018). However, we are more interested by graph based annotation methods. Generally graphs allow the consideration of spatial information as well as other interesting relations (such as semantic between labels or similarities between images)

Annotation methods	Advantages	Weaknesses
Generative based models	Conditional probabilistic distribution, Well-formed theory, Alternative number of labels	Require prior image segmentation, Expensive training and computation, Sensitive to noisy data, Parametric, Might not be optimal
Nearest Neighbor models	Conceptually simple, Non-parametric, Do not require prior image segmentation	Sensitive to small dataset, Distance metric learning, Sensitive to cluster result, Fixed number of labels
Discriminative based models	Multi-label, Graph framework usual, Computation-efficient	Sensitive to label-imbalance and classes relevance, Parametric
Tag completion based	Robust to noisy data, more scalable in dataset size, Non-train process, Non-parametric	Sparse matrix, Optimization problem, Various constraints on the relationship between text and visual components, Little guarantees that the annotations are optimal
Deep learning based methods	Deal with mass data, Learn very complicated relationships, Derive Robust features, No manual selection is required, Obtain sufficiently side information, Alternative number of labels	Local optimum, Vast training images, Training process cannot be controlled

TABLE 2.1: Advantages and limits of the five categories of AIA methods

2.7 Conclusion

Satellite images are important to monitor our environment and help reaching places that are not easily accessible by humans. They are used in several fields and applications. In this chapter, we draw the basic definitions related to the acquisition process, the resolutions and RS image types. Then, we define the image classification problem against the image annotation problem and we report the related works in the field of RS images. Therefore, we present a recent taxonomy of automatic image annotation methods and we go through a representative set of annotation methods. Motivated by the limits faced by the presented methods, we present in the Chapter 3 our approach to address the classification of hyperspectral images.

Chapter 3

Classification of Spatially Refined Hyperspectral Image based on Radiometric Indices

3.1 Introduction

Hyperspectral images (HSI) contain a large number of bands resulting in a narrow spectral resolution. Consequently, they allow a better identification of objects than multispectral images (MSI). However, the HSI classification is challenging due to the limited number of labelled training samples as well as the redundant information contained in the large number of bands. Many dimensionality reduction (DR) techniques were proposed in order to reduce the dimension of HSI while trying to keep most of the relevant information. DR can be performed either using feature extraction or feature selection. As a result, the number of bands is reduced (for feature selection) or a new set of features with lower dimension is built. In both cases, the new representation of HSI is used to perform the classification.

Eventhough the ultimate purpose of the thesis is to efficiently annotate HSI and MSI, we focus in this chapter on the classification because we are going to use the same features for classification and annotation. Also, since the annotation process involves classification and annotation computation having good performances for classification will result in a good performance for annotation. More specifically, we check, in this chapter, the discriminative power of the radiometric indices (RI) for HSI classification. Then, based on the classification results, we use the discriminative RI for the enrichment of the core ontology *SatelitesSceneOntology-V3* and for the annotation of RS images.

3.2 The proposed classification approach

Motivated by the good accuracy of vegetation indices for the monitoring of vegetation, we proceed to test a combination of RI for the classification of HSI. RI are consistent and can overcome the illumination problem and the imaging conditions. Using multispectral and hyperspectral sensors, many indices can be computed by the combination of different spectral bands in order to compute features.

Generally, RI are used for vegetation monitoring based on the NDVI or EVI indices. However, there was no combination of indices. We believe that even a negative value of an indice is a relevant information and can enhance the classification accuracy. Figure 3.1 shows the flowchart of our approach entitled 'Low Dimensional Feature Vector' (LDFV).

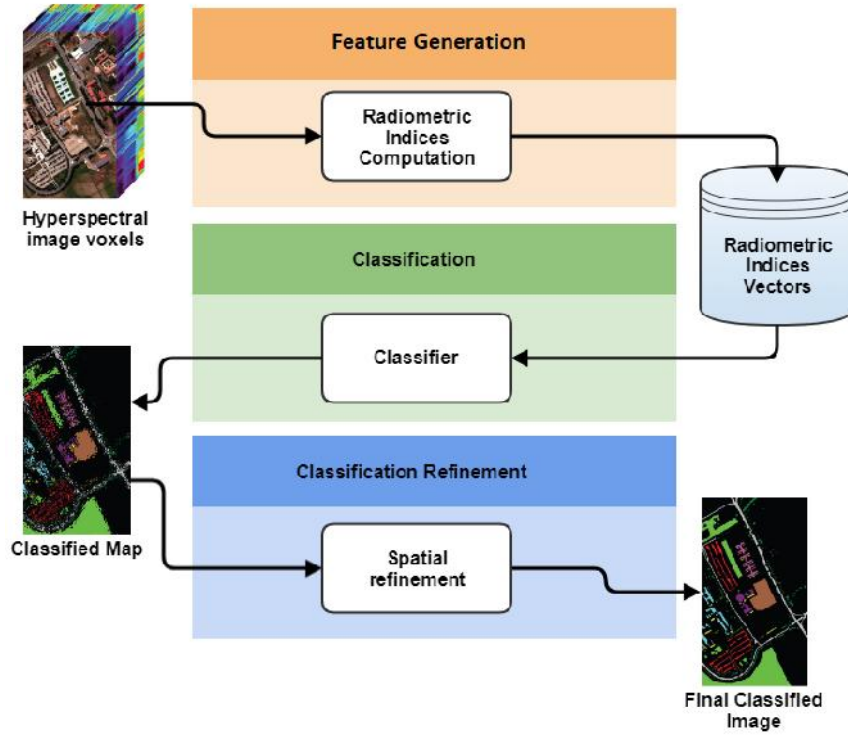


FIGURE 3.1: The flowchart of our approach for HSI classification

In order to perform the classification of HSI, we start by the computation of features which are a set of RI for each pixel in the image. Then, we perform the classification of the extracted features using classifiers such as SVM or Bagged Trees. Finally, a post-processing step of the resulting classification is performed based on the spatial neighborhood information. These steps are illustrated in Algorithm 1.

Algorithm 1 Main algorithm: LDFV

Input HSI : Hyperspectral Image

Output RML : Refined Matrix of Labels

Begin

$RI \leftarrow \text{Feature_Generation}(HSI) \langle O(N) \rangle$

$ML \leftarrow \text{Classification}(RI) \langle O(M), \text{ where } M \text{ is the number of samples} \rangle$

$RML \leftarrow \text{Spatial_Refinement}(ML) \langle O(N) \rangle$

End

We detail each step in the next subsections.

3.2.1 Radiometric indices

A radiometric indice (RI) can be defined as a quantitative measure computed by the addition, the division, and/or the multiplication of spectral values of several bands. A RI is a single value that reflects the characteristics of a given class such as vegetation, water, etc. For example, a vegetation index (NDVI, EVI, etc) can measure the biomass or vegetation health and thus allows to automatically monitor large vegetation areas.

Indices	Description	Formula
IR	Redness Index (Pouget et al., 1991)	$\frac{R^2}{B \cdot G^3}$
IC	Color Index (Pouget et al., 1991)	$\frac{R-G}{R+G}$
BI	Bare Soil Index (BI) (Jamalabad, 2004)	$\frac{(SWIR+R)-(NIR+B)}{SWIR+R+NIR+B}$
IB	Brilliance Index (Zarco-Tejada and Ustin, 2001)	$\sqrt{\frac{R^2+G^2}{2}}$

TABLE 3.1: Soil indices

Indices	Description	Formula
NDBI	Normalized Difference Built Up Index (Abdellaoui and Rougab, 1997)	$\frac{SWIR-NIR}{SWIR+NIR}$
ISU	Index Surfaces Built (Fauvel et al., 2009)	$a - \frac{b \cdot R}{NIR}$ where $a = 100, b = 25$

TABLE 3.2: Built-up indices

According to the OTB project, there are many RI. They can be divided into 1) soil indices (Table 3.1), 2) built-up indices (Table 3.2), 3) water indices (Table 3.3) and 4) vegetation indices (Table 3.4). The RI are also classified differently according to their consideration of soil, atmosphere and soil temperature effects. Thus, five classes of indices are distinguished: simple indices (DVI, RVI, NDVI), indices that consider the atmosphere (MSI and NDWI), indices considering soil influences (PVI, SAVI, TSAVI and MSAVI), indices considering both soil and atmosphere (ARVI, GEMI and EVI) and indices considering the soil temperature (TDVI).

Among all the available RI shown in Tables 3.1-3.4, we use a subset of 15 indices that we believe are discriminative, allow us to efficiently identify the classes and can almost always be computed for most of the datasets. In fact, few datasets contain more than the following four bands: Infrared, Red, Green and Blue. The set of indices that we use are: *NDVI, RVI, SAVI, MSAVI, MSAVI2, DVI, IPVI, GNDVI, IC, BI, NDTI, ISU, NDWI, CVI* and *IR*. This set can however be updated according to the image type (the available bands we have and the aim of the study).

The computation of RI (at the pixel level) is performed using Algorithm 2.

The NDVI (Normalized Difference Vegetation Index) (Rouse Jr et al., 1973) is one of the most used indices to detect vegetation areas in multispectral RS images. It is computed based on the Red and Near Infrared spectral bands and its values range from -1 to 1 where 1 corresponds to high canopies vegetation areas and -1 to non vegetation areas.

Indices	Description	Formula
SRWI	Simple Ratio Water Index (Gao, 1996)	$\frac{R}{NIR}$
NDWI	Normalized Difference Water Index (Xu, 2006)	$\frac{G-NIR}{G+NIR}$
MNDWI	Modified Normalized Difference Water Index (Lacaux et al., 2007)	$\frac{G-MIR}{G+MIR}$
NDPI	Normalized Difference Pond Index (Zha et al., 2003)	$\frac{MIR-G}{MIR+G}$
NDTI	Normalized Difference Turbidity Index (Zha et al., 2003)	$\frac{R-G}{R+G}$

TABLE 3.3: Water indices

Indice	Description	Formula
NDVI	Normalized Difference Vegetation Index (Rouse Jr et al., 1973)	$\frac{NIR-R}{NIR+R}$
RVI	Ratio Vegetation Index (Pearson and Miller, 1972)	$\frac{NIR}{Red}$
PVI	Perpendicular Vegetation Index (Richardson and Wiegand, 1977; Wiegand et al., 1991)	$\frac{1}{\sqrt{a^2+1}}(NIR - aR - b)$ a : slope of the soil line in the space (R, NIR) and b : ordinate at the origin of the line of soil
SAVI	Soil Adjusted Vegetation Index (Huete, 1988)	$\frac{NIR-R}{NIR+R+L} \times (1+L)$ where $L = 0.5$
DVI	Difference Vegetation Index	$NIR - R$
TSAVI	Transformed Soil Adjusted Vegetation Index (Baret et al., 1989), (Baret and Guyot, 1991)	$\frac{a(NIR-aR-b)}{R+NIR-ab+0.08(1+a^2)}$ a : slope of the soil line in the space (R, NIR) and b : ordinate at the origin of the line of soil
MSAVI	Modified Soil Adjusted Vegetation Index (Qi et al., 1994)	$\frac{NIR-R}{NIR+R+L} \times (1+L)$ where $L = 1 - 2aNDVI.WDVI$ and where $WDVI = NIR - aR$
MSAVI2	Modified Soil Adjusted Vegetation Index (Qi et al., 1994)	$\frac{(2*NIR+1-\sqrt{(2*NIR+1)^2-8*(NIR-R)})}{2}$
GEMI	Global Environment Monitoring Index (Pinty and Verstraete, 1992)	$n(1 - 0.25n) - (\frac{R-0.25}{1-R})$ where $n = \frac{2(NIR^2-R^2)+1.5NIR+0.5R}{NIR+R+0.5}$
WDVI	Weighted Difference Vegetation Index (Clevers, 1988) (JGPW, 1991)	$NIR - s * R$ where s is the slope of the soil line
AVI	Angular Vegetation Index (Plummer, 1994)	$Tan^{-1}(\frac{A1}{NIR-R}) + Tan^{-1}(\frac{A2}{G-R})$ where $A1 = \frac{\lambda3-\lambda2}{\lambda2}$ $A2 = \frac{\lambda2-\lambda1}{\lambda2}$ $\lambda1, \lambda2$ and $\lambda3$ are the mid-band wavelengths for the green, red and NIR bands
ARVI	Atmospherically Resistant Vegetation Index (Kaufman and Tanre, 1992)	$\frac{NIR-RB}{NIR+RB}$ where $RB = R - \gamma(B - R)$, and γ is an atmospheric auto-correction factor that depends on the type of aerosol present
EVI	Enhanced Vegetation Index (Huete et al., 1994)	$G \times (\frac{NIR-R}{NIR+C1R-C2B+L})$ where G : gain factor (equals to 2.5) L : soil adjustment factor, $C1$ and $C2$: correction coefficients of atmospheric diffusion
IPVI	Infrared Percentage Vegetation Index (Crippen, 1990)	$\frac{NIR}{NIR+R}$
TVI	Transformed Vegetation Index (Deering, 1975)	$\sqrt{\frac{NIR-R}{NIR+R+0.5}}$
CVI	Chlorophyl Vegetation Index	$\frac{NIR \times R}{G^2}$

TABLE 3.4: Vegetation indices

NDVI indice (Rouse Jr et al., 1973) is very usefulm for many applications. However, it is sensitive to atmospheric conditions. Thus, other indices were proposed to enhance it such as the Enhanced Vegetation Index (EVI) (Huete et al., 1994) and Atmospherically Resistant Vegetation Index (ARVI) (Kaufman and Tanre, 1992). The EVI indice is robust to soil effect by using a soil adjustment factor. It also uses two coefficients to correct the atmospheric diffusion. To be considered, we need to know all the parameters of the ARVI and EVI indices. Nevertheless, these parameters are not easy to fix and we need to have information on the conditions of image acquisition process. This information is not available for most of the HSI and MSI datasets.

Algorithm 2 Feature Generation

Input B, G, R, NIR : Various bands of an HSI of size $L \times C$

Output RI : Radiometric indices vector of size $L \times C \times 15$

Begin

$pix \leftarrow 1$

for $i \leftarrow 1 : L$ **do**

for $j \leftarrow 1 : C$ **do**

$RI(pix, 1) \leftarrow NDVI(i, j)$

$RI(pix, 2) \leftarrow RVI(i, j)$

$RI(pix, 3) \leftarrow SAVI(i, j)$

$RI(pix, 4) \leftarrow MSAVI(i, j)$

$RI(pix, 5) \leftarrow MSAVI2(i, j)$

$RI(pix, 6) \leftarrow DVI(i, j)$

$RI(pix, 7) \leftarrow IPVI(i, j)$

$RI(pix, 8) \leftarrow GNDVI(i, j)$

$RI(pix, 9) \leftarrow IC(i, j)$

$RI(pix, 10) \leftarrow BI(i, j)$

$RI(pix, 11) \leftarrow NDTI(i, j)$

$RI(pix, 12) \leftarrow ISU(i, j)$

$RI(pix, 13) \leftarrow NDWI(i, j)$

$RI(pix, 14) \leftarrow CVI(i, j)$

$RI(pix, 15) \leftarrow IR(i, j)$

$pix \leftarrow pix + 1$

end for

end for

End

Besides the NDVI indice, we use the Ratio Vegetation Index (RVI) (Pearson and Miller, 1972) which is the ratio between the red and the near infrared bands. As vegetation indices, we also use the Soil Adjusted Vegetation Index (SAVI) (Huete, 1988), the Modified Soil Adjusted Vegetation Index (MSAVI) (Qi et al., 1994), the Modified the Soil Adjusted Vegetation Index (MSAVI2) (Qi et al., 1994) and the Infrared Percentage Vegetation Index (IPVI) (Crippen, 1990) because they consider the atmospheric and soil effects at the same time. The Difference Vegetation Index (DVI) on the other hand is another vegetation index that is the difference between the near infrared and the red bands. The Green Normalized Difference Vegetation Index (GNDVI) is also a modified version of NDVI with the aim to be more sensitive to the variation of chlorophyl. It uses the near infrared band and the green band instead of the red band in the NDVI computation. Thus, the Chlorophyl Vegetation Index (CVI) is also used and is computed using the green, the red and the near infrared bands.

To consider the soil indices, we use the Color Index (IC) (Pouget et al., 1991) which is computed using the red and green bands and the Bare-soil Index (BI) (Jamalabad, 2004) that is computed using the blue, the near infrared and short wave infrared bands. In addition, we use the index of redness of the soil (IR) (Pouget et al., 1991) which helps in the differentiation of the soil types.

We consider the Normalized Difference Water Index (NDWI) (Xu, 2006) and the Normalized Difference Turbidity Index (NDTI) (Zha et al., 2003) as water indices which use the green, the red and the near infrared bands. Finally, we consider the Index Surfaces Built (ISU) (Fauvel et al., 2009) which allows us to identify built up areas that have positive values of ISU unlike other types of geographic objects.

3.2.2 Classification of HSI

In order to classify the HSI, we extract the already mentioned RI for each pixel (see Algorithm 2). Then, we run a set of largely used classifiers such as SVM, KNN and Bagged Trees to classify the generated feature vectors. The classification process is explained in Algorithm 3. We use 70% of each class for training and 30% for testing. In the experimental result section 3.3, we report the accuracy of the best classifier on each dataset.

Algorithm 3 Classification

Input RI : Radiometric Indices vector
Output ML : Matrix of Labels
Begin
for $i \leftarrow 1 : N$ **do** $< N \text{ nb lines of } RI >$
 for $j \leftarrow 1 : M$ **do** $< M \text{ nb columns of } RI >$
 $ML(i, j) \leftarrow SVM(RI(i, j))$
 end for
end for
End

3.2.3 Spatial refinement of the classification results

As many researchers, we believe that the spatial information is quite important for HSI classification. Therefore, considering neighboring pixels can enhance the classification results. In fact, neighbouring pixels tend to share the same characteristics thus the same labels specially when we have a high spatial resolution.

In the post processing of the obtained classification, we consider the spatial information in order to refine results by updating the pixel classes if any. Thus, for each pixel p , we consider its eight (direct) neighbours. Then we update the label of the pixel p based on the maximal frequency of labels in the 9 pixels (8 neighbours + pixel p). Note that we do not consider the pixels in the borders of the image. Our refinement solution is described in Algorithm 4.

We iteratively run the spatial refinement Algorithm 4 until convergence and we report the number of iterations required for each dataset in the Experimental results in section 3.3.

Algorithm 4 Spatial Refinement

Input: ML : Matrix of Labels
Output: RML : Refined Matrix of Labels
Begin
for $i \leftarrow 1 : L$ **do** L and C are the size of the image I >
 for $j \leftarrow 1 : C$ **do**
 $N1 \leftarrow \text{Neighbors}(ML(i, j))$
 $RML(i, j) \leftarrow \text{Max_frequency}(\text{labels}(N1))$
 end for
end for
End

3.2.4 Temporal complexity of the proposed approach

In order to check the effectiveness and the scalability of our approach, we measure its temporal complexity. In fact, we have two major steps, feature generation and spatial refinement, so we need to compute the complexity of each step.

In the feature generation step, we extract the already mentioned 15 RI for each pixel. Thus, with N pixels per image, 15 RI per pixel are computed, we have $15 \times N$ iterations (as shown in Algorithm 2). Consequently, the temporal complexity of the feature generation step is $O(N)$. The refinement algorithm is also applied for each pixel and thus has a complexity of $O(N)$. Consequently, our approach has a linear complexity with the number of pixels which is very practical for HSI with a large number of bands and pixels.

3.3 Experimental results

In this section, we present the HS datasets used to carry out the experimental results. Then, we report the classification accuracies with and without spatial refinement and we present the confusion matrices for each dataset.

3.3.1 Description of the datasets

We use six available HS datasets namely the Indian Pines, Botswana, Pavia center, Pavia university, Salinas A and Salinas scene datasets. The details of these datasets are summarized in Table 3.5.

Dataset	Size	Bands	Resolution (m)	# of Classes
Indian Pines	145×145	200	20	16and9
Botswana	1476×256	145	30	14
Pavia center	1096×1096	102	1.3	9
Pavia University	610×610	103	1.3	9
Salinas A	86×83	204	3.7	6
Salinas scene	512×217	204	3.7	16

TABLE 3.5: Hyperspectral datasets used in the evaluation of our classification approach

The *Indian Pines* data (Figures 3.2 and 3.3) is one of the most used HSI. It was recorded by the AVIRIS sensor over the Indian Pines site in North-Western of Indian.

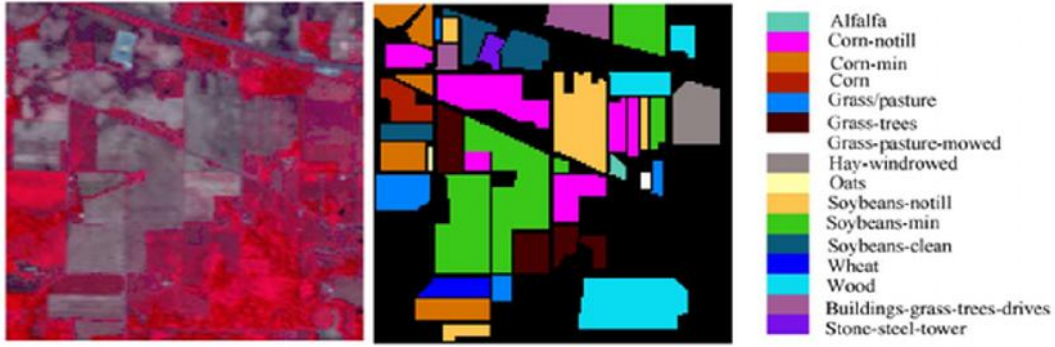


FIGURE 3.2: The Indian Pines dataset and its ground truth (16 classes)

It contains 145×145 pixels and 224 spectral reflectance bands from 0.4 to $2.5\mu m$. There are two available versions of the ground truth. In the first version, we have 16 classes (see Figure 3.2) which are not all mutually exclusive. The second version of the ground truth contains only 9 classes (Figure 3.3) by merging some classes of the first version. Generally, the HS datasets are preprocessed by their providers in order to remove unnecessary bands mainly the water absorption bands (because we do not have any information). Thus for the *Indian Pines* data, we have 200 bands remaining after the preprocessing step.

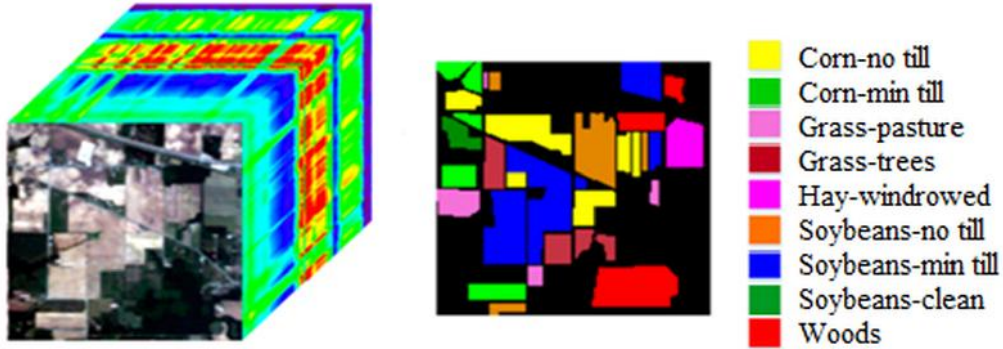


FIGURE 3.3: The Indian Pines dataset and its ground truth (9 classes)

On the other hand, the *Pavia University* (Figure 3.4) and *Pavia center* (Figure 3.5) are two datasets acquired by the ROSIS sensor during a flight campaign over Pavia in Northern Italy. There are 102 and 103 bands in the Pavia center and Pavia University dataset respectively. Both datasets contain 9 classes.

The *Salinas scene* (Figure 3.6) is acquired by the AVIRIS sensor over Salinas Valley in California, USA. It contains 16 classes and it has a spatial resolution of $3.7m$ and 224 bands. The area covers 512 lines by 217 samples. Due to water absorption, 20 bands of the 224 are removed. The *Salinas-A* is a sub scene of the Salinas image. It contains 86×83 pixels and includes six classes.

Finally, the *Botswana* dataset is acquired by NASA EO-1 satellite over the Okavango Delta, Botswana between 2001 and 2004. It has a spatial resolution of $30m$ and 242 bands ranging from 0.4 to $2.5\mu m$. The data was preprocessed by the UT Center for Space Research (Crawford et al., 2003) and 145 bands remained after preprocessing. The Botswana data comprises 14 classes.

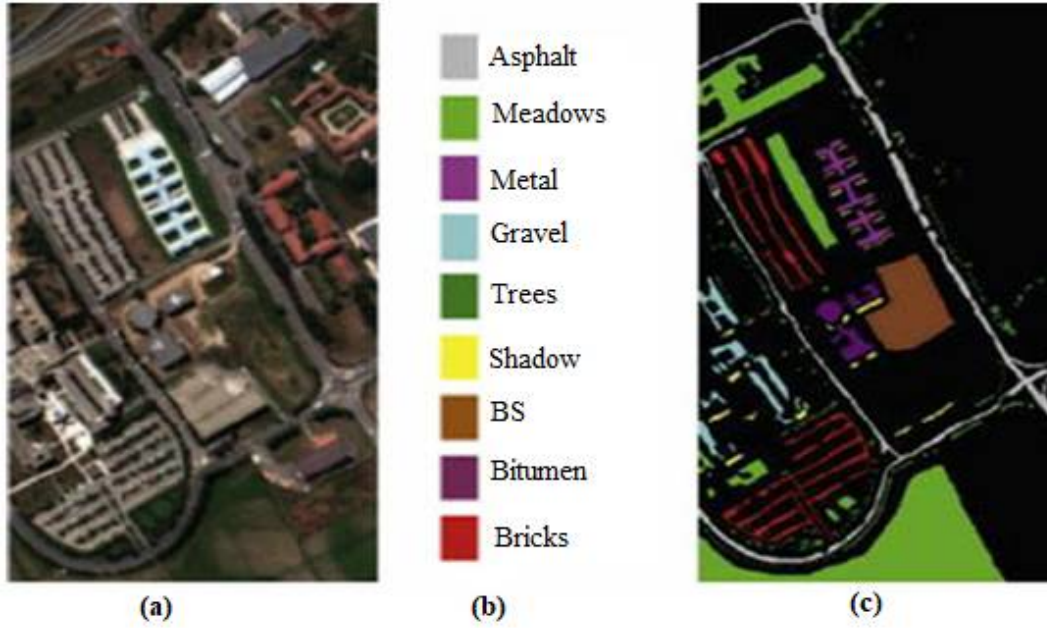


FIGURE 3.4: The *Pavia University* dataset (a) Three band color composite, (b) Reference data, (c) Color code

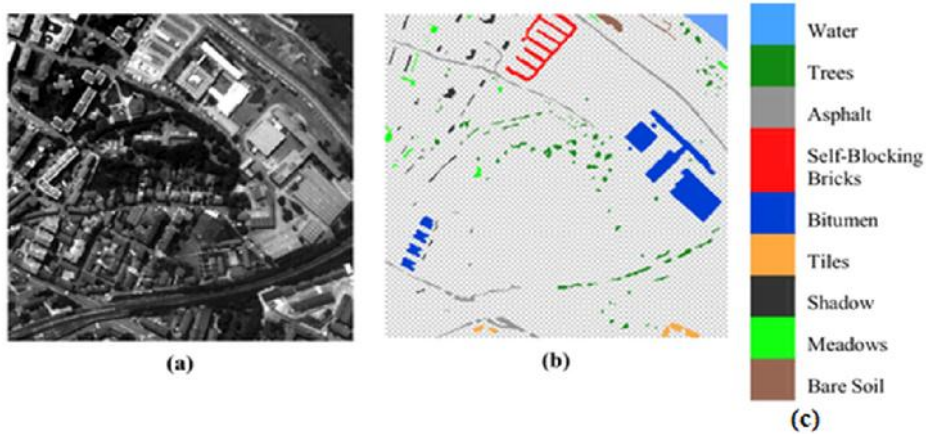


FIGURE 3.5: The *Pavia Scene* dataset (a) Three band color composite, (b) Reference data, (c) Color code

3.3.2 Results

We tested our approach on the HS datasets and we obtained very interesting classification accuracies. In the following tables, we present the performance results in terms of precision, recall, F-score, OA (overall accuracy), AA (average accuracy) and kappa index.

The precision metric is defined as the ratio between the number of correctly predicted labels and the number of predicted labels for a testing image. Let m_1 be the number of labels in the ground truth of the image, m_2 the number of predicted labels by our approach and m_3 the number of correctly predicted labels ($m_3 \leq m_2$).

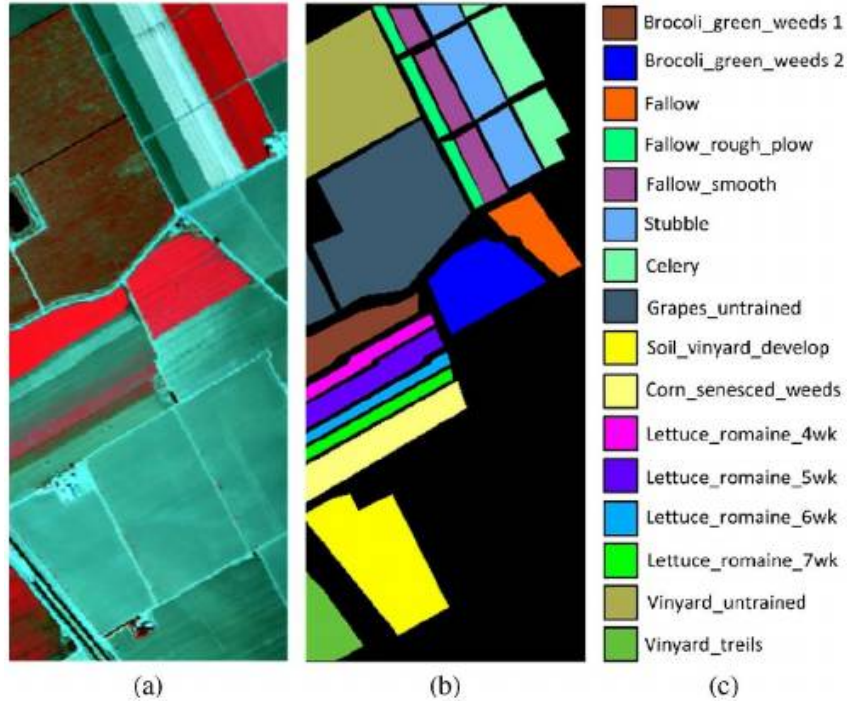


FIGURE 3.6: The *Salinas* dataset, (a) Three band color composite, (b) Reference data (c) Color code

and $m_3 \leq m_1$). Then the precision is defined by Equation 3.1:

$$P = \frac{m_3}{m_2} \quad (3.1)$$

The second metric used is the Recall and can be expressed by Equation 3.2 as follows:

$$P = \frac{m_3}{m_1} \quad (3.2)$$

The *F1* score is defined based on the precision (*P*) and the recall(*R*) metrics. It is the harmonic mean of *P* and *R*. It is a tradeoff between them and is described by Equation 3.3:

$$P = \frac{2PR}{P + R} \quad (3.3)$$

The precision, recall and *F1* scores are generally used for the evaluation of the performance of image retrieval methods.

On the other hand, the classification is generally assessed based on the overall and the average accuracies. The OA (Guo et al., 2018a) measures the number of correctly classified hyperspectral pixels divided by the number of all test samples. The AA (Guo et al., 2018a) computes the mean of the classification accuracies of all classes. Kappa (*K*) represents a statistical measurement of agreement based on the confusion matrix of the various classes.

a) The Botswana dataset

The classification results on the *Botswana* dataset are described in Table 3.6. The best OA and AA values are highlighted in bold, before and after the Spatial Refinement (SR) phase.

Botswana	Initial classification (%)	SR: Run 1 (%)	SR: Run 2 (%)	SR: Run 3 (%)	SR: Run 4 (%)
Recall	85.67	93.66	94.10	94.59	94.82
Precision	82.20	92.64	93.42	94.05	94.41
F-score	83.90	93.15	93.76	94.32	94.62
Kappa	79.99	90.84	91.62	92.28	92.65
OA	81.55	91.55	92.26	92.86	93.21
AA	82.20	92.64	93.42	94.05	94.41

TABLE 3.6: Classification results of our approach on the Botswana dataset

We notice that after 4 iterations of the spatial refinement, our algorithm converges with an OA of 93.21% and an AA of 94.41%. The confusion matrices of the *Botswana* data before and after the SR are presented in Figures 3.7 and 3.8:

		Target Class													
		Water	HG	FG1	FG2	Reeds	Riparian	Firecar	II	AW	AS	AG	SM	MM	ES
Output Class	Water	100.0% 70	7.7% 2	0.0% 0	0.0% 0	1.4% 1	0.0% 0	7.5% 5	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	12.5% 3
	HG	0.0% 0	88.5% 23	0.0% 0	0.0% 0	1.4% 1	1.4% 1	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	FG1	0.0% 0	0.0% 0	86.2% 56	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	1.6% 1	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	FG2	0.0% 0	0.0% 0	0.0% 0	64.3% 36	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Reeds	0.0% 0	3.8% 1	0.0% 0	30.4% 17	88.6% 62	41.4% 29	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	2.1% 1	2.9% 2	0.0% 0
	Riparian	0.0% 0	0.0% 0	0.0% 0	3.6% 2	8.6% 6	57.1% 40	0.0% 0	1.9% 1	30.9% 25	0.0% 0	1.3% 1	2.1% 1	5.8% 4	0.0% 0
	Firecar	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	92.5% 62	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	II	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	80.8% 42	0.0% 0	0.0% 0	0.0% 0	0.0% 0	8.7% 6	0.0% 0
	AW	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	64.2% 52	0.0% 0	0.0% 0	0.0% 0	8.7% 6	0.0% 0
	AS	0.0% 0	0.0% 0	13.8% 9	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	82.8% 53	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	AG	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	1.9% 1	0.0% 0	14.1% 9	98.7% 78	0.0% 0	0.0% 0	0.0% 0
	SM	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	91.5% 43	5.8% 4	0.0% 0
	MM	0.0% 0	0.0% 0	0.0% 0	1.8% 1	0.0% 0	0.0% 0	0.0% 0	15.4% 8	4.9% 4	1.6% 1	0.0% 0	4.3% 2	68.1% 47	0.0% 0
	ES	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	87.5% 21

FIGURE 3.7: Confusion matrix of our approach on the *Botswana* dataset without refinement

From the confusion matrix in Figure 3.7, we can notice that the classes *Riparian* and *Reeds* are not easy to identify. In fact these classes are so close and have almost the same spectral curve. Also the class *Riparian* and *AW* are also mixed. Finally the classes *FG2* and *Reeds* tend also to be misclassified.

b) The Indian Pines dataset

However, after performing the spatial refinement, the classification is significantly enhanced. And only the classes (*AW*, *Reeds*) and classes (*Riparian*, *Reeds*) are still but less misclassified.

We report the performances on the *Indian Pines* dataset in the Table 3.7.

The confusion matrices for the *Indian Pines* data before and after applying the refinement algorithm are presented in Figures 3.9 and 3.10.

Output Class	Water	100.0% 70	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	11.9% 8	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	HG	0.0% 0	100.0% 26	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	FG1	0.0% 0	0.0% 0	100.0% 65	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	FG2	0.0% 0	0.0% 0	0.0% 0	100.0% 56	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	Reeds	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 70	42.9% 30	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	Riparian	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	57.1% 40	0.0% 0	0.0% 0	23.5% 19	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	Firecar	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	88.1% 59	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	II	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 52	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	AW	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	76.5% 62	0.0% 0	0.0% 0	0.0% 0	0.0% 0	
	AS	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 64	0.0% 0	0.0% 0	0.0% 0	
	AG	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 79	0.0% 0	0.0% 0	
	SM	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 47	0.0% 0	
	MM	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 69	
	ES	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 24
		Water	HG	FG1	FG2	Reeds	Riparian	Firecar	II	AW	AS	AG	SM	MM	ES
	Target Class														

FIGURE 3.8: Confusion matrix of our approach on the *Botswana* dataset (after 4 runs of SR)

Indian pines	Initial classification	SR : Run 1	SR : Run 2	SR : Run 3	SR : Run 4	SR : Run 5	SR : Run 6	SR : Run 7	SR : Run 8	SR : Run 9
Recall	64.11	79.94	80.58	80.95	81.32	81.55	81.67	81.86	81.90	81.92
Precision	60.14	70.32	71.29	71.94	72.65	73.18	73.41	73.71	73.84	73.93
F-score	62.06	74.82	75.65	76.18	76.74	77.14	77.31	77.57	77.66	77.72
Kappa	99.84	99.82	99.82	99.81	99.81	99.81	99.81	99.81	99.81	99.81
OA	60.45	70.99	72.20	72.83	73.41	73.75	73.91	74.16	74.29	74.37
AA	60.14	70.32	71.29	71.94	72.65	73.18	73.41	73.71	73.84	73.93

TABLE 3.7: Classification results of our approach on the *Indian Pines* dataset

We notice that the AA and OA are poor on the *Indian Pines* data. This can be explained by the similarity of the spectral signatures of the concepts present since all of them represent variations of the vegetation class. Even after nine iterations, the results are not very satisfactory.

c) The Pavia Center dataset

The performance results on *Pavia Center* are presented in Table 3.8.

The confusion matrices on the *Pavia Center* data are presented in Figures 3.11 and 3.12 before and after SR respectively.

From the confusion matrix of the *Pavia Center* presented in Figure 3.11, the classes *Self-Building Bricks*, *Bitumen* tend to be mixed. The spatial refinement reduced this confusion as shown in Figure 3.12 and results in an OA of 98.53% and an AA of 93.18%.

d) The Pavia University dataset

The performances on the *Pavia University* dataset are illustrated by Table 3.9 and the confusion matrices by Figures 3.13 and 3.14 before and after the spatial refinement respectively.

The confusion matrix of Figure 3.13 shows that the pairs of classes (*Self-Building*

		Target Class								
		CornN	CornM	GrassP	GrassT	HayW	SoybeanN	SoybeanM	SoybeanC	Woods
Output Class	CornN	22.3% 83	13.9% 29	0.0% 0	0.0% 0	0.0% 0	1.2% 3	15.9% 101	46.8% 72	0.0% 0
	CornM	29.0% 108	22.0% 46	0.0% 0	0.0% 0	0.0% 0	2.4% 6	12.5% 79	14.9% 23	0.0% 0
	GrassP	0.0% 0	0.0% 0	72.2% 91	0.0% 0	2.4% 3	0.0% 0	0.2% 1	0.0% 0	2.1% 7
	GrassT	0.0% 0	0.0% 0	7.1% 9	97.4% 185	4.0% 5	0.0% 0	0.2% 1	0.6% 1	0.0% 0
	HayW	0.0% 0	0.0% 0	4.0% 5	0.5% 1	91.1% 113	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	SoybeanN	1.1% 4	0.0% 0	0.0% 0	0.0% 0	0.0% 0	62.8% 159	0.5% 3	7.1% 11	0.0% 0
	SoybeanM	41.1% 153	57.4% 120	5.6% 7	0.5% 1	2.4% 3	33.2% 84	68.9% 437	23.4% 36	0.6% 2
	SoybeanC	6.5% 24	6.7% 14	0.8% 1	0.0% 0	0.0% 0	0.4% 1	1.9% 12	7.1% 11	0.0% 0
	Woods	0.0% 0	0.0% 0	10.3% 13	1.6% 3	0.0% 0	0.0% 0	0.0% 0	0.0% 0	97.3% 321

FIGURE 3.9: The confusion matrix of the *Indian Pines* dataset without refinement

		Target Class								
		CornN	CornM	GrassP	GrassT	HayW	SoybeanN	SoybeanM	SoybeanC	Woods
Output Class	CornN	33.9% 126	11.5% 24	0.0% 0	0.0% 0	0.0% 0	0.0% 0	11.8% 75	55.2% 85	0.0% 0
	CornM	32.0% 119	35.9% 75	0.0% 0	0.0% 0	0.0% 0	0.0% 0	1.6% 10	5.8% 9	0.0% 0
	GrassP	0.0% 0	0.0% 0	100.0% 126	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	GrassT	0.0% 0	0.0% 0	0.0% 0	100.0% 190	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	HayW	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 124	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	SoybeanN	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	92.1% 233	0.0% 0	7.8% 12	0.0% 0
	SoybeanM	34.1% 127	52.6% 110	0.0% 0	0.0% 0	0.0% 0	7.9% 20	86.6% 549	14.3% 22	0.0% 0
	SoybeanC	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	16.9% 26	0.0% 0
	Woods	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 330

FIGURE 3.10: The confusion matrix of the *Indian Pines* after refinement

Bricks, Gravel), (*Trees, Meadows*), (*Bare soil, Meadows*) and (*Self-Building Bricks, Bitumen*) are confused. The spatial refinement slightly enhanced the results and we have

Pavia Center	Initial classification (%)	SR: Run 1 (%)	SR: Run 2 (%)	SR: Run 3 (%)	SR: Run 4 (%)	SR: Run 5 (%)
Recall	90.16	95.08	95.33	95.37	95.41	95.45
Precision	88.61	93.00	93.13	93.14	93.16	93.18
F-score	89.38	94.03	94.22	94.24	94.27	94.30
Kappa	95.44	97.80	97.88	97.89	97.91	97.92
OA	96.78	98.45	98.51	98.51	98.53	98.53
AA	88.61	93.00	93.13	93.16	93.16	93.18

TABLE 3.8: Classification results of our approach on the Pavia Center dataset

		Target Class								
		Water	Trees	Asphalt	SBB	Bitumen	Tiles	Shadows	Meadows	BareSoil
Output Class	Water	100.0% 17157	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.1% 1	0.0% 0	0.0% 0
	Trees	0.0% 0	90.3% 1788	15.8% 125	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Asphalt	0.0% 0	9.6% 191	82.8% 657	0.0% 0	0.4% 7	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	SBB	0.0% 0	0.0% 0	0.0% 0	65.3% 457	2.9% 49	1.3% 32	4.3% 82	0.3% 28	0.0% 0
	Bitumen	0.0% 0	0.0% 0	1.4% 11	27.6% 193	87.0% 1493	0.0% 1	0.7% 13	0.1% 12	0.0% 0
	Tiles	0.0% 0	0.0% 0	0.0% 0	2.3% 16	8.0% 138	96.3% 2318	2.8% 53	0.5% 52	0.0% 0
	Shadows	0.0% 0	0.0% 0	0.0% 0	4.0% 28	0.0% 0	2.2% 54	92.2% 1751	0.0% 0	0.0% 0
	Meadows	0.0% 0	0.0% 0	0.0% 0	0.9% 6	1.7% 30	0.0% 1	0.0% 0	99.2% 11043	0.0% 0
	BareSoil	0.0% 0	0.1% 1	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 746

FIGURE 3.11: The confusion matrix of the *Pavia Center* dataset without refinement

an OA of 77.83% and an AA of 70.86%.

e) The Salinas-A dataset

The classification accuracies of *Salinas-A* as well as the confusion matrices before and after refinement are presented in Table 3.10 and Figures 3.15 and 3.16 respectively.

The confusion matrix on the *Salinas-A* data illustrated by Figure 3.15 presents small confusions and the spatial refinement (Figure 3.16) resolved this issue and we got an OA and AA of 100%.

f) The Salinas Scene dataset:

The classification results on the *Salinas Scene* dataset are summarized in Table 3.11.

The Figures 3.17 and 3.18 present the confusion matrices of the *Salinas Scene*

	Target Class								
	Water	Trees	Asphalt	SBB	Bitumen	Tiles	Shadows	Meadows	BareSoil
Output Class	Water	100.0% 17157	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Trees	0.0% 0	95.7% 1895	11.0% 87	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Asphalt	0.0% 0	4.3% 85	89.0% 706	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	SBB	0.0% 0	0.0% 0	0.0% 0	73.1% 512	1.7% 30	0.9% 22	1.7% 32	0.0% 0
	Bitumen	0.0% 0	0.0% 0	0.0% 0	26.9% 188	98.3% 1687	0.0% 0	0.3% 6	0.0% 0
	Tiles	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	98.7% 2375	0.0% 0	0.0% 0
	Shadows	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.4% 9	98.0% 1862	0.0% 0	0.0% 0
	Meadows	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 11135	0.0% 0
	BareSoil	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 746

FIGURE 3.12: The confusion matrix of the *Pavia Center* dataset after refinement

Pavia University	Initial classification (%)	SR: Run 1 (%)	SR: Run 2 (%)
Recall	67.64	80.43	81.04
Precision	66.92	71.00	70.86
F-score	67.28	75.42	75.61
Kappa	60.67	69.07	69.16
OA	70.83	77.72	77.83
AA	66.92	71.01	70.86

TABLE 3.9: Classification results of our approach on the *Pavia University* dataset

Salinas-A	Initial classification (%)	SR: Run 1 (%)	SR: Run 2 (%)
Recall	96.82	99.86	100
Precision	96.92	99.81	100
F-score	96.87	99.83	100
Kappa	96.41	99.81	100
OA	97.14	99.85	100
AA	96.92	99.81	100

TABLE 3.10: Classification results of our approach on the *Salinas-A* dataset

dataset without refinement (initialization) and after running the refinement algorithm (after 4 iterations, we obtained stable results).

From the confusion matrix of *Salinas Scene* (Figure 3.17), we can notice that the pairs of classes (*Fallow*, *Corn senesced weeds (csgw)*), (*Vinyard untrained(VU)*, *Grapes*

		Target Class								
		Asphalt	Meadows	Gravel	Trees	PMS	BareSoil	Bitumen	SBB	Shadows
Output Class	Asphalt	75.2% 1291	0.0% 0	13.7% 74	0.0% 0	0.0% 0	0.3% 4	34.4% 119	1.4% 13	0.0% 0
	Meadows	0.2% 3	88.4% 4213	0.0% 0	39.3% 313	0.0% 0	67.8% 889	0.0% 0	0.2% 2	0.0% 0
	Gravel	7.8% 133	0.0% 0	40.7% 220	0.0% 0	0.0% 0	1.7% 22	0.3% 1	43.1% 414	0.0% 0
	Trees	0.0% 0	3.7% 177	0.0% 0	60.3% 480	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	PMS	0.0% 0	0.0% 0	0.0% 0	0.0% 0	99.1% 347	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	BareSoil	0.9% 15	7.9% 378	0.6% 3	0.4% 3	0.9% 3	25.3% 332	0.0% 0	0.4% 4	0.0% 0
	Bitumen	12.8% 220	0.0% 0	10.2% 55	0.0% 0	0.0% 0	0.4% 5	65.3% 226	7.0% 67	0.0% 0
	SBB	3.0% 52	0.0% 0	34.9% 189	0.0% 0	0.0% 0	4.5% 59	0.0% 0	47.9% 460	0.0% 0
	Shadows	0.1% 2	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 247

FIGURE 3.13: The confusion matrix of the *Pavia University* dataset without refinement

		Target Class								
		Asphalt	Meadows	Gravel	Trees	PMS	BareSoil	Bitumen	SBB	Shadows
Output Class	Asphalt	88.2% 1513	0.0% 0	15.2% 82	0.0% 0	0.0% 0	0.0% 0	24.3% 84	0.6% 6	0.0% 0
	Meadows	0.0% 0	99.5% 4746	0.0% 0	56.0% 446	0.0% 0	70.3% 921	0.0% 0	0.0% 0	0.0% 0
	Gravel	5.1% 88	0.0% 0	56.2% 304	0.0% 0	0.0% 0	0.0% 0	0.0% 0	48.9% 469	0.0% 0
	Trees	0.0% 0	0.5% 22	0.0% 0	44.0% 350	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	PMS	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 350	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	BareSoil	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	29.7% 390	0.0% 0	0.7% 7	0.0% 0
	Bitumen	6.7% 115	0.0% 0	1.8% 10	0.0% 0	0.0% 0	0.0% 0	75.7% 262	5.4% 52	0.0% 0
	SBB	0.0% 0	0.0% 0	26.8% 145	0.0% 0	0.0% 0	0.0% 0	0.0% 0	44.4% 426	0.0% 0
	Shadows	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 247

FIGURE 3.14: The confusion matrix of the *Pavia University* dataset after refinement

untrained (GU) and (*Grapes untrained (GU)*, *Vinyard treils (VVT)*) are not easily distinguishable. After the spatial refinement (Figure 3.18), the confusion between the mentioned classes is reduced and we have an OA of 88.17% and an AA of 91.71%.

	Target Class					
	BGW1	CSGW	LR4wk	LR5wk	LR6wk	LR7wk
Output Class	BGW1	CSGW	LR4wk	LR5wk	LR6wk	LR7wk
BGW1	98.9% 87	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
CSGW	1.1% 1	98.5% 326	2.6% 4	0.0% 0	0.0% 0	0.0% 0
LR4wk	0.0% 0	0.6% 2	94.7% 144	0.0% 0	0.0% 0	0.0% 0
LR5wk	0.0% 0	0.0% 0	2.6% 4	100.0% 385	0.0% 0	0.0% 0
LR6wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	96.5% 165	0.0% 0
LR7wk	0.0% 0	0.9% 3	0.0% 0	0.0% 0	3.5% 6	100.0% 202

FIGURE 3.15: The confusion matrix of the *Salinas-A* dataset without refinement

	Target Class					
	BGW1	CSGW	LR4wk	LR5wk	LR6wk	LR7wk
Output Class	BGW1	CSGW	LR4wk	LR5wk	LR6wk	LR7wk
BGW1	100.0% 88	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
CSGW	0.0% 0	100.0% 331	0.0% 0	0.0% 0	0.0% 0	0.0% 0
LR4wk	0.0% 0	0.0% 0	100.0% 152	0.0% 0	0.0% 0	0.0% 0
LR5wk	0.0% 0	0.0% 0	0.0% 0	100.0% 385	0.0% 0	0.0% 0
LR6wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 171	0.0% 0
LR7wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 202

FIGURE 3.16: The confusion matrix of the *Salinas-A* dataset after refinement

Salinas scene	Initial classification (%)	SR: Run 1 (%)	SR: Run 2 (%)	SR: Run 3 (%)	SR: Run 4 (%)	SR: Run 5 (%)	SR: Run 6 (%)
Recall	89.22	93.64	94.01	94.18	94.26	94.29	94.32
Precision	86.89	91.15	91.53	91.65	91.68	91.07	91.71
F-score	88.04	92.38	92.75	92.90	92.95	92.97	93.00
Kappa	99.94	99.94	99.94	99.94	99.94	99.94	99.94
OA	82.89	87.36	87.78	87.98	88.09	88.13	88.17
AA	86.89	91.15	91.53	91.65	91.68	91.70	91.71

TABLE 3.11: Classification results of our approach on the *Salinas Scene* dataset

		Target Class															
		BGW1	BGW2	Fallow	FRP	FS	Stubble	Celery	GU	SVD	CSGW	LR4wk	LR5wk	LR6wk	LR7wk	VU	VVT
Output Class	BGW1	99.6% 513	0.1% 1	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	BGW2	0.4% 2	99.9% 971	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Fallow	0.0% 0	0.0% 0	55.1% 284	0.0% 0	14.0% 98	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	FRP	0.0% 0	0.0% 0	0.0% 0	99.2% 360	0.4% 3	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	FS	0.0% 0	0.0% 0	0.0% 0	0.6% 2	84.4% 589	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Stubble	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	99.9% 1031	0.0% 0	0.0% 0	0.1% 1	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Celery	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	98.6% 920	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	GU	0.0% 0	0.0% 0	0.0% 0	0.3% 1	0.9% 6	0.0% 0	82.6% 2415	1.9% 30	7.8% 66	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	53.1% 991	18.3% 83
	SVD	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	83.8% 1355	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	CSGW	0.0% 0	0.0% 0	44.3% 228	0.0% 0	0.3% 2	0.1% 1	0.9% 25	10.3% 167	91.1% 772	16.3% 45	3.0% 15	0.0% 0	0.0% 0	0.4% 1	0.5% 9	1.3% 6
	LR4wk	0.0% 0	0.0% 0	0.6% 3	0.0% 0	0.0% 0	0.0% 0	0.0% 0	3.9% 63	0.5% 4	83.3% 230	0.8% 4	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	LR5wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.6% 5	0.4% 1	96.2% 478	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.9% 4
	LR6wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	95.8% 227	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	LR7wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	4.2% 10	99.6% 275	0.0% 0	0.0% 0	0.0% 0
	VU	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	16.3% 478	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	46.3% 864	4.6% 21	0.0% 0
	VVT	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	1.4% 13	0.2% 7	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.2% 4	74.8% 339	0.0% 0

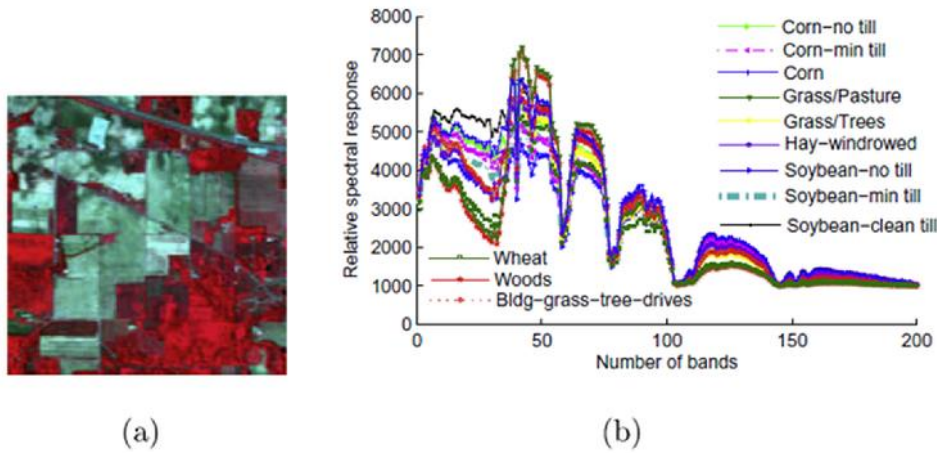
FIGURE 3.17: The confusion matrix of the *Salinas Scene* dataset without refinement

3.3.3 Discussion and comparison of the classification results

The classification accuracies obtained for the *Indian Pines* dataset, as stated earlier, are not satisfactory. This can be explained by two major factors namely the class imbalances and the close spectral signatures. Therefore, the *Indian Pines* data contains almost only vegetation classes with very close spectral signatures and thus very close FV of RI as shown in Figure 3.19.

In the *Salinas-A* data, we have a precision of 100%. This can be explained by the large differences between the spectral signatures of the classes and thus the RI are discriminative enough to classify the image classes efficiently. Besides, we don't have the problem of class imbalance and we have a sufficient number of pixels in each class. The same observation can be seen in the *Salinas scene* data.

	Target Class															
	BGW1	BGW2	Fallow	FRP	FS	Stubble	Celery	GU	SVD	CSGW	LR4wk	LR5wk	LR6wk	LR7wk	VU	VVT
Output Class	BGW1	100.0% 515	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	BGW2	0.0% 0	100.0% 972	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Fallow	0.0% 0	0.0% 0	63.1% 325	0.0% 0	6.3% 44	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	FRP	0.0% 0	0.0% 0	0.0% 0	100.0% 363	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	FS	0.0% 0	0.0% 0	0.0% 0	0.0% 0	93.7% 654	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Stubble	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 1032	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	Celery	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 933	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	GU	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	92.1% 2694	1.8% 29	4.8% 41	0.0% 0	0.0% 0	0.0% 0	0.0% 0	49.3% 920	15.2% 69
	SVD	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	90.3% 1459	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	CSGW	0.0% 0	0.0% 0	36.9% 190	0.0% 0	0.0% 0	0.0% 0	0.0% 0	7.4% 120	95.2% 806	2.5% 7	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	LR4wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.5% 8	0.0% 0	97.5% 269	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	LR5wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 497	0.0% 0	0.0% 0	0.0% 0	0.0% 0
	LR6wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 237	0.0% 0	0.0% 0	0.0% 0
	LR7wk	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	100.0% 276	0.0% 0	0.0% 0
	VU	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	7.9% 231	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	50.7% 948	0.0% 0
	VVT	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	0.0% 0	84.8% 384

FIGURE 3.18: The confusion matrix of the *Salinas Scene* dataset after refinementFIGURE 3.19: *Indian Pines* data (a) Three band false color composite (b) Spectral signatures

In order to evaluate our approach, we compare it to the best known state-of-the-art methods. It is worth noting that we are the first to use all the available hyperspectral datasets. To have a fair comparison, we first compare our method to a representative set of handcrafted feature generation based methods. Then, we compare it to another set of deep learning based methods.

Since our proposed approach for HSI classification is handcrafted based model, it is worthy to compare it with the handcrafted feature generation based method. We performed the classification on almost all the available HS datasets. However, this is not the case for the-state-of-the-art methods. Hence, the comparison will be

Methods	OA (%)	AA (%)	K (%)
MMP (He et al., 2008)	55.92	60.42	49.67
PCA (Fauvel et al., 2009)	63.71	67.53	58.16
KDA (Li et al., 2018)	63.90	67.07	58.68
LDA (Chang and Ren, 2000)	65.35	67.62	59.86
NWFE (Kuo and Landgrebe, 2004)	68.86	70.37	63.86
RFCF (Hosseini and Ghassemian, 2016)	74.37	76.96	70.39
LDFV (Our approach)	74.37	73.93	69.42
MFCF (Li et al., 2018)	75.28	78.68	71.45

TABLE 3.12: Comparison of classification results on the *Indian Pines* dataset

Methods	OA (%)	AA (%)	K(%)
PCA	96.99	93.56	95.76
KPCA	97.32	94.40	96.23
LDFV (Our approach)	98.53	93.18	97.92
EMP (KPCA)	98.87	98.25	98.41

TABLE 3.13: Comparison of classification results on the *Pavia Center* dataset

Refs/Metrics	OA (%)	AA (%)	K (%)
LDFV (our approach)	74.37	73.93	69.42
(Ma et al., 2016a)	86.46	79.30	84.63
(Hu et al., 2015b)	90.16	-	-
(Pan et al., 2017)	90.65	95.48	89.36
(Paul and Kumar, 2018)	96.66	97.42	96.19
(Paoletti et al., 2017)	98.37	99.27	98.15

TABLE 3.14: Comparison of classification results with deep learning based methods on the *Indian Pines* dataset

done considering the datasets used by the related works.

The *Indian Pines* is largely studied and we present a comparison of our approach and the related work in Table 3.12 (in ascending order of overall and average accuracies).

The results on the *Indian Pines* dataset are presented in (Li et al., 2018). We can see from Table 3.12 that our method gives very comparable result ($OA = 74.37\%$) to the MFCF which has the best OA (75.28%). As we stated earlier, the performance of our method on the *Indian Pines* data is less appealing because the spectral signatures of the classes are very close. Consequently, the RI lose a part of their discriminative power.

For the *Pavia Center* dataset (see Table 3.13), the best OA is given by the Extended Morphological Profile (EMP KPCA) approach proposed by (Fauvel et al., 2009) which makes use of Kernel PCA. On the other hand, our approach gives a fairly comparable OA (98.53%) but the best score is (98.87%).

Refs/Metrics	OA (%)	AA (%)	K (%)
LDFV (our approach)	74.37	73.93	69.42
(Ma et al., 2016a)	86.46	79.30	84.63
(Hu et al., 2015b)	90.16	-	-
(Pan et al., 2017)	90.65	95.48	89.36
(Paul and Kumar, 2018)	96.66	97.42	96.19
(Paoletti et al., 2017)	98.37	99.27	98.15

TABLE 3.15: Comparison of classification results with deep learning based methods on the *Indian Pines* dataset

Refs/Metrics	OA (%)	AA (%)	K (%)
LDFV (our approach)	77.83	70.86	69.16
(Ma et al., 2016a)	82.61	85.10	78.07
(Hu et al., 2015b)	92.56	-	-
(Paul and Kumar, 2018)	96.66	96.21	95.57
(Paoletti et al., 2017)	97.80	98.29	97.09

TABLE 3.16: Comparison of classification results with deep learning based methods on the *Pavia University* dataset

Refs/Metrics	OA (%)	AA (%)	K (%)
LDFV (our approach)	93.21	94.41	92.65
(Paul and Kumar, 2018)	97.61	97.74	97.41

TABLE 3.17: Comparison of classification results with deep learning based methods on the *Botswana* dataset

The proposed DL based methods gave interesting classification accuracies (Tables 3.14 -3.17). However, they are complicated to design and computationally expensive. For example, for the *Botswana* dataset, our approach gave an overall accuracy of 93.21% which is only 4.4% lower than the best result found by (Paul and Kumar, 2018), and is still a good result regarding the linear temporal complexity and the simplicity of our approach.

3.4 Conclusion

In this chapter, we presented our approach for the classification of hyperspectral images. Our approach is composed of two steps. In the first step, we extract a set of easy-to-compute radiometric indices. Then, motivated by the importance of the spatial information in image analysis, we perform a spatial refinement in the second step. We tested our approach on various HS benchmarks and we reported promising results on datasets having different classes of objects (e.g. the *Botswana* dataset). Nevertheless, its performance is low on datasets containing classes having similar or close spectral signatures (e.g. on the *Indian Pines* dataset). The objective of this chapter was to validate the choice of the RI used to perform the automatic annotation of RS images and the enrichment of the *SatelliteSceneOntology-V3* ontology which will be presented in Chapter 4.

Chapter 4

Graph Model for the Annotation of RS Images

4.1 Introduction

After performing the classification of HSI based on Radiometric Indices (RI), we present, in this chapter, a graph of concepts for the annotation of multispectral and hyperspectral images. Our annotation method considers most of the available information including spatial, spectral and contextual information. In order to annotate the images with relevant keywords, we start by the enrichment of the largest existing *SatelliteSceneOntology-V3* based on the already tested RI presented in Chapter 3. Then, we proceed to annotate a set of hyperspectral and multispectral images based on a similarity metric between the Feature Vector (FV) of image regions and the FV of the concepts stored in the ontology.

4.2 The Annotation Process

The basic image annotation process, generally, starts by a segmentation step. Then, a feature extraction step is performed and is sometimes followed by a feature selection step. And based on a distance metric, images are clustered or classified. Therefore, the annotation model is instantiated to annotate images.

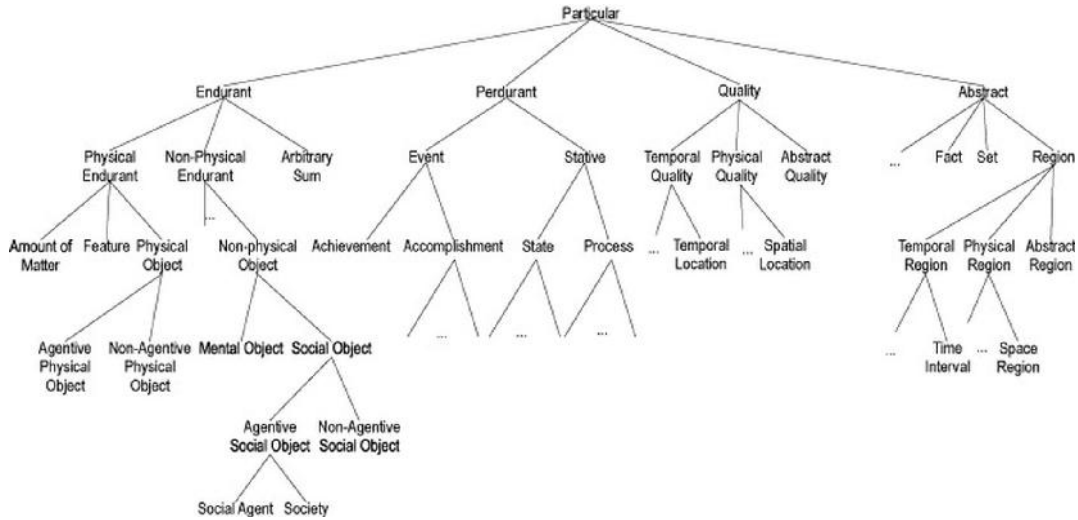
For the semantic annotation of images, the concepts are more and more stored in knowledge structures such as taxonomies, vocabularies and domain ontologies. Domain ontologies are becoming popularly used due to several reasons. They allow to formally represent the knowledge in a way which is understandable by humans as well as computers.

In order to efficiently annotate HS and MS images, we start by the enrichment of the *SatelliteSceneOntology-V3* as explained in section 4.3.

4.3 SatelliteSceneOntology-V4: Enrichment of Satellite Scene Ontology-V3

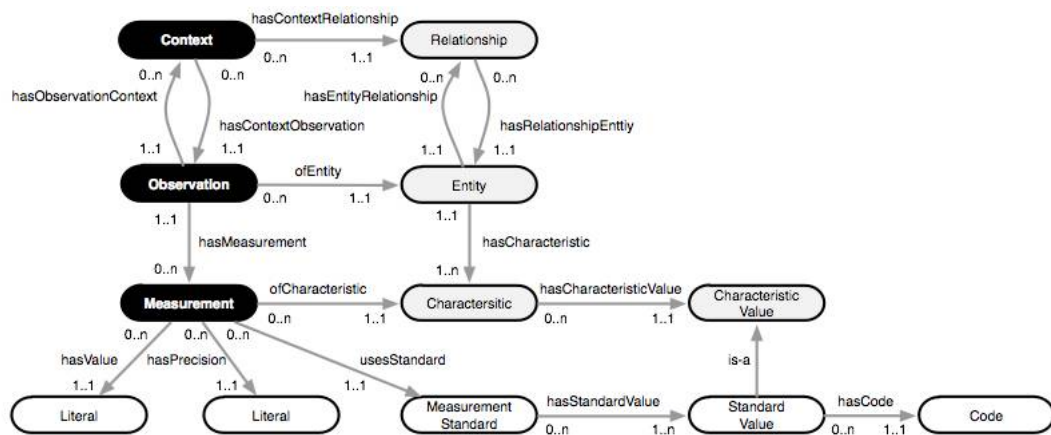
Formulating knowledge in a comprehensive way by humans and machines is very challenging and time consuming. However, knowledge structures have many advantages such as the re-utilisability and the automation of several tasks.

In order to perform the semantic annotation of RS images, we proceed to enrich the existing *SatelliteSceneOntology-V3*. The main limits of this ontology is that it does not contain instances. It is only a hierarchy of concepts related to satellite scene objects. However, it constitutes a good starting point.

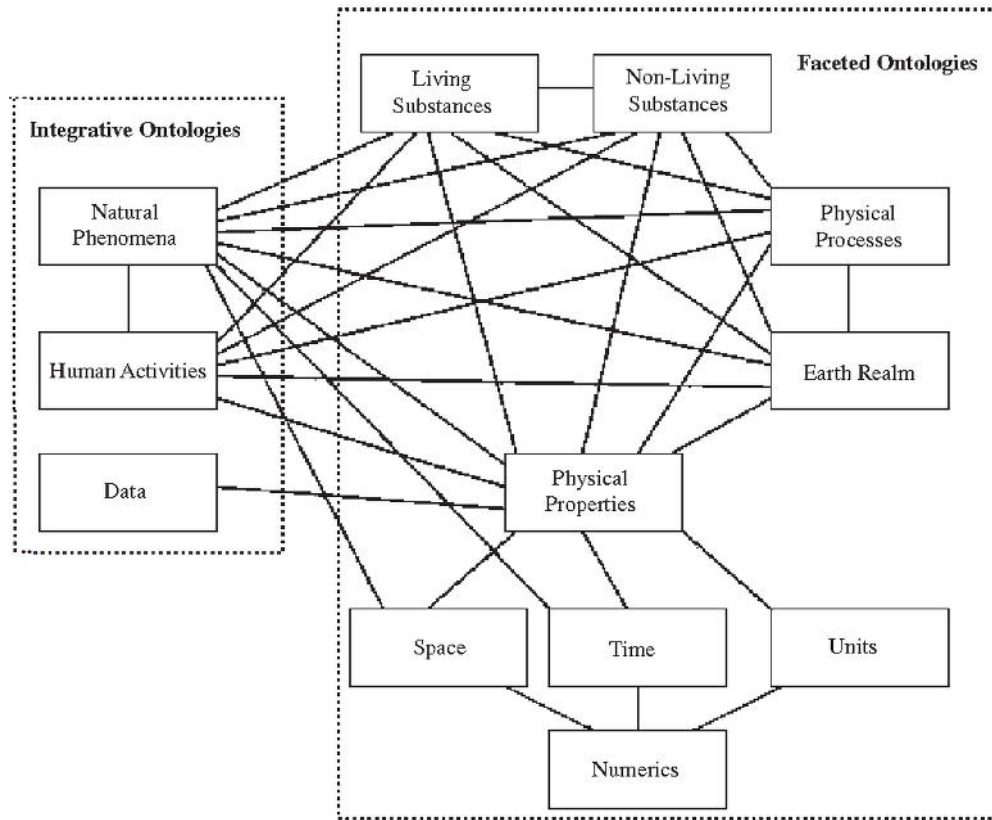
FIGURE 4.1: A subset of the *DOLCE* ontology

An ontology is defined by (Gruber, 1995) as an explicit specification of a conceptualization. This conceptualization includes the set of objects, concepts and other entities that are considered relevant in a specific context and the set of relations between them. The objective of the domain ontology is to represent the concepts, the relationships and any other distinctions that are useful and relevant for modeling the studied domain. The representation of an entity is called "individual" according to (Nardi and Brachman, 2003). And a concept can be seen as unary predicate that represents a set of individuals.

Ontologies are generally divided into core ontologies (also called upper level ontologies) and domain ontologies. Core ontologies represent abstract and general concepts. However, domain ontologies, as their name suggests, are specialized and represent concepts that are relevant to a given domain. Under the upper-level ontologies we can find *DOLCE* (Figure 4.1), *OBOE* (Figure 4.2) and *SWEET* (Figure 4.3).

FIGURE 4.2: A subset of the *OBOE* ontology

There are many domain ontologies in our context, such as *FODOMUST* (Durand et al., 2007) which is an ontology that represents urban objects in satellite images. It has a depth of 6, contains 91 concepts resulting in 66 leaf concepts characterized by 20 attributes. Each concept in the *FODOMUST* ontology has contextual, spatial and spectral attributes that are enriched with a priori values. Also, *DAFAOE*

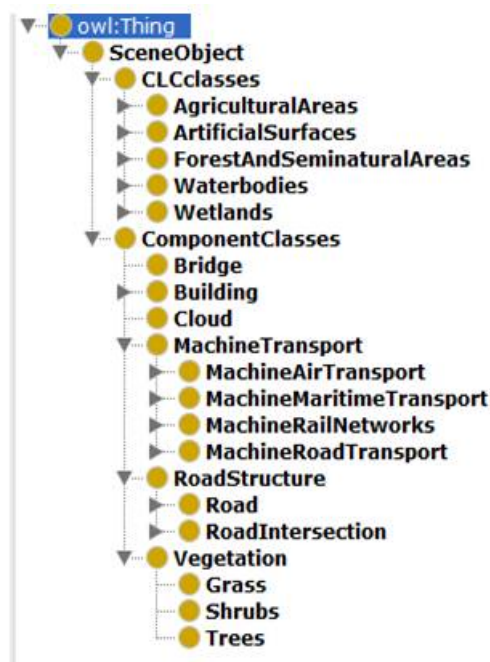
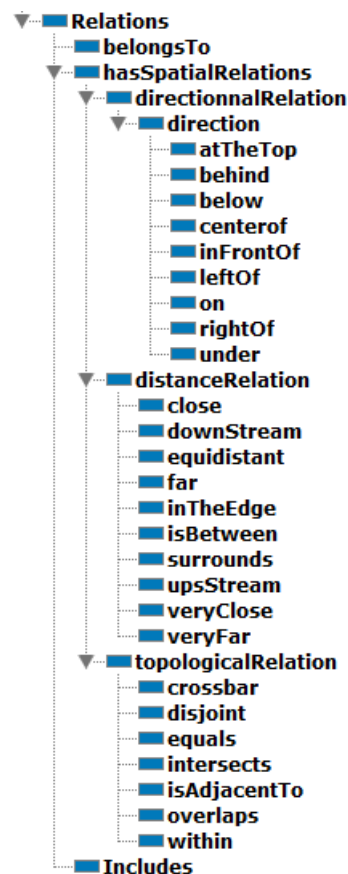
FIGURE 4.3: A subset of the *SWEET* ontology

(Charlet et al., 2010) ontology covers concepts related to land use, agriculture, water areas, etc. It contains a set of concepts that are associated with data stemming from the European database of land biophysical occupation “Corine-Land-Cover”. DAFAOE also allows the consideration of spatial relations between concepts and offers three families of spatial relations: topological, distance and directional relations. *SatelliteSceneOntology-V3* is another domain ontology and is illustrated by Figure 4.4. It contains 608 concepts and a depth of 9. It considers the same spatial relations as DAFAOE (topological, relational and directional relations) as depicted in Figure 4.5.

After reviewing the literature, we found that the available RS ontologies contain only textual descriptions of concepts. Thus, we cannot use them directly to perform image annotation or image retrieval. In fact, we cannot compare an image to a concept directly. RS images are rich in information. However, several useful characteristics are not considered while developing the ontology such as the spectral characteristics, the sensor and temporal characteristics, etc.

Building an ontology from scratch is not obvious nor an easy task and requires an expert in the target domain. Another alternative would be to enrich an existing core ontology. The second option seems to be adequate, however the choice of the core ontology is very critical. Thus, we choose to enrich the *SatelliteSceneOntology-V3* because it contains the largest number of concepts relative to satellite scene domain. There are mainly four types of ontology enrichment:

- The conceptual enrichment which consists of adding new concepts to the core ontology

FIGURE 4.4: A subset of the core ontology *SatelitesSceneOntology-V3*FIGURE 4.5: Spatial relations between objects in *SatelitesSceneOntology-V3*

- The relational enrichment that aims to add relations to the core ontology. Relations can link concepts and objects or individuals

- The intentional enrichment which consists of adding a priori values to the ontology concepts
- The axiomatic enrichment that aims to add axioms and rules to help inference of new knowledge

We perform the conceptual and the intentional enrichment using the available resources in order to make it suitable for semantic RS image annotation and retrieval. In order to add new concepts to the ontology, we use the USGS spectral library (Kokaly et al., 2017) jointly with the WordNet lexical database. And to perform the intentional enrichment, we proceed to add feature vectors to the concepts based on the spectral signatures given in the USGS library. We also add representative crops of the concepts using the available hyperspectral images (ground truth).

4.3.1 Conceptual enrichment of the *SatelliteSceneOntology-V3*

As stated earlier, the conceptual enrichment consists of adding new concepts to the core ontology. We use the USGS spectral library (Kokaly et al., 2017) as a knowledge resource and we check the presence of concepts in the WordNet lexical database. WordNet allows us to add definitions, antonyms, synonyms and higher level concepts (father of the concept for example). The USGS spectral library contains lots of information other than the spectral signature values for numerous sensors (Aster, Landsat8, Sentinel2 and WorldView3). It contains descriptions of the concepts, the recording conditions, the sensors used, an image representing the spectral signature plot, etc. We use the last version of the USGS library *splib07b* proposed in 2017. The spectral library is organized into 7 chapters and the reflectance values of a total number of 2450 concepts are given. The vegetation chapter is composed of 286 concepts. The artificial materials chapter contains 290 spectral signatures, the coating chapter contains 12 spectral signatures, the liquids chapter contains 24 spectral signatures, the minerals chapter contains 1276 spectral signatures, the organic compounds chapter contains 360 spectral signatures and finally the soils and mixtures chapter contains 209 spectral signatures.

In order to compute RI, we use the resampled spectra to the ASTER sensor (we can also use any other sensor). Consequently, we have 9 bands for ASTER spectral signature ranging from 0.556 to $2.397\mu m$.

In the USGS library, there are many records that represent the same concept as shown in Figure 4.6 for the "Actinolite" concept presented in the chapter Minerals. We find 19 records with the name "Actinolite" of type "Inosilicate".

From the example stated in Figure 4.6, we will keep only one concept "Actinolite" to be added to the ontology. We use WordNet in order to analyze the addition of concepts from the USGS library (remove duplications).

We start by the mineral chapter since we do not have any mineral concept in the core ontology. We found that the intersection between WordNet and the chapter Minerals is 670 concepts. After removing duplicates (as illustrated by the concept "Actinolite" in Figure 4.6), only 89 concepts are kept. Also, there are 132 concepts that exist in the minerals chapter and not in WordNet.

Consequently, we add $221(89 + 132)$ concepts to the *SatelliteSceneOntology* under the class minerals which is under the root node "Thing". For the 89 common concepts, we add their definitions from WordNet to the ontology as a description of the concepts. We can also add the set of synonyms and antonyms. For the 132 concepts, we use the metadata given in the USGS library. A subset of the enriched ontology is represented in Figure 4.7.

Actinolite	HS116.1B ASDFRb AREF
Actinolite	HS116.2B ASDFRb AREF
Actinolite	HS116.3B BECKb AREF
Actinolite	HS116.3B ASDFRb AREF
Actinolite	HS116.4B ASDFRb AREF
Actinolite	HS22.1B ASDFRb AREF
Actinolite	HS22.2B ASDFRb AREF
Actinolite	HS22.3B BECKb AREF
Actinolite	HS22.3B ASDFRb AREF
Actinolite	HS22.3B NIC4bbb RREF
Actinolite	HS22.4B ASDFRb AREF
Actinolite	HS315.1B ASDFRb AREF
Actinolite	HS315.2B ASDFRb AREF
Actinolite	HS315.4B BECKb AREF
Actinolite	HS315.4B ASDFRb AREF
Actinolite	HS315.4B NIC4bbb RREF
Actinolite	NMNH80714 BECKb AREF
Actinolite	NMNR16485 BECKb AREF
Actinolite	NMNR16485 NIC4bbb RREF

FIGURE 4.6: Illustration of the "Actinolite" concept from the chapter Minerals of the USGS library (Kokaly et al., 2017)

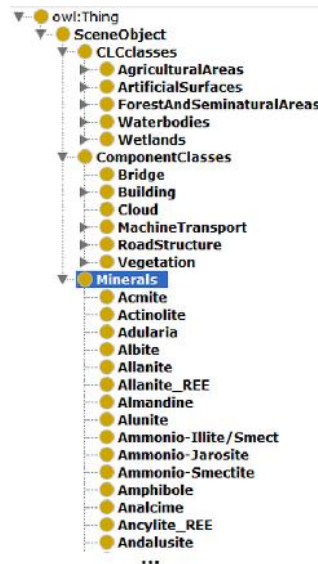


FIGURE 4.7: Subset of the enriched ontology (conceptual) enrichment

We perform the same analysis for all the chapters. Considering the vegetation chapter, we found 181 common concepts between USGS and WordNet. 21 concepts remain after removing duplicates such as *Willow*, *Flower* and *Grass*. There are 54 concepts existing only in the vegetation chapter including *Aspen_Leaf_A* and *Chamise*. Consequently, 75 (21 + 54) concepts are added to the ontology. The analysis of the soil and mixture chapter gives 4 common concepts and 116 concepts

present only in the USGS library. A total of 120 concepts are added to the ontology. There are also 37 common concepts between the organic compound chapter and WordNet and 138 concepts that only exist in the USGS library. Finally, we found 31 common concepts (between the artificial material and WordNet) and 116 concepts that exist only in the artificial material chapter. As a result, we add a total number of 738 concepts to the *SatelliteSceneOntology-V3* as summarized in Figure 4.8.

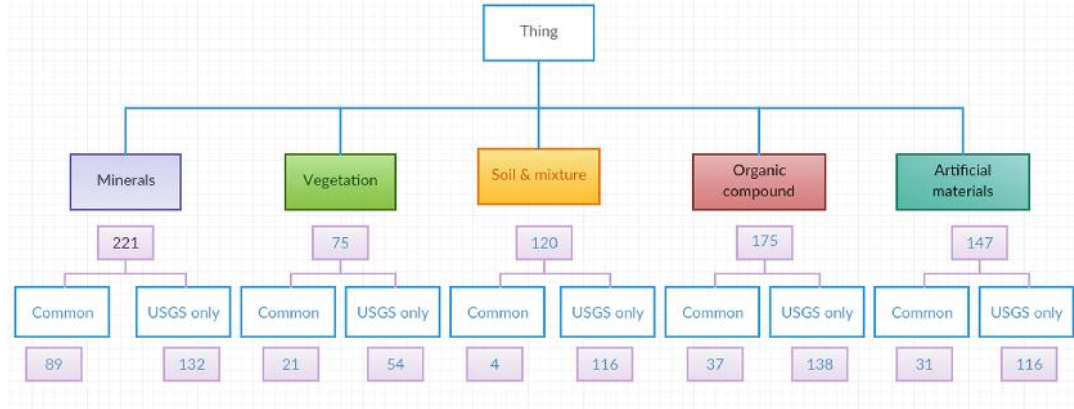


FIGURE 4.8: Summary of the added concepts to the *SatelliteSceneOntology-V3*

4.3.2 Intentional enrichment of *SatelliteSceneOntology-V3*

After performing the conceptual enrichment, we add feature vectors (FVs) to the concepts which corresponds to the RI. In fact, these RI have high discriminative capacities for the classification of hyperspectral datasets as explained in Chapter ?? and allow achieving very interesting accuracies while having a linear complexity. Consequently, we proceed to use RI as FVs of the concepts in the ontology. To compute the RI, we use the USGS spectral library. This type of enrichment is very useful for the semantic annotation of RS images.

We consider several data and object properties. The properties of objects represent the spatial relationships between objects or classes. We define 3 main types of data properties: **i) *hasImage***, **ii) *hasSpectralSignature***, and **iii) *hasRadiometricIndices***.

The property *hasImage* is an URL to one or multiple hyperspectral crops representing the considered concept. These crops are created from the available hyperspectral images according to the ground truth. Actually, we don't have the crops corresponding to all the concepts since we only have a small set of hyperspectral images covering few concepts in the ontology. Table 4.1 shows the HSI images that are used to create the crops to be considered for the enrichment (the concepts illustrated with crops are highlighted in green color).

On the other hand, the property *hasSpectralSignature* represents the spectral signature of the corresponding concept. These values are taken from the USGS spectral library.

As explained in Chapter ??, RIs allow the distinction between vegetation, soil, water and built up areas. Therefore, the *hasRadiometricIndices* property includes the following set of 10 indices: NDVI, RVI, MSAVI, DVI, IPVI, GNDVI, CVI, IC, BI, ISU, NDWI and NDTI. These indices are computed using the Aster spectral signature values and allow to faithfully characterize the concepts. We reduced the number of RI, thus SAVI, MSAVI2 and IR are no longer used. Instead of using SAVI, MSAVI

Dataset	Bands	Size (pixels)	Ground truth
Indian pines	200	145*145	Alfalfa, Corn-notill, Corn-mintill, Corn, Grass-pasture, Grass-trees, Grass-pasture-mowed, Hay-windrowed, Oats, Soybean-notill, Soybean-mintill, Soybean-clean, Wheat, Woods, Buildings-Grass-Trees-Drives, Stone-Steel-Towers
Samson data	156	95 x 95	Soil, tree, water
Jasper ridge	198	100*100	Road, soil, water, tree
Urban data	162	307 x 307	Asphalt, grass, tree, roof
Washington DC MALL	191	1208*307	Roofs, Street, Path (graveled paths down the mall center), Grass, Trees, Water, and Shadow.
University of Pavia	103	610*610	Asphalt, meadows, gravel, trees, painted metal sheets, bare soil, Bitumen, self-blocking bricks, shadows
Cuprite	188	250 x 190	Alunite, Andradite, Buddingtonite, Dumortierite, Kaolinite1, Kaolinite2, Muscovite, Montmorillonite, Nontronite, Pyrope, Sphene, Chalcedony
Botswana	145	1476*256	
Salinas scene	204	512*217	Brocoli_green_weeds_1, Brocoli_green_weeds_2, Fallow, Fallow_rough_plow, Fallow_smooth, Stubble, Celery, Grapes_untrained, Soil_vinyard_develop, Corn_senesced_green_weeds, Lettuce_romaine_4wk, Lettuce_romaine_5wk, Lettuce_romaine_6wk, Lettuce_romaine_7wk, Vinyard_untrained and Vinyard_vertical_trellis
Salinas_A scene	204	86*83	Brocoli_green_weeds_1, Corn_senesced_green_weeds, Lettuce_romaine_4wk, Lettuce_romaine_5wk, Lettuce_romaine_6wk, Lettuce_romaine_7wk

TABLE 4.1: Hyperspectral images used for the intentional ontology enrichment

and MSAVI2, we used only MSAVI since they are highly correlated and reflects the same information. The Redness Index (IR) is not used because we do not have the blue band in the given ASTER signature.

We illustrate the properties of the *Aspen* concept in Figure 4.9.

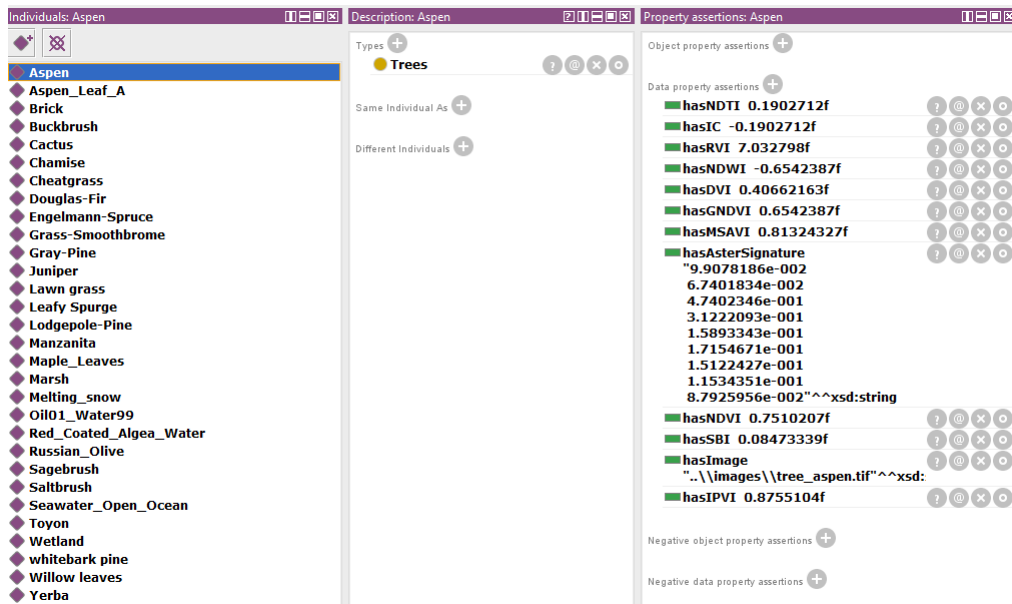


FIGURE 4.9: *Aspen* properties (using the USGS spectral library)

In the ontology, we provide the RI as well as the corresponding spectral signatures in order to allow the definition and the computation of new indices that may

be useful for other tasks. After performing the ontology enrichment, we perform the annotation of RS images as explained in the next section.

4.4 Automatic annotation of RS images

In order to annotate RS images, we perform several steps that are illustrated in Figure 4.10 (Amiri and Farah, 2019).

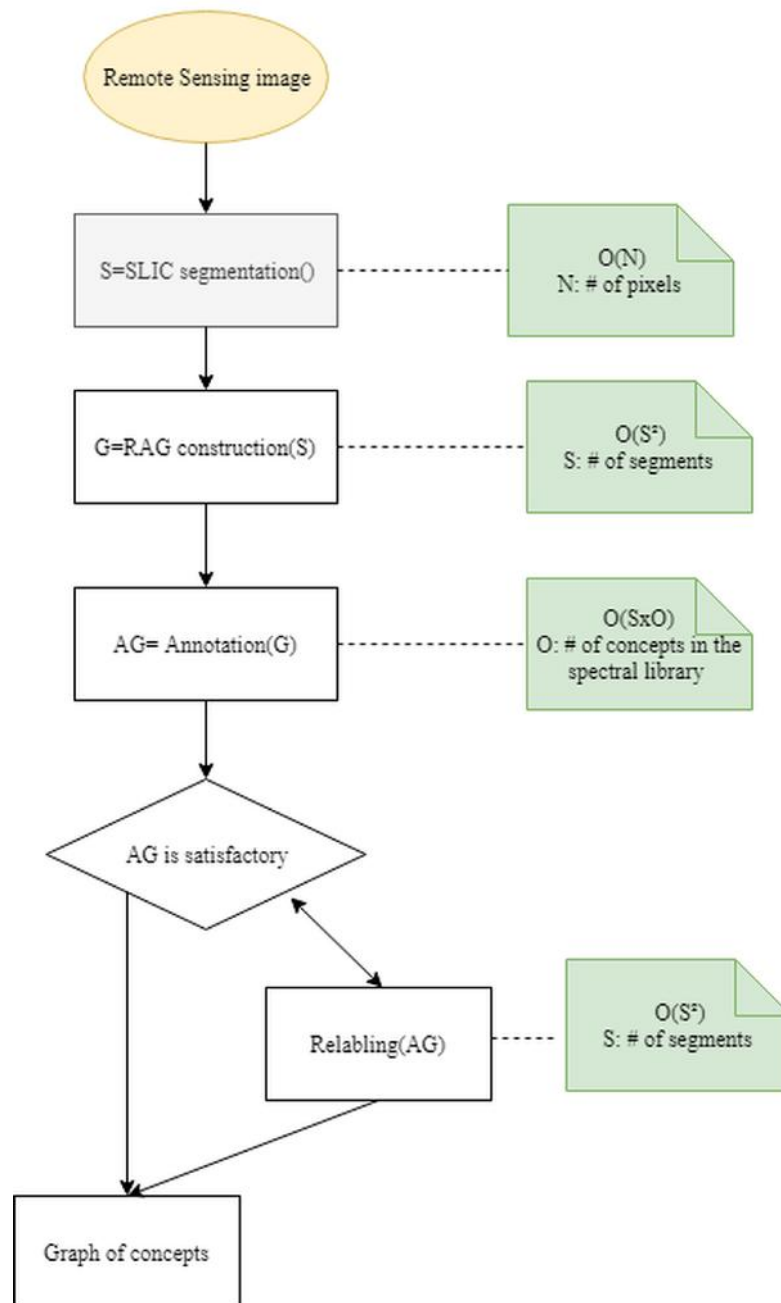


FIGURE 4.10: The flowchart of our annotation approach

For a given RS image, we start by the over-segmentation step. Then, we represent the segmented image by a region adjacency graph (RAG) that contains spatial and spectral attributes. Therefore, we perform the annotation by comparing the FV

of image segments to the FV of concepts of the USGS spectral library. Finally, if the resulting annotation is not satisfactory, we perform relabelling by exploiting the spatial information encoded by the RAG. The annotation is finally displayed as a graph of concepts. We explain each step of the proposed annotation in the following subsections.

4.4.1 Superpixel based segmentation

Algorithm 5 shows the Simple Linear Iterative Clustering (SLIC) proposed by (Achanta et al., 2010) in order to segment an image into a set of K regions called superpixels (SP). The temporal complexity of the SLIC algorithm is linear to the number of pixels, i.e. $O(N)$ where N is the number of pixels in the image.

Algorithm 5 SLIC superpixel segmentation

Input I : a RS image, S : the grid step
Output SP : superpixels resulting from segmentation
Begin
 Initialize cluster centers $C_k = [L_k, a_k, b_k, x_k, y_k]^T$ by sampling pixels at regular grid steps S
 Perturb cluster centers in an $n \times n$ neighborhood to the lowest gradient position
 Repeat
 for each cluster center C_k **do**
 Assign the best matching pixels from a $2S \times 2S$ square neighborhood around the cluster center according to the distance measure
 end for
 Compute new cluster centers and residual error E (L_1 distance between previous centers and recomputed centers)
 $E \leq threshold$
 Enforce connectivity (relabelling disjoint segments)
End

The resultant SPs for an airplane image can be seen in Figure 4.11:

The SPs can be mixed and also homogeneous regions can be split according to the desired size or number of SPs to produce.

4.4.2 The Graph Model

The segmented image produced by the SLIC algorithm (Achanta et al., 2010) is represented by a RAG in which nodes represent segments or SPs and edges reflect the adjacency relation between segments.

Each node has a set of attributes and is linked to its direct neighbors ($N1$) based on the 8-connectivity.

Formally the graph is defined by:

$$G = (V, E) \quad (4.1)$$

Where: $V = \{v_1, v_2, \dots, v_n\}$ is the set of vertices corresponding to regions of the image. Each vertex $v_i, i = 1, \dots, n$, is characterized by a label l_i , a set of spectral descriptors $spec_i$ and a set of spatial descriptors $spat_i$.

l_i corresponds to a concept or keyword assigned to the region i . It is initially set to the value 'unknown' before being matched with the concepts of the USGS spectral library.

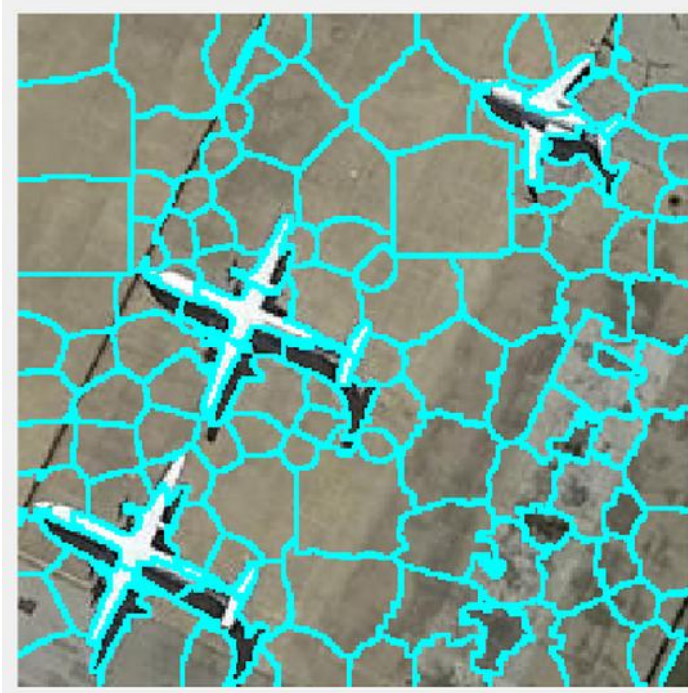


FIGURE 4.11: The SPs generated by SLIC algorithm (Achanta et al., 2010) for an airplane sample image

$spec_i$ defines the RI of the region i presented in Chapter ??.

$spat_i$ describes the spatial descriptors of the region i corresponding to the geographic attributes : the spatial coordinates (x_i, y_i) of the centroid, the area, the perimeter, the miller index, the elongation and the compactness of the region.

$E = \{e_1, e_2, \dots, e_m\}$ represents the set of edges of the graph G . Each two regions i and j are linked only and only if they are adjacent.

We summarize the RAG construction phase in Algorithm 6 .

4.4.3 Annotation of the RS image

After representing an image by a RAG, we have to annotate each region by a semantic concept by matching its low-level features with those computed using the spectral library via a similarity measure. This step will replace the 'unknown' value of the label by the closest concept from the spectral library.

Let C be the set of the concepts of the spectral library O , A represents the set of the considered attributes or features, $a \in A$ be an attribute, $c \in C$ be a concept, $V_{a,c}$ be the value of a for concept c . R is the set of regions, a_i be the value of attribute a for region i , and $w_{a,c}$ be the weight corresponding to the relevance of attribute a to concept c .

We suppose that the values of spectral attributes are normally distributed (Equation 4.2):

$$\eta_{a,c} = (\mu_{a,c}, \sigma_{a,c}) \quad (4.2)$$

Where

$$\mu_{a,c} = \frac{\min_{a,c} + \max_{a,c}}{2} \text{ and } \sigma_{a,c} = \frac{\mu_{a,c} - \min_{a,c}}{2}$$

Therefore, a region v_i with a value a_i upon an attribute a would correspond to the concept c with a credibility $f_{a,c}(a_i)$ (as illustrated by Equation 4.3):

Algorithm 6 RAG construction

```

Input  $R$  : set of regions
Output  $GL$  : RAG
Begin
label = "unknown"
 $G = \text{emptyGraph}()$ 
for  $i \leftarrow 1 : |R|$  do
     $Spat_i = \text{computeSpatialDescriptors}(v_i)$ 
     $Spec_i = \text{computeRadiometricIndices}(v_i)$ 
     $v_i = \text{buildNode}(spat_i; spec_i; label)$ 
     $\text{addNode}(G, v_i)$ 
end for
for  $i \leftarrow 1 : |R| - 1$  do
    for  $j \leftarrow (i + 1) : |R|$  do
        if  $\text{adjacent}(v_i, v_j)$  then
             $e_j = \text{buildEdge}(v_i, v_j)$ 
             $\text{addEdge}(G, e_k)$ 
        end if
    end for
end for
End

```

$$f_{a,c}(a_i) = \frac{1}{\sigma_{a,c}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{a_i - \mu_{a,c}}{\sigma_{a,c}}\right)^2} \quad (4.3)$$

Then, we compute a similarity score $\text{sim}(v_i, c)$ between the region v_i and the concept c in the spectral library as follows (Equation 4.4):

$$\text{sim}(v_i, c) = \frac{\sum_{a \in A} w_{a,c} f_{a,c}(a_i)}{\sum_{a \in A} w_{a,c}} \quad (4.4)$$

Finally, we annotate each region with the most suitable concept from the spectral library. We choose the concept having the highest similarity score (Equation 4.5):

$$\text{Concept} = \arg_max(\text{sim}(v_i, c)) \quad (4.5)$$

The annotation of the image by concepts is described by Algorithm 7.

Algorithm 7 Annotation algorithm

```

Input  $G$ : graph  $O$ : spectral library
Output  $AG$ : Annotated graph
Begin
 $AG = G$ 
for  $i \leftarrow 1 : n$  do
    for  $j \leftarrow 1 : \text{size}(O)$  do
         $\text{Concept} = \arg\_max(\text{sim}(v_i, c))$ 
    end for
     $\text{updateNodeLabel}(AG, v_i, \text{Concept}, \text{sim}(v_i, \text{Concept}))$ 
end for
End

```

4.4.4 Spatial information

The spatial information is very important for understanding image semantics. Lots of formulations were proposed in order to consider them. We can cite the 2D topological relations of the RCC8 (Cohn et al., 1997) depicted in 4.12 and the fuzzy spatial ontology proposed in (Bloch et al., 2007) shown in Figure 4.13.

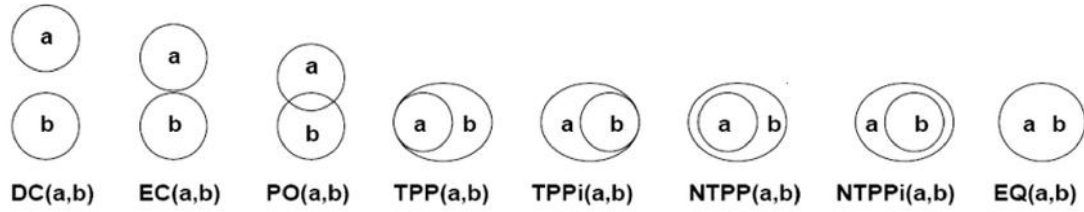


FIGURE 4.12: Illustrations of topological relations of RCC-8 (Cohn et al., 1997)

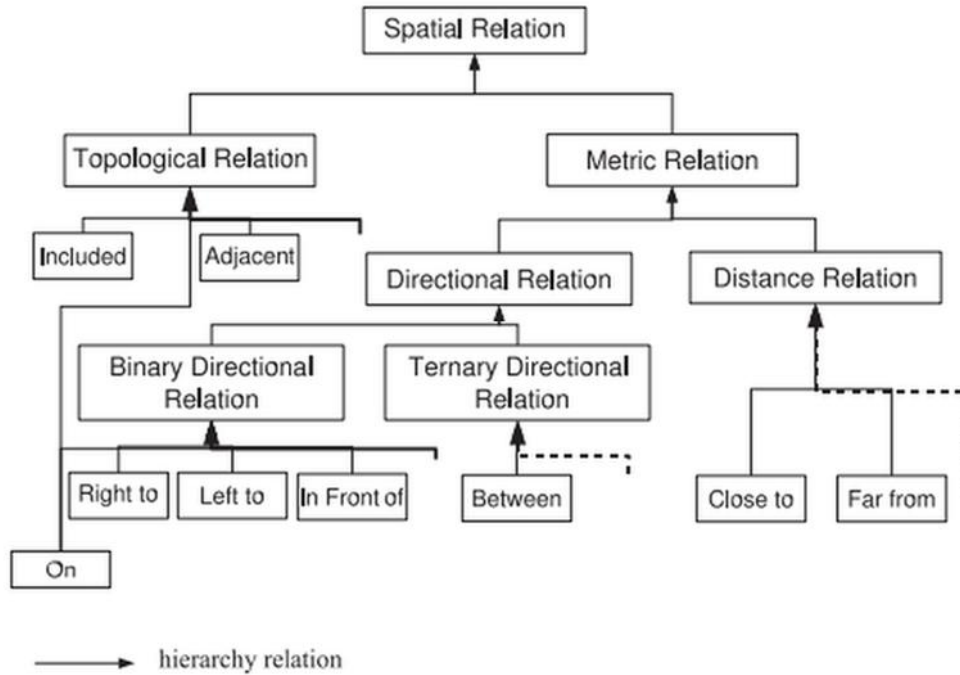


FIGURE 4.13: Excerpt of the hierarchical organization of spatial relations (Bloch et al., 2007)

In order to infer the directional relations based on the centroid values stored in graph nodes, we use the angle α with the x-axis as proposed by (Miyajima and Ralescu, 1994). This is depicted in Figure 4.14 and illustrated by Algorithm 8.

4.4.5 Refinement of the annotation results

In order to smooth the annotation produced by Algorithm 7, we proceed to relabel regions by considering the labels of their adjacent regions. Since, we oversegment the image into a set of superpixels (SPs), the adjacent segments or SPs tend to share the same labels. So before merging them (the adjacent regions sharing the same

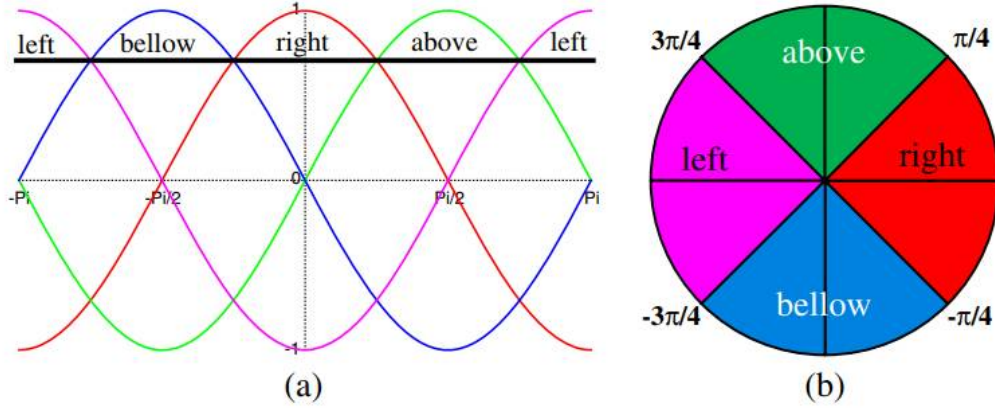


FIGURE 4.14: Directional relationships are computed according to an angle α with the x-axis (Miyajima and Ralescu, 1994)

Algorithm 8 Directional relationships

Input (x_i, y_i) and (x_j, y_j) : coordinates of the centroids of regions v_i and v_j

Output Left-to, Right-to, Above, Bottom: binary relations

Begin

$\alpha = \text{ComputeAngle}(X_{axis}, \text{segment of centroids of } v_i \text{ and } v_j)$

Left-to, Right-to, Above, Bottom = false

if $\alpha \in [\frac{\pi}{4}, \frac{3\pi}{4}]$ **then**

 Above = True // v_i is above v_j

else if $\alpha \in [-\frac{3\pi}{4}, -\frac{\pi}{4}]$ **then**

 Below = True // v_i is below v_j

end if

if $\alpha \in [-\frac{3\pi}{4}, \frac{3\pi}{4}]$ **then**

 Left-to = True // v_i is left to v_j

else if $\alpha \in [\frac{-\pi}{4}, \frac{\pi}{4}]$ **then**

 Right-to = True // v_i is right to v_j

end if

End

label), we proceed to perform a relabeling step based on the adjacency relation between segments and on the difference of similarity values. The relabeling step is summarized in Algorithm 9.

We update the labels based on two factors: the adjacency relation between the segments and the difference of their similarity values.

As shown in Algorithm 9, in order to update the labels of the regions, we define the **confidence node** as the node having the highest similarity value among the set of nodes (thus we are sure of its label). We mark the confidence node and check its direct neighbors. If all the neighbors share the same label with the confidence node, then we keep the actual labels. However, if the neighbors have different labels, then we compare the confidence node to the best neighbor defined as the neighboring node that has the highest similarity value that we call the **local confidence node**. Thereafter, we compare the absolute difference of similarity values (confidence node, local confidence node) to a threshold h and we update the labels of the neighbors accordingly.

Algorithm 9 Relabing Algorithm

Input h : threshold $AG = (V, C, Sim)$: Annotated Graph resulting from Algorithm 7

Output $updated_labels$: vector of updated labels

Begin

$TAG = AG$ // temporary variable

while ($TAG \neq \emptyset$) **do**

$sorted_nodes = sort_nodes(TAG, Sim)$

$confidence_node = sorted_nodes[1]$

$N1 = neighbors(TAG, confidence_node)$

$sorted_neighbors = sort(N1, 'ascend')$

$lcn = sorted_neighbors[1]$ //lcn stands for local confidence node

$labels = label_neighbors()$

if ($\#labels > 1$) **then**

for $i \leftarrow 1 : size(N1)$ **do**

if ($label(confidence_node) \neq label(N1(i))$ and ($sim(confidence_node) - sim(N1(i)) \leq Th1$) **then**

$label(N1[i]) = label(confidence_node)$

end if

end for

end if

$TAG = TAG \setminus N1$ // delete all the neighbours

end while

End

4.5 Complexity of the Proposed Algorithm

To perform the over-segmentation of an image, we use the SLIC algorithm which has a linear complexity $O(N)$, where N is the number of pixels in the image. Then, we construct the RAG using two nested loops over the segments, so if we set S to be the number of segments found using SLIC algorithm, then the complexity of the computation will be $O(S^2)$. Once the graph is constructed, we proceed to the annotation step. Thereby, we compare the feature vectors of the segments S to those of the spectral library and we have a complexity of $O(S \times C)$ where C is the number of concepts in the library. Finally, we relabel the graph nodes. The maximal degree in the graph is 8, thus the complexity of relabeling algorithm is $O(S^2)$. Hence the temporal complexity is $(O(N) + O(S^2) + O(S \times C)) = O(S \times C)$. In general, the number of concepts in the spectral library is bigger than the number of segments in the image.

4.6 Experimental Results: RAG-based Annotation of HSI and MSI RS Images

4.6.1 Datasets

In order to test the effectiveness of our approach, we use a set of available multispectral and hyperspectral data. Five HSI are used which are *Jasper Ridge*, *Urban* dataset, the *Samson* dataset, the *Indian Pines* dataset and the *Cuprite* dataset. We also use of subset of five multispectral images from Zurich dataset.

Jasper Ridge is a popular hyperspectral image (Figure 4.15) used in the literature. In this paper, we only consider a sub-image from *Jasper Ridge* of size 100×100 pixels. There are four end-members in this data: *Road*, *Soil*, *Water* and *Tree*.

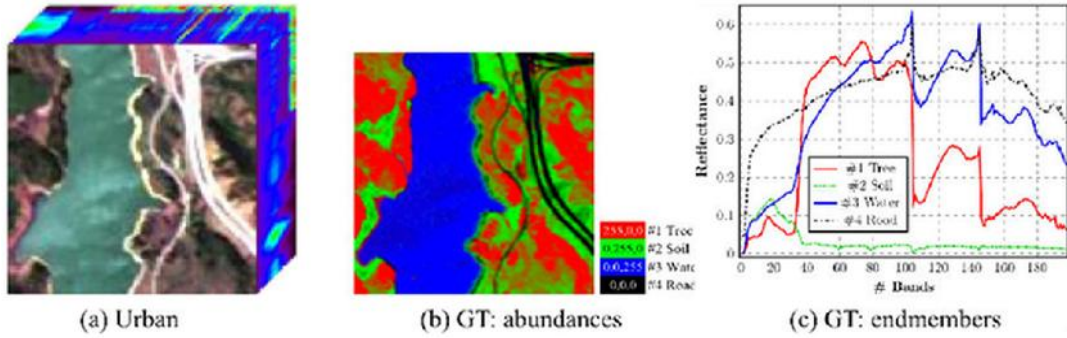


FIGURE 4.15: The Jasper Ridge data and its ground truth

The Indian Pines data is also a largely used HSI which is already presented in the Chapter ?? by Figures 3.2 and 3.3.

The urban dataset, on the other hand, is a HSI generally used for unmixing task. It contains 307×307 pixels and 210 channels ranging from 0.4 to $2.5 \mu m$ resulting in a high spectral resolution of $10nm$. The pre-processing of the Urban dataset removes 48 channels (1 – 4, 76, 87, 101 – 111, 136 – 153 and 198 – 210) and consequently 162 channels remain. There are three ground truth versions of the Urban dataset with 4 ("Asphalt", "Grass", "Tree" and "Roof"), 5 ("Asphalt", "Grass", "Tree", "Roof" and "Dirt") and 6 ("Asphalt", "Grass", "Tree", "Roof", "Metal", and "Dirt") endmembers. The ground truth with 4 endmembers is illustrated in Figure 4.16.

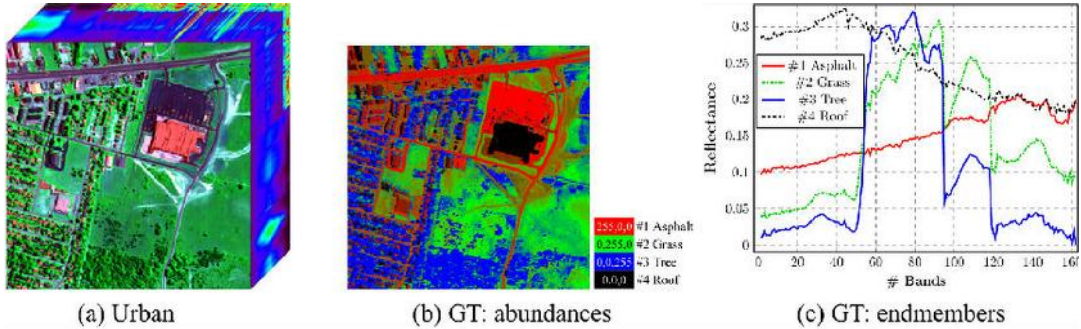


FIGURE 4.16: The Urban dataset its ground truth (with 4 endmembers)

The Cuprite dataset is acquired over the Cuprite site in Las Vegas, USA. It contains 224 channels from 0.37 to $2.48 \mu m$. The removal of water absorption bands gives us 188 bands. The considered region has a size of 250×190 pixels and contains 12 classes of minerals including "Alunite", "Andradite", "Buddingtonite", "Dumortierite", "Kaolinite1", "Kaolinite2", "Muscovite", "Montmorillonite", "Nontronite", "Pyrope", "Sphene" and "Chalcedony".

We also consider a subset of the Samson dataset. It has a size of 95×95 pixels from an original image of size 952×952 pixels. The Samson data contains 156 channels ranging from $401 nm$ to $889 nm$ resulting in a very high spectral resolution of

3.13 nm. This dataset contains 3 classes which are "Tree", "soil" and "water" as shown in Figure 4.17.

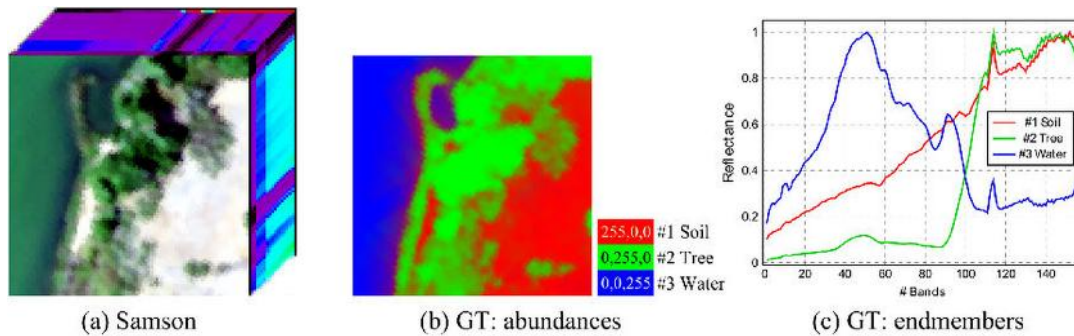


FIGURE 4.17: The Samson dataset its ground truth

We also test on a subset of the Zurich dataset which is composed of 20 crops acquired by the QuickBird sensor over the city of Zurich, Switzerland in August 2002. The images are composed of 4 channels (blue, green, red and near infrared) and they include 8 different classes namely "road", "building", "tree", "grass", "bare soil", "water", "railways" and "swimming pools".

4.6.2 Annotation Results

In order to evaluate our approach and to compare it with the state-of-the-art methods, we use the precision (P), Recall (R) and $F1$ and $N+$ scores already introduced in Chapter ?? in subsection 3.3.2 with equations 3.1, 3.2 and 3.3. The $N+$ score is the number of labels that are correctly assigned to at least one test image (in other words, the number of labels with recall greater than zero).

We summarize the annotation results in Table 4.2 on the set of the datasets.

Dataset/ measure	P	R	F1	N+	m_1	m_2
Indian pines	0.33	0.06	0.1	1	16	"Lawn Grass", "decane", "Polyester Pile"
Samson data	0.66	0.66	0.66	2	3	"Decane", "Lawn Grass", "Cactus"
Jasper Ridge	0.57	1	0.73	4	4	"Water+ Montmor", "Malachite", "Blue Efflorscent Min", "Chrysocolla", "Verdigris", "Bare Red Brick", "Pink Quartzite"
Urban data	0.6	0.75	0.66	3	4	"Bare soil", "Grass", "Conifer-Meadow-Mix", "Aspen leaf", "Chrysocolla"
Zh1	1	0.37	0.54	3	8	"Tree", "Bare soil", "Grass"
Zh2	0.5	0.25	0.33	1	4	"Bare soil", "Tree"
Zh3	1	0.28	0.44	2	7	"Water", "Tree"
Zh18	1	0.33	0.4	2	6	"Bare soil", "Tree"
Zh20	1	0.4	0.57	2	5	"Bare soil", "Grass"

TABLE 4.2: Annotation results of our method on all the datasets used

For the *Jasper Ridge* data, we consider 'Chrysocolla', 'Malachite' and 'Pink Quartzite' which are 3 types of rocks as correct labels since we do not have the concepts 'road' and 'soil' in the USGS spectral library.

For the *Samson* data, we consider 'cactus' and 'law grass' as correct labels since they represent vegetation concepts and we do not have the tree concept neither in the USGS library nor in the Aster library.

For the *Indian Pines* dataset (in the first row of Table 4.2), we notice that the F1 score (0.1) is considerably low. This can be explained by the fact that we do not have almost all the concepts of the Indian Pines ground truth in the USGS library, so we cannot identify the classes in the *Indian Pines* image. We can make a correspondence between labels in the ground truth and labels in the spectral library so that 'law grass' for example could be considered as a correct predicted label. Another reason could be the low discriminative power of the RI as explained in Chapter ?? in subsection 4.6.4.

For the *Zurich* images (Zh1, Zh2, Zh3, Zh18 and Zh20), we compare the FVs of the segments produced by SLIC algorithm to the FV of the eight classes. The FV of the eight classes is composed of the intervals of values of 11 RI (NDVI, RVI, MSAVI, DVI, IPVI, GNDVI, IR, IC, SBI, NDWI and NDTI). These indices are computed for each class based on the ground truth of the Zurich dataset. Then, in order to annotate image regions, we compare the FV of the regions to those that are already computed and we assign the closest concept to each region. We are not using the USGS or the Aster spectral libraries because we do not have the reflectance values of the Zurich data. Thus, we cannot compare the FV of the radiance values extracted from the Zurich images to the RI computed based on the reflectance values of the spectral libraries.

Considering the spatial relationships between regions to update the labels assigned to image region (based on Algorithm 9) will not update the results in terms of precision, recall, F1 and N+ scores. However, the spatial coordinates of labels could change. So, if the merged regions are homogeneous then we consider that the relabeling algorithm enhances the results and gives a more faithful annotation of the image.

As we can notice in Figure 4.18, the annotation results on the *Jasper Ridge* data are enhanced. Thus, homogeneous regions tend to be merged after the relabeling phase.

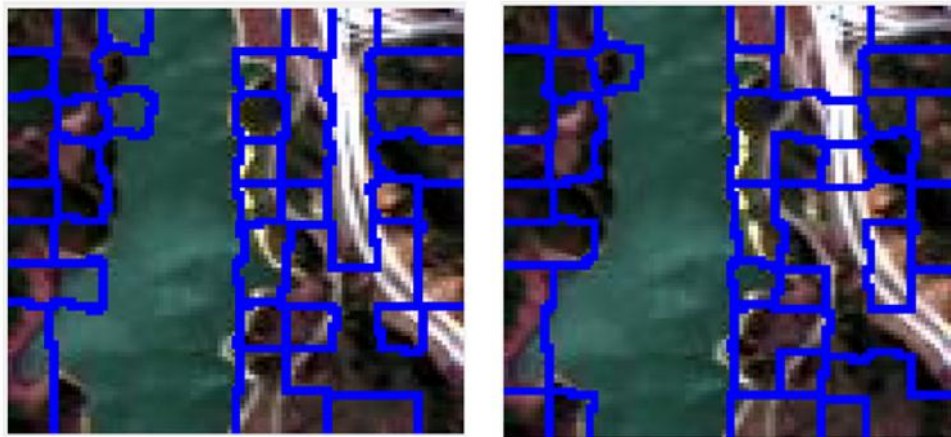


FIGURE 4.18: The original annotation result (left) and the updated regions (right) of the *Jasper Ridge* dataset (using Algorithm 4.2)

The graph of concepts is represented by Figure 4.19 and detailed in Table 4.3. It shows the spatial distribution of the labels on the *Jasper Ridge* data and hence constructs a very intuitive representation of image semantics.

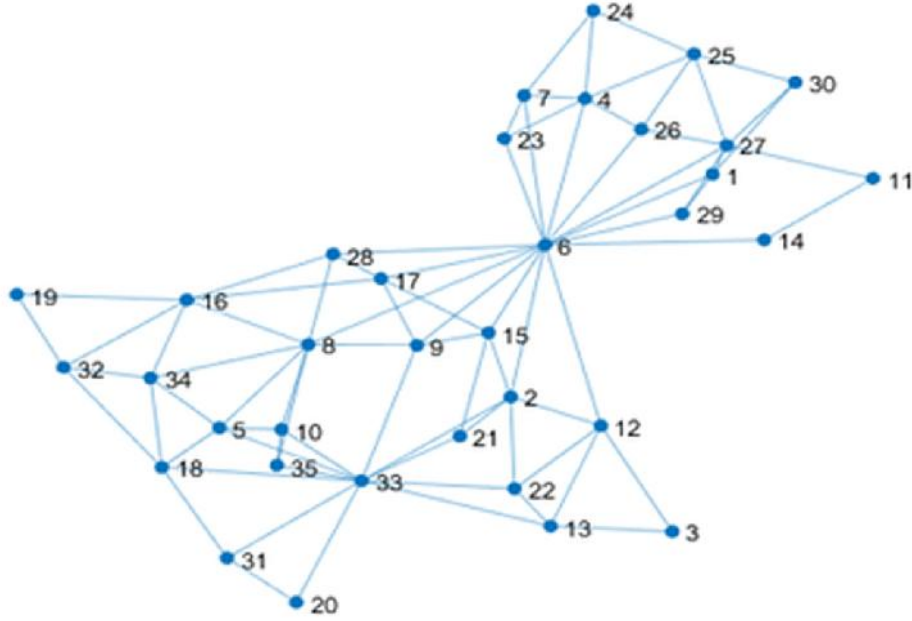


FIGURE 4.19: The associated graph with the annotation produced by our approach for the *Jasper Ridge* dataset

Concept	Node(s)
Water+ Montmor	6, 25
Malachite (several sub types)	30, 23, 3, 14, 7, 11, 28, 29, 30
Chrysocolla	27, 2, 4, 10, 15, 16, 17, 20, 26, 34
Verdigris	8, 18, 19, 22, 33
Bare Red Brick	12, 13, 9, 21, 31, 32, 35
Pink Quartzite	5
Blue Efflorescent Min	1, 24

TABLE 4.3: Spatial arrangement of labels in the *Jasper Ridge* data

This can be illustrated on the *Jasper Ridge* data as shown in Figure 4.20

For the *Jasper Ridge* data, there are many subtypes of 'Malachite' and 'Chrysocolla'. Thus if we merge the adjacent regions having these labels, we can have the graph of Figure 4.21 corresponding to the concepts existing in this dataset as well as their relations. The representation of the annotation of the RS image is very intuitive and can help us make deductions visually that can be investigated using the spatial attributes (area and perimeter) and the spatial relationships between the regions.

4.6.3 Comparison with the-state-of-the-art Methods

Most state-of-the-art annotation methods do not produce a label per region so we cannot compare directly our results to them. Nevertheless, we can compare our results to the recent work of (Verma and Jawahar, 2017) . We report in Table 4.4 the

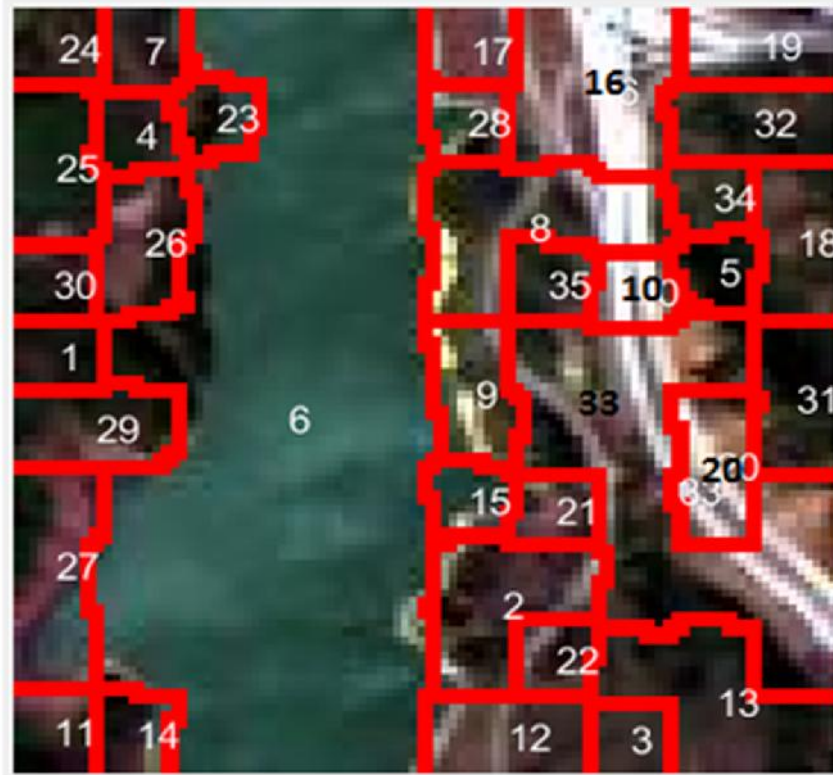


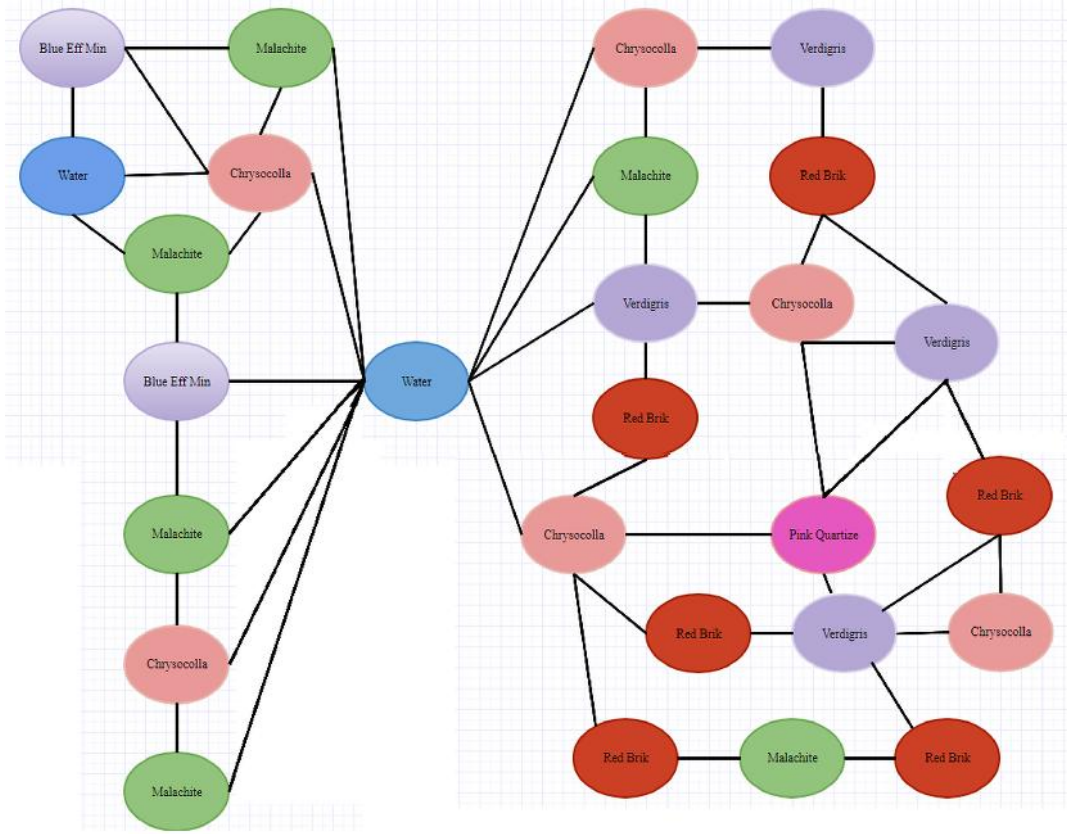
FIGURE 4.20: The spatial arrangement of labels on the *Jasper Ridge* image

comparison that they obtained between their method (2 Pass K Neighrest Neighbours 2PKNN) and other state-of-the-art methods. In order to compare the methods, they use the already mentioned evaluation metrics namely precision (P), recall (R), $F1$ and $N+$ scores. The evaluation metrics are placed in columns and the state-of-the-art methods in rows. The datasets used are also mentioned which are Corel-5k, ESP-Game and IAPR-TC12. The best results are highlighted in bold in Table 4.4. We notice that we get good results compared to the best ones shown in the comparative study reported in Table 4.4. The 2PKNN of (Verma and Jawahar, 2017) get the best $F1$ scores of 51.9% on the MIRFlickr-25K dataset, 49.9% on the Corel-5K dataset, 41.1% on the ESP-GAME dataset and 45.1% on IAPR-TC12 dataset. However, the best $F1$ score that we obtained is 73% on the *Jasper Ridge* dataset.

Also, it is worth noting that our approach may have low performance when the spectral signatures of the classes in the image are too similar. Another issue that we intend to address is the granularity level of concepts of the spectral libraries. In fact, the concepts in the USGS and Aster libraries are so thin, so we might need to abstract the concepts in order to be able to annotate the image with better keywords. There are also some lacking concepts in the spectral libraries that we used. This can also limit the performance of our approach.

4.6.4 Discussion

The existing annotation methods gave, generally, keywords related to the whole image and do not represent the spatial relations between the concepts even if they are sometimes considered in the processing. In our method, we exploit the spatial

FIGURE 4.21: Graph of concepts associated to the *Jasper Ridge* data

Dataset/ Method	Corel-5k				ESP-Game				IAPR-TC12			
	P	R	F1	N+	P	R	F1	N+	P	R	F1	N+
CRM	16	18	17.4	107	-	-	-	-	-	-	-	-
MBRM	24	25	24.5	122	18	19	18.5	209	24	23	23.5	223
InfNet	17	24	19.9	112	-	-	-	-	-	-	-	-
NPDE	18	21	19.4	114	-	-	-	-	-	-	-	-
SML	23	29	25.7	137	-	-	-	-	-	-	-	-
TGLM	25	29	26.9	131	-	-	-	-	-	-	-	-
JEC	27	32	29.3	139	23	19	20.8	227	25	16	19.5	196
MRFA	31	36	33.3	172	-	-	-	-	-	-	-	-
CCD(SVRMKL + KPCA)	36	41	38.3	159	36	24	28.8	232	44	29	35.0	251
GroupSparsity	30	33	31.4	146	-	-	-	-	32	29	30.4	252
BS-CRM	22	27	24.2	130	-	-	-	-	24	22	23.0	250
Random Forest	29	40	33.6	157	41	26	31.8	235	44	31	36.4	253
KSVM-VT	32	42	36.3	179	33	32	32.5	259	47	29	35.9	268
TagProp-SD	30	33	31.4	136	48	19	27.2	212	50	20	28.6	215
TagProp-q SD	28	35	31.1	145	39	24	29.7	232	41	30	34.6	259
TagProp-ML	31	37	33.7	146	49	20	28.4	213	48	25	32.9	227
TagProp-q ML	33	42	37.0	160	39	27	31.9	239	46	35	39.8	266
Fast Tag	32	43	36.7	166	46	22	29.8	247	47	26	33.9	280
SKL-CRM	39	46	42.2	184	41	26	31.8	248	47	32	38.1	274
SVM-DMBRM	36	48	41.1	197	55	25	34.4	259	56	29	38.2	283
2PKNN	48	50	49.0	190	49	35	40.8	255	53	39	44.9	279
2PKNN+ML	48	52	49.9	196	48	36	41.1	255	50	41	45.1	275

TABLE 4.4: Comparative results of 2PKNN and the state of the art approaches (Verma and Jawahar, 2017)

information during the annotation and also in the representation of the resulting annotation by drawing a graph of concepts (representing adjacent concepts in the

image). We also use the spectral information via the use of radiometric indices. The annotation task is very challenging, and even if the results are encouraging, there are several questions to deal with such as the computation of the FVs of concepts that do not exist in the USGS library. Thus, we used the USGS library for the annotation and not the enriched ontology because we cannot compare the FVs of image segments to concepts of the ontology that are not having FVs. An aggregation step of the FVs should be carried on in order to assign FVs to the father concepts based on the FVs of their children. Another question would be how can we exploit the produced graph of concepts in order to propose a semantic textual description of the RS image.

4.7 Conclusion

In this chapter, we tackled the problem of AIA. Unlike other methods, we do not make a separation between image annotation and image representation. Motivated by the promising performance of graph-based methods we proposed a graph of concepts for semantic annotation of hyperspectral and multispectral images using USGS and ASTER spectral libraries and radiometric indices. The proposed graph allows to represent regions of the image with their labels, their spatial and spectral attributes together. This gives a more detailed representation of the relationships between regions and labels and thus helps the understanding of image semantics. The experimental results on the-state-of-the-art of HSI datasets show promising performances. In order to explore the potential of our proposed graph of concepts, we test a modified version of our graph in the classification of large scale RS datasets (such as PatternNet and NWPU-RESISC45) that will be described in Chapter 5.

Chapter 5

Bag of Visual SubGraphs for the Classification of large scale RS scenes

5.1 Introduction

Image classification is very important (is involved in many applications) and constitutes an effective way to organize images. Aerial RS images are recently becoming available and several datasets were collected and freely accessible such as NPWU-RESISC45 and PatternNet datasets. To classify these images, our proposed approach of Chapter 4 cannot be applied because we cannot compute the radiometric indices since they contain only three bands (R,G,B). Therefore, the classification of RS scenes is very challenging due to two main reasons: the inter-class similarity and the intra-class dissimilarity. The inter-class similarity reflects the case when different classes share similar characteristics such as having the same color or the same texture. However, the intra-class dissimilarity is when we have different characteristics for images belonging to the same class. These two challenges are very important especially for RS images and make the classification challenging.

In order to classify these images, we should focus on the spatial information and low level features. However, unlike the HSI, the spectral information is not very rich. Thus, we present in this chapter our approach for ensuring a reliable classification of RS scenes with a reduced temporal complexity compared to machine learning based approaches. We mainly use a RAG to represent the segmented image and then we mine the frequent subgraphs and use them for the classification. Applying our method, on three datasets of RS scenes, we got very interesting accuracies. We describe our approach in detail in the next sections.

5.2 Proposed Classification approach

We propose a new approach called BoVSG (Bag of Visual SubGraphs) for the classification of RS scenes. It begins with superpixel extraction using the SLIC algorithm presented in Chapter 3 (Algorithm 5). This segmentation leads to the representation of the image with much small number of elements (superpixels) while retaining most of the information. In the second phase, a feature vector is computed for each superpixel based on the colour and texture descriptors. Then, the resulting feature vectors are clustered using the k-means algorithm. Each cluster is supposed to correspond to a land cover type or a scene class. Therefore, each image is represented by a graph where each node corresponds to a superpixel and the neighbourhood information is represented by edges. Afterwards, each image is represented by the

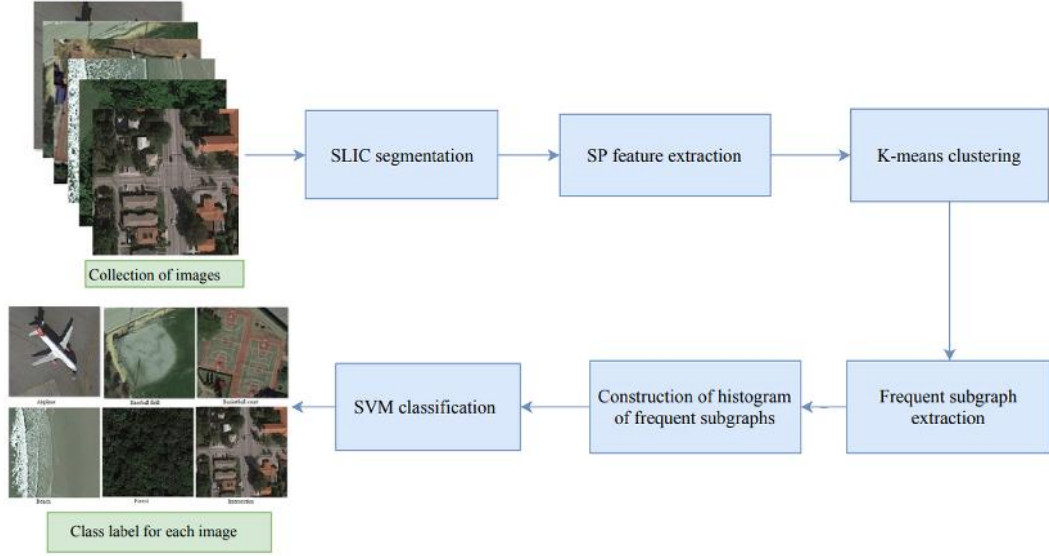


FIGURE 5.1: Flowchart of the proposed BoVSG approach

histogram of frequent subgraphs in order to capture frequent patterns in some scene classes. The flowchart of the BoVSG is illustrated by Figure 5.1 and the details of each step is explained in the following sub-sections.

5.2.1 Over Segmentation using the SLIC Algorithm

This is the first step in the BoVSG approach. The segmentation step is very important and can greatly influence the performance of classification methods. Consequently, we proceed to over-segment the image using the SLIC algorithm (Achanta et al., 2010) which is summarized in Chapter 4 by Algorithm 5. We keep using the SLIC algorithm because its temporal complexity is low and produces the escompted results. An illustration of the resulting segmentation of the SLIC algorithm on a 'dense residential' sample image is shown in Figure 5.2. Thereafter, low level features of the resulting superpixels are extracted (Subsection: 5.2.2).

5.2.2 Feature Extraction

After segmenting the image into a set of SPs, we extract the low level features of each superpixel. We compute colour moments including mean, standard deviation and entropy of red, green and blue channels of the SPs. Color moments can be used in order to distinguish images based on their colour features. We compute the mean using Equation 5.1 which reflects the average color of the SP.

$$E_i = \sum_{j=1}^N \frac{P_{ij}}{N} \quad (5.1)$$

where N is the number of pixels in the SP, P_{ij} is the value of pixel j at the i^{th} color channel and i is either R for red, G for Green or B for blue.

The standard deviation of each RGB channel is computed using Equation 5.2. The standard deviation is obtained using the square root of the variance of the color distribution.



FIGURE 5.2: The SPs generated by the SLIC algorithm (Achanta et al., 2010) for a dense residential image

$$\sigma_i = \sqrt{\left(\frac{1}{N} \sum_{j=1}^N (P_{ij} - E_i)^2\right)} \quad (5.2)$$

where E_i is the first color moment (the mean value) of each channel in the RGB color space.

Finally, we compute the entropy moment S of each RGB channel based on Equation 5.3 where I is a grayscale image, p the histogram of counts.

$$S(I) = - \sum p * \log_2(p) \quad (5.3)$$

The entropy is a statistical metric that measures the randomness of the grayscale image I and that can be used to characterize the texture of the image.

Then, we compute the texture descriptor by calculating Gabor filters. This is done using 8 orientations and 6 scales and produces a feature vector of size 40. Thus, we create a feature vector, containing the color moments and the Gabor features of the SPs of the input image. The FVs are normalized to zero mean and unit variance and will be used in the clustering of similar patches of the image.

5.2.3 Clustering of the Feature Vector

The clustering of the computed FVs of the SPs is based on the *k-means* algorithm proposed by (MacQueen, 1967). The *k-means* algorithm allows to cluster data into a given number of clusters (k should be fixed) by the definition of k centroids corresponding to the k clusters. The locations of the k centroids should be well chosen and it would be better to choose them as far as possible from each other. Then, each data point is assigned to the nearest centroid resulting in a first clustering. Therefore, the

k centroids are re-computed based on the barycenters of the clusters obtained in the previous step. The data points are then re-assigned to the new closest centroids. At each step, the locations of the k centroids change iteratively until convergence (no changes in the centroids' locations). K-means minimizes the squared error objective function defined by Equation 5.4:

$$\sum_{i=1}^k \sum_{x \in C_i} \|x - \mu\|^2 = \operatorname{argmin} \sum_{i=1}^k |C_i| \operatorname{Var} C_i \quad (5.4)$$

Where $(x_1, x_2, \dots, x_n)$ is the set of observations and each observation is a vector of dimension d , $C = \{C_1, C_2, \dots, C_k\}$ is the set of k clusters and μ_i is the mean of data points in the cluster C_i .

Therefore, having as input, the FV of the SPs of all the images in the dataset and a fixed clusters number k , the k-means algorithm gives the set of k clusters.

5.2.4 Graph Representation of Image Superpixels

In the previous step (subsection 5.2.3), the FVs of all the SPs from the dataset of images are clustered using the k-means algorithm. We expect that the SPs sharing the same or similar material/land cover tend to be clustered together. As a result, we have a label per SP, although we do not explicitly know what land cover it corresponds to. Then, each image is represented by an adjacency graph of SPs $G = (V, E)$ where V represents the set of vertices corresponding to the SPs in the image and E represents the set of edges. The edges E reflect the adjacency relation between the SPs of the image based on the eight-connectivity (as explained in Chapter 4).

Using the SP extraction and the SP clustering steps, we managed to create a much concise representation of the image in the form of a graph that summarizes the structure of the image. Thus, the land cover types in small patches and their geometrical relationship are captured. The corresponding graph for the 'dense residential area' sample image of the Figure 5.2 is shown in Figure 5.3.

The node labels or numbers shown in Figure 5.3 correspond to the cluster numbers found using the k-means algorithm. At this stage, we create an adjacency graph of SPs as output. Then, we mine the frequent subgraphs (in Subsection 5.2.5) in order to use them to classify RS scenes.

5.2.5 Frequent Subgraph Mining (Maximal Cliques)

In order to generate candidate subgraphs (SGs), we use a maximal clique-based method. Hence, in each image, we search for the maximal cliques. A clique is defined as a subset of vertices of an undirected graph such that every two distinct vertices in the clique are adjacent, thus forming a complete subgraph. In the over-segmented image, already represented as an adjacency graph of SPs, a clique of size two carries few information. On the other hand, we rarely find cliques of size larger than three. In fact, cliques of size three is the case that occurs the most in the over-segmented images. Thus our generated subgraphs are of size three. We generate all the cliques from each image (without duplication) based on the labels of the SP using the Bron-Kerbosch algorithm (Bron and Kerbosch, 1973) which is recursive and backtracking. It gives as output a matrix A of size $V \times N$ where V is the number of vertices and N the number of distinct cliques found, where $A_{ij} = 1$ if SP_i belongs to clique C_j . From the maximal clique matrix A , we create another matrix A' of size $N \times 3$ where A'_{i1} , A'_{i2} and A'_{i3} contain the first, second and third node numbers belonging to clique i . Then we replace in A' node numbers with node labels which are

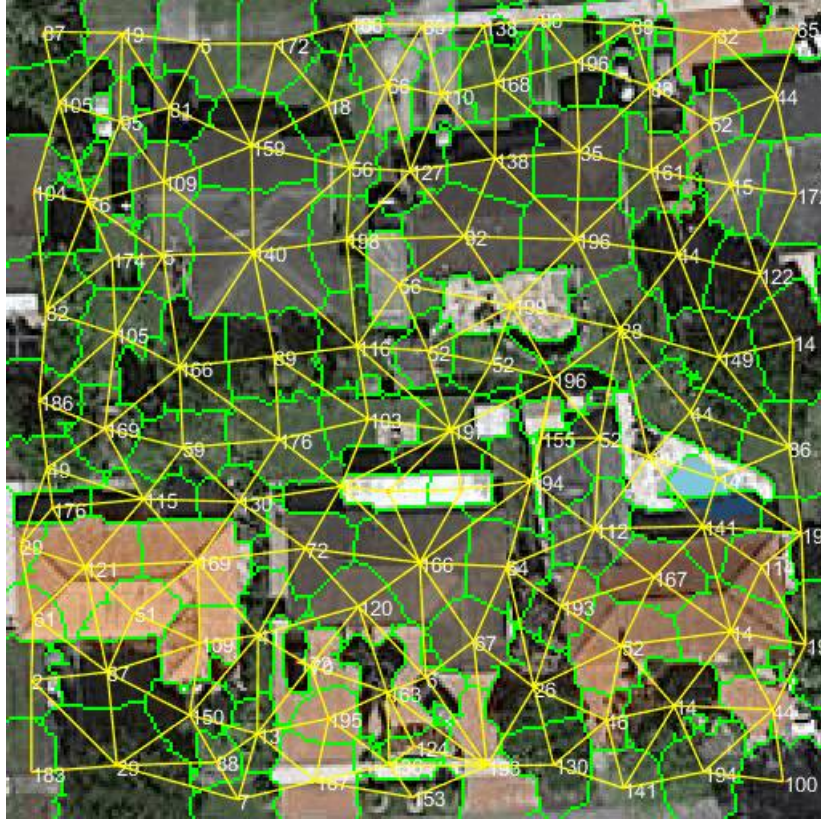


FIGURE 5.3: The adjacency graph of SPs (shown in Figure 5.2) of a dense residential sample image

found using the k-means clustering. The resulting matrix C would have duplicate entries because of nodes having the same labels and different numbers. These duplications will be used to count the frequencies of the cliques or the subgraphs. Finally, we build a matrix SG of size $(M \times 4)$ with $M \leq N$, where SG_{i1} , SG_{i2} and SG_{i3} contain the first, second and third node labels belonging to SG_i and SG_{i4} corresponds to the frequency of SG_i in the image.

After computing the frequent SGs from each image in each category and we store all the unique SGs from all categories in one global matrix. Then, the frequency is computed using the whole dataset and not using only a specific category. Two examples of the SGs containing the labels (116, 56, 198) and (176, 130, 54) are highlighted in red and green colours respectively in Figure 5.4.

In Figure 5.5, the SGs composed of land cover types (116, 56, 198) is shown overlaid on the aerial image. The label 56 corresponds to patches that are dominated by concrete, label 198 corresponds to mixed grass, shadow and roof and label 116 to mixed grass and tree. This combination of patches can be frequent in residential areas but can rarely be found in, for example, wetlands, harbours or deserts, so it can be a discriminative feature in the classification of the RS scenes.

5.2.6 Histogram of Subgraphs

The histogram of subgraphs encodes the global frequency of subgraphs in the whole dataset. In order to classify images, we tested with several numbers of SGs. Thus, the dimension of the histogram ranges from a minimum of 50 SGs to a maximal value of 1000 SGs. The SGs are sorted in a descending order of their frequencies and

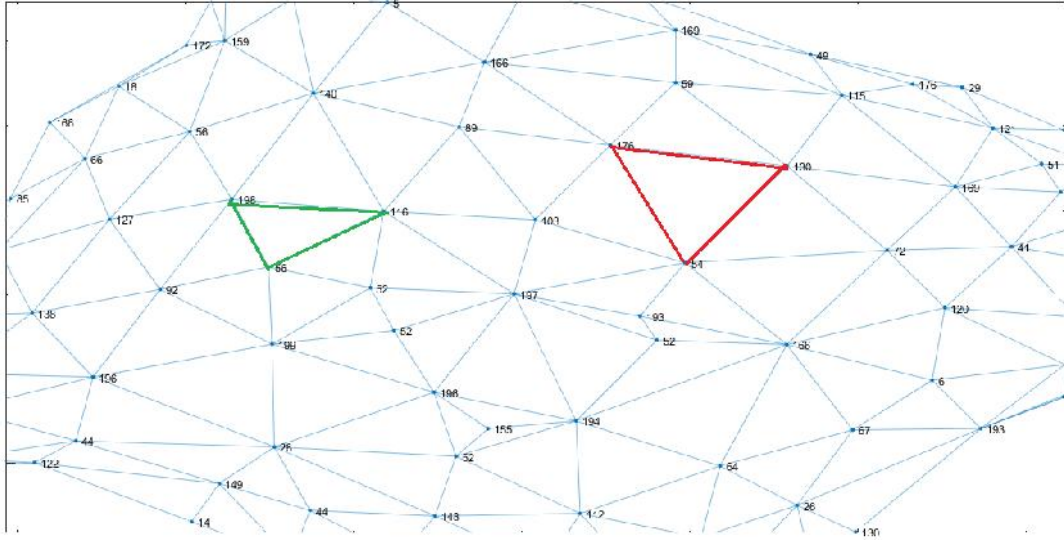


FIGURE 5.4: Samples of the SG having the node labels (116,56,198) and (176,130,54) illustrated in red and green colours



FIGURE 5.5: Illustration of the subgraph (116,56,198) on the "dense residential" image

we pick the first 50 SGs for example to perform the classification. In the FV of size 50, the first indice represents the number of occurrences of the most frequent SG in the dataset for a given image i belonging to a category j . This constitutes the histogram of frequent SGs for each image as illustrated by Figure 5.6 in which the SGs 19 and 35 are the most frequent (2 times).

5.2.7 Classification of RS Scenes

After choosing the number of SGs to use, each image is now represented by a histogram containing the local frequencies of "frequent SGs" in the image. Finally, we randomly split the histogram of SGs dataset (several times) into training and testing sets. We tested with different training and testing ratios in order to test whether our method is sensitive to the size of the training set and whether it generalizes well to the testing set. Then, we perform the classification using multiple classifiers including SVM and Bagged Trees. The classifiers are tested and the best accuracies are retained. Mainly, varieties of SVM (except the Linear SVM), Fine KNN and Random forest give better results than Weighted KNN, Medium KNN.

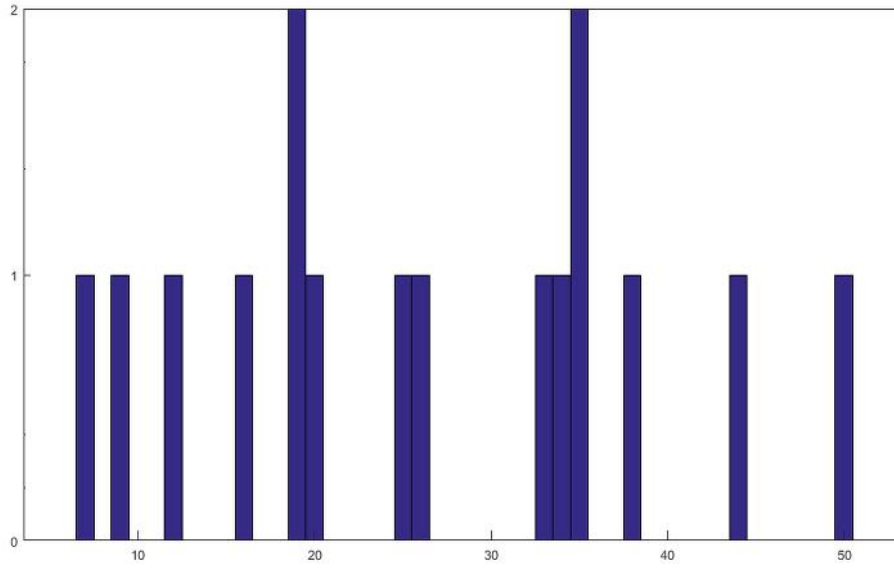


FIGURE 5.6: Illustration of the histogram of frequent SGs of the 'dense residential' image with 50 SGs

5.3 Complexity of the Proposed Algorithm

Let C be the number of categories (21 for UC Merced dataset), N be the number of images per category ($N = 100$ in UC Merced dataset), M be the number of pixels in each image ($M = 256 \times 256$ in UC Merced dataset), P be the number of superpixels in each image, let S be the total number of subgraphs in the whole dataset with no repetitions, S_m be the maximal number of subgraphs found among the C categories. Let also D be the total size of the dataset with $D = N \times C \times M$, and L be the total size of the feature vector of colour moments and Gabor texture, $L = N \times C \times P$. We set also T to be the total number of images. $T = N \times C$. So the temporal complexity is expressed by the Equation 5.5:

$$O(\text{BoVSG}) = O(D + (FV)^2 + \frac{T \times 3^P}{3} + S \times C \times S_m + N^2 \times C \times S)) \quad (5.5)$$

where $O(N \times C \times P)$ is the complexity of the SLIC over-segmentation step, $O((N \times C \times P)^2)$ is the complexity of k-means algorithm, while $N * C * P$ is the size of feature vector,

$O(N \times C \times \frac{3^P}{3})$ is the cost of maximal clique computation where $O(\frac{3^P}{3})$ is the worst case complexity of Bron–Kerbosch algorithm (**Moon 1965**).

$O(S \times C \times S_m)$ is the complexity of the computation of subgraphs frequencies, since the number of categories is small, we can state that $O(S \times C \times S_m) = O(S \times S_m)$, while S_m is always less than S , thus if we suppose that $S_m = S$, then the complexity of subgraph frequencies computation will be $O(S^2)$.

$O(N^2 \times C \times S)$ is the cost of the computation of histogram of frequent subgraphs.

So in the worst case, our approach will have a quadratic temporal complexity either of the size of the feature vector of colour and texture or of the number of subgraphs.



FIGURE 5.7: Class representatives of the UC Merced data set. (1) Agriculture (2) Airplane (3) Baseball diamond (4) Beach (5) Buildings (6) Chaparral (7) Dense residential (8) Forest (9) Freeway (10) Golf course (11) Harbor (12) Intersection (13) Medium residential (14) Mobile home park (15) Overpass (16) Parking lot (17) River (18) Runway (19) Sparse residential (20) Storage tanks (21) Tennis court

5.4 Experimental results

5.4.1 Datasets

In order to evaluate our approach, we consider three challenging datasets namely *UC Merced* dataset, the *NWPU-RESISC45* dataset and the *PatternNet* dataset. *UC Merced* covers many regions in United States and is composed of 21 categories each containing 100 images. Some image samples from this dataset are illustrated in Figure 5.7.

The *NWPU-RESISC45* is a publicly available benchmark for REMote Sensing Image Scene Classification (RESISC), created by Northwestern Polytechnical University (NWPU). It contains 45 classes each containing 700 RGB images of size 256×256 pixels. A sample from the *NWPU-RESISC45* dataset is shown in Figure 5.8.

The *PatternNet* dataset contains 38 classes each composed of 800 images as shown in 5.9.

These datasets are representative for RS scene classification. In the sub-section 5.4.2, we present the classification accuracies of our method and a comparison with the-state-of-the-art methods in the sub-section 5.4.3.

5.4.2 Classification performance

In order to test our approach, we used several configurations based on the length of the FV. We kept increasing the number of frequent SGs used ranging from 50 SGs to 1000 SGs. We trained various classifiers on the FVs including Linear SVM (LSVM), Quadratic SVM (QSVM), Cubic SVM (CSVM), and Bagged Trees (BT). Using the most frequent SG on *NWPU-RESISC45* dataset (training 75%, testing 25%).



FIGURE 5.8: Sample of images in *NWPU-RESISC45* dataset (Cheng et al., 2017b)

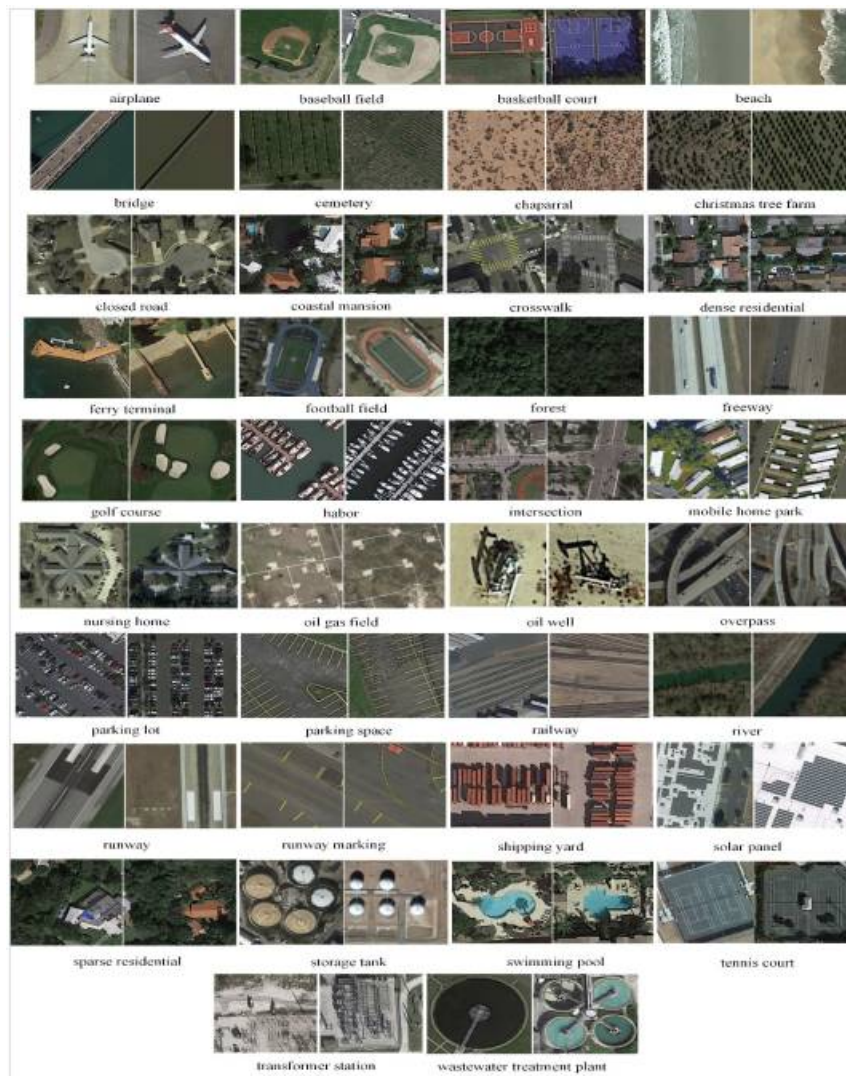


FIGURE 5.9: Sample of images in *PatternNet* dataset

# SG Classifier	BT	CSVM	QSVM	LSVM	WKNN	FKNN	MKNN	CT
1000	99.60%	99.10%	99.00%	91.11%	99.50%	99.60%	99.00%	99.60%
500	99.40%	98.70%	98.90%	98.40%	99.30%	99.30%	98.80%	99.40%
300	98.80%	98.50%	98.30%	97.80%	98.80%	98.60%	98.30%	98.70%
200	97.90%	98.00%	97.80%	89.61%	98.20%	97.90%	97.50%	97.40%
100	86.40%	87.30%	87.00%	73.70%	84.30%	83.60%	83.60%	87.50%
50	66.30%	70.30%	68.00%	65.50%	65.50%	61.90%	64.60%	72.40%

TABLE 5.1: Comparison of training performance of various classifiers using the histogram of frequent subgraphs on the *NWPU-RESISC45* dataset

From the results in Table 5.1, we got a training accuracy of 99.60%. From Table 5.2, we can conclude that a number of 300 SGs is sufficient to have the best classification accuracy 99.80% which is the best accuracy at testing phase using the BT classifier.

The classification accuracies on the *UCMerced* dataset for the training and testing phases are summarized in Tables 5.3 and 5.4 respectively. We can notice that the best accuracy is obtained using the BT classifier with a number of 1000 SGs in training and testing phases. Moreover, a number of 500 and 300 SGs gave very comparable results with a significant reduction of the size of the FV.

# SG Classifier	BT	CSVM	QSVM	LSVM	WKNN	FKNN	MKNN	CT
1000	99.87%	91.92%	93.10%	91.11%	94.51%	94.84%	94.08%	91.40%
500	99.54%	92.32%	92.24%	88.88%	93.82%	93.49%	93.61%	87.86%
300	97.35%	93.60%	93.68%	91.59%	90.90%	91.78%	90.55%	88.65%
200	98.26%	91.80%	92.72%	89.61%	85.60%	84.81%	85.45%	84.20%
100	78.39%	75.19%	76.85%	73.70%	70.46%	68.57%	69.84%	54.74%
50	55.14%	56.14%	59.92%	56.46%	52.86%	50.03%	51.17%	54.74%

TABLE 5.2: Comparison of testing performance of various classifiers using the histogram of frequent subgraphs on the *NWPU-RESISC45* dataset

# SG Classifier	BT	CSVM	QSVM	LSVM	WKNN	FKNN	MKNN	CT
1000	99.70%	98.30%	97.70%	97.00%	98.20%	99.50%	96.60%	98.30%
500	99.40%	98.00%	97.50%	96.70%	98.20%	99.30%	96.00%	98.30%
300	99.60%	98.30%	97.80%	96.10%	98.00%	99.40%	96.30%	97.70%
200	99.40%	98.30%	97.80%	96.40%	97.80%	99.30%	96.00%	97.30%
100	98.70%	95.20%	95.70%	94.70%	96.40%	98.60%	93.90%	95.60%
50	98.70%	95.20%	95.70%	93.90%	91.50%	94.50%	87.00%	95.60%

TABLE 5.3: Comparison of training performance of various classifiers using the histogram of frequent subgraphs on the *UC Merced* dataset

# SG Classifier	BT	CSVM	QSVM	LSVM	WKNN	FKNN	MKNN	CT
1000	99.62%	98.86%	98.86%	97.33%	98.48%	99.43%	97.33%	99.05%
500	99.43%	99.43%	97.90%	96.57%	98.67%	99.62%	97.52%	99.24%
300	99.81%	99.43%	97.90%	97.14%	99.43%	99.81%	98.29%	98.67%
200	99.81%	99.05%	98.10%	97.33%	99.05%	99.30%	98.29%	98.48%
100	99.05%	96.57%	95.62%	95.00%	98.33%	98.81%	95.24%	95.81%
50	99.05%	96.57%	95.62%	95.00%	98.33%	98.81%	95.24%	95.81%

TABLE 5.4: Comparison of testing performance of various classifiers using the histogram of frequent subgraphs on the *UC Merced* dataset

Performing the same tests on *PatternNet* dataset gives interesting results that are illustrated in Tables 5.5 and 5.6. The best results on the *PatternNet* are highlighted in bold in both tables generally given by the Bagged Trees classifier. It is always preferable to obtain good results with small size FV, which corresponds to a number of 300 SGs.

5.4.3 Comparison of the classification results

We compare the performance of our approach to the-state-of-the-art methods on the *NWPU-RESISC45* and the *UC Merced* datasets and we report the results in Tables 5.7 and 5.8.

The *PatternNet* dataset is a recent dataset that is primarily proposed for image retrieval, thus there are few works dealing with its classification. We cite the recent work of (Scott et al., 2018). Our approach outperforms, however, their work which has an accuracy of 99.70% against 99.84% for our approach.

# SG Classifier	BT	CSVM	QSVM	LSVM	WKNN	FKNN	MKNN	CT
1000	99.80%	99.60%	99.40%	99.00%	99.70%	99.60%	99.40%	97.20%
500	99.80%	99.60%	99.40%	99.10%	99.70%	99.00%	99.40%	96.30%
300	99.70%	99.50%	99.40%	99.10%	98.40%	97.80%	98.20%	96.80%
200	99.70%	99.50%	99.40%	99.10%	98.00%	97.10%	97.70%	96.90%
100	99.70%	99.40%	99.30%	98.90%	96.70%	96.80%	95.70%	94.80%
50	99.60%	99.20%	99.00%	98.30%	94.40%	95.40%	92.30%	91.70%

TABLE 5.5: Comparison of training performance of various classifiers using the histogram of frequent subgraphs on the *PatternNet* dataset

# SG Classifier	BT	CSVM	QSVM	LSVM	WKNN	FKNN	MKNN	CT
1000	99.82%	99.67%	99.53%	99.13%	99.66%	99.63%	99.66%	97.08%
500	99.82%	99.64%	99.54%	99.13%	99.70%	99.09%	99.62%	96.64%
300	99.79%	99.61%	99.53%	99.29%	97.67%	97.92%	97.50%	96.55%
200	99.74%	99.62%	99.46%	99.22%	96.97%	97.16%	96.80%	97.18%
100	99.79%	99.57%	99.12%	99.12%	96.53%	96.91%	95.80%	94.76%
50	99.63%	99.41%	99.22%	98.53%	94.53%	95.66%	92.86%	91.45%

TABLE 5.6: Comparison of testing performance of various classifiers using the histogram of frequent subgraphs on the *PatternNet* dataset

Considering the classification accuracies shown in Tables 5.2, 5.4 and 5.6 on the three considered datasets, we clearly see that our approach outperforms the-state-of-the-art methods. It gives interesting results even with a small number of SGs.

In order to evaluate whether our approach is sensitive to the amount of training data, we test different training set ratios for the three datasets. We report the classification accuracy of the *UC Merced* dataset while varying the number of SGs and the training ratio in Figure 5.10. We use 1000, 500, 200 and 50 SGs and a training ratio ranging from 25% to 75%

We notice from Figure 5.10 that our approach performs very well even with a training ratio of 30% and 50 SGs with an accuracy of 98.03%. This result is confirmed by the tests on *PatternNet* dataset as shown in Figure 5.11.

Figure 5.11 shows that our approach is able to learn from small training set and is able to generalize very well to the testing set. This confirms that the feature vector or the frequent subgraphs have a great discriminative power that allows us to separate scene classes efficiently.

Method	Overall accuracy
BoCF (Cheng et al., 2017b)	84.32%
BoVW (Cheng et al., 2017b)	44.97%
(Cheng et al., 2017a)	90.36%
Hydra (Minetto et al., 2018)	95.00%
MSCP (He et al., 2018)	88.93%
MSDF(Yuan et al., 2018)	84.30%
(Jiang et al., 2018b)	83.35%
MSFF (Yang et al., 2018)	99.33%
IORN (Wang et al., 2018)	91.30%
HML (Goel et al., 2017)	84.00%
ADSSM (Zhu et al., 2018b)	94.29%
DNN (Buscombe and Ritchie, 2018)	89.00%
LGFF (Zhu et al., 2018a)	96.37%
DCCA (Zheng et al., 2018)	93.47%
TrAdaBoost (Yang et al., 2018)	87.13%
SSGF (Han et al., 2017)	88.60%
Our approach (BoVSG)	99.87%

TABLE 5.7: Comparative table of the classification performance of the state of the art methods on the NWPU-RESISC45 dataset

Method	Overall accuracy
BOVW (Liu et al., 2018)	72.05%
SPCK++ (Yang and Newsam, 2011)	77.38%
PLSA (Liu et al., 2018)	80.71 %
SIFT+SC (Cheriyadat, 2014)	81.67 %
LDA (Liu et al., 2018)	81.92 %
SPM+SIFT (Zhao et al., 2016a)	82.30 %
S-UFL (Zhang et al., 2015)	82.72 %
LLC (Zhao et al., 2016a)	82.85 %
SAL-PTM (Zhong et al., 2015)	88.33 %
LPCNN (Zhong et al., 2016)	89.90 %
FK-S (Zhao et al., 2016c)	91.63 %
SSBFC (Zhao et al., 2016a)	91.67 %
DMTM (Zhao et al., 2016b)	92.92 %
Multiview CNN (Luus et al., 2015)	93.48 %
GBRCN (Zhang et al., 2016)	94.53 %
Multiscale ADPM (Liu et al., 2016a)	94.86 %
Fine-tuned CaffeNet (Castelluccio et al., 2015)	95.48 %
SPP-net+MKL (Liu et al., 2016b)	96.38 %
MCNN(Liu et al., 2018)	96.66 %
Fine-tuned GoogleNet (Castelluccio et al., 2015)	97.10 %
VGG-S + IFK (VGG VD16) (Hu et al., 2015a)	98.49 %
Our approach (BoVSG)	99.81%

TABLE 5.8: Comparative table of the classification performance of the state of the art methods on the UC Merced dataset

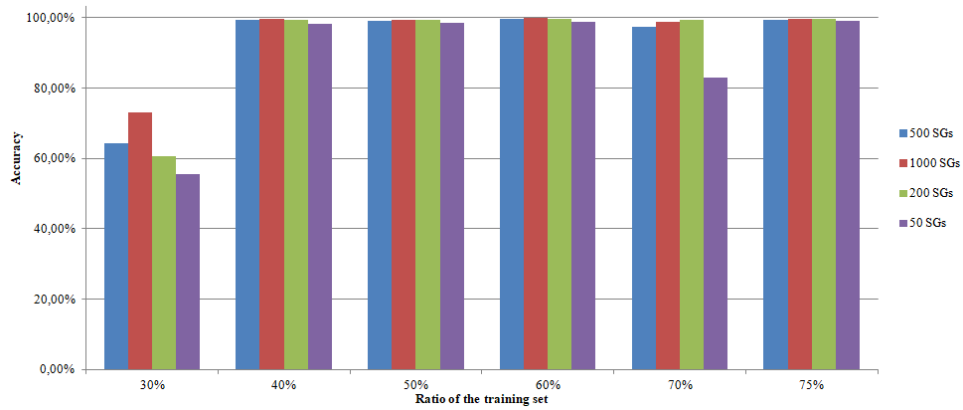


FIGURE 5.10: Classification accuracy of the UC Merced dataset with different training set ratios and SGs

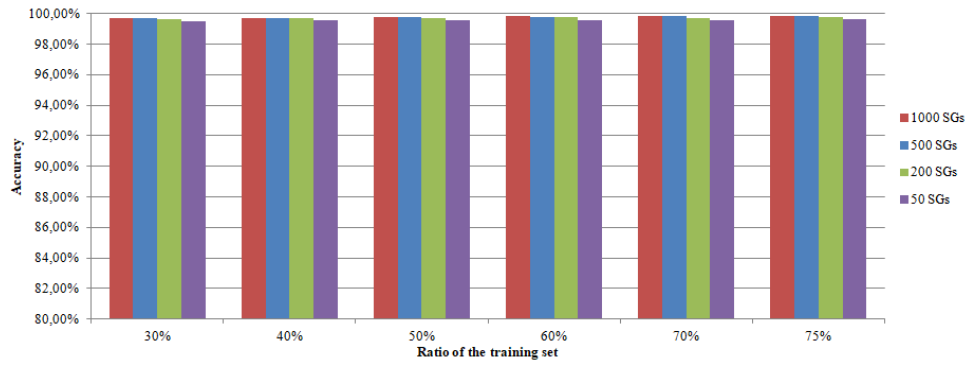


FIGURE 5.11: Classification accuracy of the *PatternNet* dataset with different training set ratios and SGs

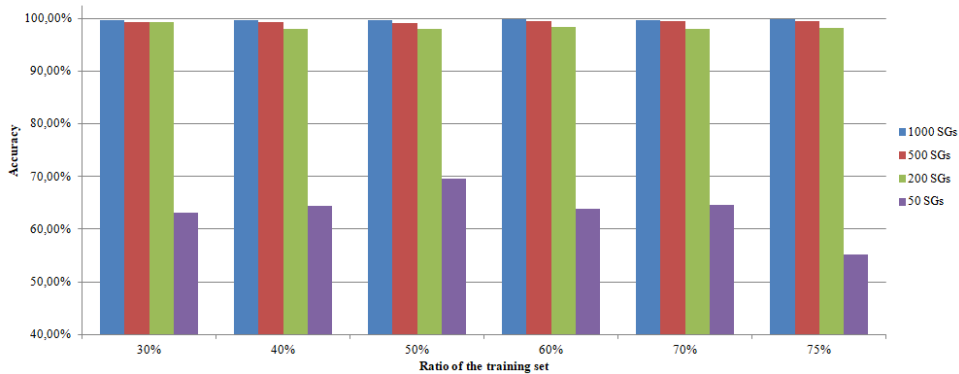


FIGURE 5.12: Classification accuracy of the *NPWU-RESISC45* dataset with different training set ratios and SGs number

The same results are confirmed on the *NPWU-RESISC45* dataset with less performance especially for the case of 50 SGs. Thus, we can clearly see that the classification accuracy decreases considerably with 50 SGs whatever is the size of the training set, when compared with a number of 200 SGs and above. We obtained an accuracy of 60.35% with 50 SGs against 99.84% for 1000 SGs and 75% training ratio in both

cases. This can be explained by the fact that in *NPWU-RESISC45* dataset, the object representing the class is smaller than the background. Thus, the frequent SGs found are common to many classes and this decreases their discriminative power.

5.5 Conclusion

In this chapter, we presented a new scene classification approach that combines the SPs and the BoVW representation. We start by segmenting the image into a set of superpixels using the SLIC algorithm. Then, the features of SPs are extracted. The colour features represent the spectral properties of the SP area, while the Gabor features represent their texture. The FV of all SPs from all the images in the dataset are clustered using k-means. Consequently, each SP is replaced by a label (representing a land cover type) corresponding to the centroid number of the cluster. The image is converted into a graph where adjacent SPs are connected. This way, each image is represented as a graph with minimal loss of information. Thereafter, the classification is performed based on the histograms of frequent subgraphs that are computed with the Bron-Kerbosh algorithm. In the BoVW model, SGs are equivalent to words.

We showed that the proposed approach is accurate when compared to the-state-of-the-art methods in terms of overall and average accuracies. The closest competitive methods are generally deep learning based methods. However, our proposed method requires less computation time. Additionally, our method is based on intuitive features that make sense and can be interpreted unlike deeply learned features which are "black box" and very difficult to interpret, if not impossible.

Although the results are satisfactory, there is still rooms for further improvements. We are currently investigating algorithms for the selection of sub-graphs based on their discriminative power rather than their frequency. Superpixel algorithms other than SLIC can be tested along with new features for the clustering of the superpixels.

Chapter 6

Conclusion and Perspectives

In this research, we shed the light on the profits and challenges of the annotation of remotely sensed images. This problem is challenging for humans as well as for machines. Generally, humans use their own knowledge and expertise to semantically annotate images. Thus, they can use objects that are visually present in the image, but also objects or features that do not exist visually (such as time, location, events, etc). Automatic image annotation is gaining an increasing interest for the scientific community. Although the improvements that have been done, there are still rooms for enhancement and improvement of annotation accuracy. In this last chapter, we summarize the research and the contributions explained in the previous chapters and we present some future works and questions.

6.1 Summary of annotation research

In this thesis, we proposed a new approach to reduce the semantic gap problem in RS images via automatic image annotation. However, it is sometimes confused with image classification. Thus, we presented in Chapter 2 the formal definitions of the two problems. Then, we reported a set of relevant annotation and classification methods proposed for RS images. Generally, the existing methods combine low level features with the rich spectral information offered by RS images to deal with the semantic gap problem. Several methods also considered the spatial information in order to give richer annotations. The annotation methods start by the segmentation step which has a big effect on the annotation (as well as the classification) performance. There are many existing segmentation methods in the literature and several methods were tested by the annotation and classification methods. However, there exist no perfect segmentation, and it depends on the image characteristics and the chosen scale, etc. Since the spatial resolution of the available RS images is low, the segmentation methods tend to merge the objects. We believe that the over-segmentation can result in a better annotation performance. The segmentation is generally followed by a feature extraction step which aims to summarize image characteristics without loss of the relevant information. Feature extraction is a very important step that greatly influence the performance. On the other hand, the main limit of existing annotation methods, is the small number of produced keywords. Keywords reflect the image content which consists most of the time in a set of objects but with no indication of high level details such as location or time. Consequently, several works propose to use the available image details as well as a priori knowledge to enhance the produced descriptions or representations .

In order to efficiently annotate RS images, we propose to exploit the rich spectral information of images jointly with the spatial information by using an extended region adjacency graph as illustrated in Chapter 4. Thus, we start by the extraction

of a set of radiometric indices (RI) in order to incorporate the spectral characteristics of objects. Then, we test the discriminative power of the extracted RI for the classification of hyperspectral images (HSI) in Chapter 3. As shown in the experimental results section, the obtained classification accuracy is very interesting on a set of well known datasets. Motivated by the good accuracy of RI based classification, we use them for the enrichment of the largest existing satellite scene ontology and also for the annotation of HSI and MSI images (Chapter 4).

Due to the complexity of the annotation task, using a priori knowledge seems to be reasonable and can reduce the semantic gap and enrich the annotation results. A priori knowledge are generally encoded in several structures such as semantic networks, taxonomies, vocabularies, domain ontologies, etc. In the recent years, knowledge structures, especially domain ontologies, are gaining popularity. In fact, knowledge encoding can have great effect on many applications. With the growing availability of RS data and of spectral libraries, we can notice that there is an interesting quantity of information that can be included in RS or geographical ontologies. Thus, we propose in Chapter 4 to enrich the *SatelliteSceneOntology-V3* conceptually by adding new concepts and intentionally by adding a priori values. Once the RS ontology is enriched, we can use it to infer new knowledge and to improve the annotation results. We also tested the RI for the annotation of HSI jointly with the USGS spectral library. Using precision, recall, $F1$ and $N+$ scores, we assess the annotation performance and we got a competitive annotation accuracy. In addition, we produce a graph of concepts that depicts the spatial relations between image regions. The adjacency relation is directly encoded, however, the directional and topological relations can be deduced from the coordinates of region centroids stored as node attributes. This representation of image annotation is very intuitive and visually easy to interpret by humans. Also, further tasks can be easily performed on this graph such as knowledge inference via the use of ontologies (both structures are graphs) and image retrieval based on spatial queries.

Always convinced by the graph modeling and since the RI cannot be computed for all types of RS images, we propose to use RAG for the classification of aerial images. Based on color moments and Gabor filter features, we cluster image superpixels using k-means algorithm then we represent the image by a RAG. Therefore, we mine the frequent subgraphs (based on maximal cliques) and compute their global frequencies in the whole dataset. Finally, we use the histogram of subgraphs for the classification of RS scenes. We assess the classification accuracy using the overall and the average accuracies. The obtained results are very promising and outperform the existing methods.

6.2 Perspectives

Despite the good performance that we achieved for both RS image classification and annotation, there are several enhancements that can be performed. We can start with the enrichment of the satellite scene ontology. Future works include:

- Computing the feature vectors of the intermediate nodes: we can aggregate the feature vectors of the children concepts and use them for the parent nodes.
- Also, since we have few available hyperpectral images, we might add crops representing the concepts from the RGB datasets. In fact, these datasets are more available and represent many largely used concepts. Thus we can represent efficiently more concepts.

- We can add also the details of sensors either by including them in the enriched ontology or by using existing sensor ontologies such as Semantic Sensor Network (SSN) and Sensor, Observation, Sample, and Actuator (SOSA) ontologies.
- The addition of the temporal details about images
- The addition of spectral signature plots from the USGS library
- Finally, we can add rules in order to infer and reason about the stored knowledge in the satellite scene ontology.

These enhancements in the enrichment of the satellite scene ontology will have a direct effect on the annotation results. We expect that the annotation produced will be richer and closer to human annotations. Further work can be performed on the subgraphs used for the RS scene classification presented in Chapter 5.

- We work on the proposal of an algorithm to filter the produced subgraphs in order to reduce their number and the temporal complexity
- We intend also to test other types of subgraphs instead of using maximal cliques. We might use paths, the order of nodes or other characteristics of graphs, etc
- We also plan to use these subgraphs in order to annotate RS scenes efficiently. In fact, the available annotated datasets that we used for the classification, are annotated with only one label per image. However, the images may contain many objects and reflect more than one concept. Thus, by using the produced subgraphs, we might find the pairs (subgraph, class) and consequently be able to annotate the RS scenes.

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